HYPERBOLIC MODELS AND STABILITY RECOGNITION FOR COXETER GROUPS

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Abstract. Given a finite rank Coxeter system, we construct a hyperbolic space by coning off the wide parabolic subcomplexes of its Davis complex. The resulting space is "stability recognizing", in the sense that stable subspaces of the Davis complex are exactly the quasigeodesically connected subspaces whose images in the coned-off space are quasiisometrically embedded. Equivalently, the coned-off space is "Morse recognizing": a quasigeodesic in the Davis complex is Morse if and only if it is quasigeodesic in the coned-off space.

Similar results are true for the mapping class group of a hyperbolic surface acting on its curve graph. We strengthen this analogy with additional results. First, the action of the Coxeter group on the coned-off space is acylindrical and gives the largest acylindrically hyperbolic structure for the Coxeter group. Second, the stable subgroups of a Coxeter group are characterized as the finitely generated subgroups that are undistorted and purely loxodromic with respect to the action on the coned-off space.

The coned-off space, being constructed directly from the Coxeter structure, facilitates concrete characterizations of Morseness and stability for two important Coxeter-native families of objects: combinatorial geodesics and special subgroups.

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Date: September 25, 2026.

2020 *Mathematics Subject Classification.* Primary 20F65; Secondary 20F55, 20F67.

Key words and phrases. Coxeter group, Davis complex, Morse recognition, stable subgroup, acylindrical action.

This research was supported in part by the Austrian Science Fund (FWF) 10.55776/PAT7799924, NSF DMS-2304920, and NSF DMS-2441034.

1. Introduction

This paper is about the coarse geometry of finitely generated Coxeter groups, as revealed by their wall structure and parabolic subgroup structure. The CAT(0) geometry of a finite rank Coxeter system (W, S) is described by its Davis complex Σ, which admits a piecewise Euclidean, CAT(0) metric d_2. Our goal is to understand the large scale hyperbolic aspects of the geometry, which include the *Morse property* and *stability* for subspaces. The Morse property requires that a quasigeodesic with both endpoints in the subspace stays close to the subspace. A subspace is stable if it contains uniformly Morse quasigeodesics between every pair of points. In hyperbolic spaces these properties are equivalent to each other and to quasiconvexity. A geodesic metric space is hyperbolic if and only if its geodesics are uniformly Morse, so we think of identifying the Morse geodesics as identifying the "hyperbolic directions" in the space.

The main thesis of this paper is that hyperbolic aspects of Coxeter groups can be described and studied in close analogy with those of the mapping class group (MCG) of a hyperbolic surface.

We recall several features of MCGs that motivate our Coxeter group constructions. MCG has a geometric action on its marking complex and a proper, but not cocompact, action on its Teichmüller space. Work of Minsky shows Teichmüller geodesics are Morse if and only if they avoid the thin parts of Teichmüller space [83], which have approximate product structures [82], making them the obvious sources of non-hyperbolicity. Masur and Minsky [80, 81] showed that the curve graph for the surface is a hyperbolic graph that is quasiisometric to the space obtained by coning off the thin parts of Teichmüller space and to the space obtained from the marking complex by coning off subcomplexes corresponding to cosets of curve stabilizers, one for each orbit of essential simple closed curve in the surface. The elements of MCG that act loxodromically on the curve graph are precisely the pseudo-Anosov mapping classes. Bowditch [30] proved that the action of MCG on its curve graph is acylindrical. Abbott, Balasubramanya, and Osin [1] showed that this action gives the largest acylindrically hyperbolic structure on MCG. Kent and Leininger [73] proved a result that, in an equivalent formulation established by Durham and Taylor [58], says that the curve graph is a stable subgroup recognizing space for MCG, in the sense that a subgroup of MCG is stable if and only if the orbit map into the curve graph is a quasiisometric embedding. Bestvina, Bromberg, Kent, and Leininger [23] gave an element-wise, dynamical characterization of stability: an undistorted subgroup of MCG is stable if and only if its action on the curve graph is purely loxodromic, that is, all infinite order elements act loxodromically.

In the Coxeter group setting, the *wide parabolic subgroups* are the obvious sources of non-hyperbolicity. Here, "wide" means the canonical direct product splitting either contains multiple infinite factors or contains a factor that is of higher rank affine type. Wide parabolic subgroups are not hyperbolic, and Moussong's characterization of hyperbolicity in Coxeter groups [85, 84] can be rephrased: W is hyperbolic if and only if it has no wide parabolic subgroups.

We construct a geodesic hyperbolic metric space $(\widehat{\Sigma}, \hat{d})$, the *coned-off space*, by coning off subcomplexes of the Davis complex Σ corresponding to cosets of wide parabolic subgroups.

Theorem (Corollary 4.20 and Proposition 4.15)**.** *There is a hyperbolic, geodesic metric space $(\widehat{\Sigma}, \hat{d})$ obtained by coning off the wide parabolic subcomplexes of Σ. Moreover, $(\widehat{\Sigma}, \hat{d})$ is "stability recognizing", in the sense that a quasigeodesically connected subspace of Σ is stable if and only if its natural map into $\widehat{\Sigma}$ is a quasiisometric embedding. Equivalently, $(\widehat{\Sigma}, \hat{d})$ is "Morse recognizing", in the sense that a quasigeodesic $\gamma\colon I \to (\Sigma, d_2)$ is Morse if and only if its natural composition into $\widehat{\Sigma}$ is also a quasigeodesic.*

An important ingredient in the "Morse recognizing" property is a complete characterization of the Morse property for a restricted class of quasigeodesic paths, Theorem 3.30, which is of independent interest, but whose statement is long and technical, so we do not detail it here.

Niblo and Reeves constructed a CAT(0) cube complex X_{NR} whose wall structure is isomorphic to that of Σ [86]. The action of W on X_{NR} is proper, but is cocompact if and only if W has no irreducible higher rank affine parabolic subgroups [40]. Still, our proof that $(\widehat{\Sigma}, \hat{d})$ is hyperbolic proceeds through the wall structure. We first construct a non-geodesic, quasiruled hyperbolic metric d_f in terms of the wall system, so it applies equally well to X_{NR} and to Σ, and then show that $(\widehat{\Sigma}, \hat{d})$ and (Σ, d_f) are quasiisometric, Theorem 4.19. We observe that the metric d_f collapses the difference between Σ and X_{NR}, Proposition 4.9, just as passage to the curve graph collapses the difference between the marking complex and Teichmüller space.

The cone-off construction introduces infinite valence vertices with infinite W–stabilizers. A weakening of the notion of a proper group action appropriate to such settings is acylindricity.

Theorem (Theorem 5.26). *The action $W \curvearrowright (\widehat{\Sigma}, \hat{d})$ is acylindrical.*

Moreover, this acylindrical action is *universal*, Corollary 5.29, meaning that $\widehat{\Sigma}$ sees all *generalized loxodromics* in the sense of Osin [89]. In fact, we prove a stronger statement:

Theorem (Theorem 5.30). *$\widehat{\Sigma}$ defines the largest acylindrically hyperbolic structure for W, in the sense of Abbott-Balasubramanya-Osin* [1].

Finally, as in the MCG case, stability detection in terms of a uniform subgroup-level condition can be replaced by a dynamical element-wise condition:

Theorem (Theorem 5.43). *A finitely generated subgroup of W is stable if and only if it is undistorted and purely loxodromic for the action $W \curvearrowright \widehat{\Sigma}$.*

1.1. **Benefits of the Coxeter-native construction.** The construction of the hyperbolic *fine chain metric* d_f is a mild variant of a construction of Genevois [62]. Genevois actually defines a parameterized family of hyperbolic metrics on a CAT(0) cube complex, such that quasigeodesics with a common Morse gauge are eventually recognized in a hyperbolic space with large enough parameter. Notably, our result says that for a given Coxeter group there is a single parameter at which we can already recognize all Morse quasigeodesics, so we only need one specific hyperbolic metric, not a parameterized family or a limit of such a family. There are several recent results [92, 93, 103] extending Genevois's idea by constructing, in great generality, hyperbolic spaces that detect strongly contracting/rank-one/Morse behavior. These results provide an abstract hyperbolic Morse recognizing space for Σ. Our approach is complementary. We transfer hyperbolicity of (Σ, d_f) to the coned-off space, which is Coxeter-native, in the sense of being defined directly in terms of the Coxeter structure. This is important for Theorems 5.26, 5.30, and 5.43, whose proofs take place in $(\widehat{\Sigma}, \hat{d})$, but it has additional direct Coxeter group payoffs—it enables us to give concrete characterizations of the Morse and stable properties for combinatorial geodesics and special subgroups.

Theorem (Theorem 3.31). *There are effective quantitative equivalences between the following items, uniformly over all combinatorial geodesics γ in Σ.*

1. *There exists μ such that γ is μ–Morse.*
2. *There is an upper bound on the lengths of subsegments γ' of γ for which there exists a wide parabolic subcomplex Υ with $\gamma' \subset \Upsilon$.*
3. *There is an upper bound on the lengths of subsegments γ' of γ for which there exists $T \subset S$ such that W_T is a wide special subgroup and all edges of γ' have label in T.*

Cordes and Levcovitz proved the special case of Theorem 3.31 for Coxeter groups that are "affine-free", meaning they have no irreducible higher rank affine parabolic subgroups [49, Theorem D]. They conjectured the result should be true without the affine-free hypothesis. Theorem 3.31 confirms that conjecture.

Theorem (Theorem 4.17). *For $T \subset S$, let Γ_T be the subdiagram of the Coxeter diagram of the finite rank Coxeter system (W, S) spanned by the vertices T. Call Γ_T "spherical" if W_T is spherical.*

W_T is stable if and only if for every wide special subgroup W_U the diagram $\Gamma_{T\cap U}$ is spherical.

W_T is Morse if and only if for every wide special subgroup W_U, one of the following is true:

1. *The intersection of every connected component of Γ_U with Γ_T is spherical.*
2. *Every nonspherical connected component of Γ_U is contained in Γ_T.*

Theorem 4.17 generalizes the corresponding results from the right-angled case, compare [102, Theorem 1.11], [62, Proposition 4.9], [97, Theorem 7.5].

1.2. **Why is this strategy successful for Coxeter groups?** There are two essential properties of Coxeter groups that enable the comparison to MCGs:

- Parabolic subcomplexes are gated.
- There is a local-to-global result relating the wall structure to the parabolic subgroup structure.

The first property means that the combinatorial closest point projection in the 1-skeleton of Σ is a gate map when the target is a parabolic subcomplex. This gives strong versions of projection and parallelism between parabolic subcomplexes that are very similar to the situation for convex subsets of CAT(0) cube complexes/median graphs. This is a special feature of Coxeter groups that arises from the combinatorial

group theory of reflection groups, not from CAT(0) polyhedral geometry. Indeed, even the CAT(0) geodesics of Σ are not gated in this way. This gate property is what makes our version of subsurface projection viable.

The second property is a result of Caprace [34], recalled in Section 2.1.3, that roughly says that a sufficiently complicated local pattern of wall intersections can only arise if the reflections in those walls are contained in a wide parabolic subgroup. This is a deep result proved using the Tits symmetric bilinear form on the canonical root system for (W, S), so it is very much a Coxeter-specific result. It is another case in which the effective behavior of the Coxeter group is like a cube complex, even when Σ is not cubical; specifically, in a CAT(0) cube complex large "grids" composed of transverse chains of walls identify flat/product-like behavior [66, 62]. In Coxeter groups two chains of walls can be transverse but not intersect at right-angles, so we do not immediately get a product-type parabolic subcomplex, but Caprace's result says that the only thing that can go wrong is that we might instead get an affine-type parabolic subcomplex. Both types of subcomplexes are wide, so our cone-off construction is collapsing exactly the regions where the wall structure gets complicated, leaving us to work with simpler wall combinatorics to understand large distances in $\widehat{\Sigma}$.

Beyond mapping class groups, stable subgroup recognition theorems are known for groups hyperbolic relative to peripherals with linear divergence [12], right-angled Artin groups (RAAGs) [76], and some hierarchically hyperbolic groups (HHGs) [2]. HHGs are a common generalization of toral relatively hyperbolic groups, MCGs, RAAGs, and right-angled Coxeter groups (RACGs) [20, 22]. Our coned-off space coarsely agrees with the construction of a universal acylindrical hyperbolic space for RACGs by Abbott, Behrstock, and Durham [2]. However, not all Coxeter groups are HHGs [91, 16][1], so simply being an HHG is not the mechanism for our results. The novelty of the approach in this paper is to handle the case of general Coxeter groups, including those with affine parabolics.

It is not known if general Coxeter groups act geometrically on a hierarchically hyperbolic space (HHS), and Coxeter groups are conjectured to be HHGs if and only if they are cocompactly cubulated [3]. It seems that any attempt to build an HHS structure for arbitrary Coxeter groups would require some choices in how to decompose affine parabolics, but the existence of non-HHG Coxeter group examples shows that such choices cannot always be made in a W-invariant way. We take a different approach. While we do develop a partial MCG/HHS-like toolbox for Coxeter groups, we do not have the full HHS suite. For example, we do not have hierarchy paths or a two-sided distance formula. Moreover, there are some differences in our approach from the typical MCG/HHS constructions. An example is that in the MCG setting, if $X \subsetneq Y \subsetneq Z$ are (nonannular) nested subsurfaces, they all have curve graphs, defined in the same way, and progress supported entirely in X is bounded in both curve graphs $C(Y)$ and $C(Z)$. In our Coxeter group setting, the coned-off graph for a proper parabolic subgroup is obtained by coning off all proper parabolic cosets, rather than only the wide ones, as in the case of the parent. The construction must be different because the two coned-off graphs play different roles in the proofs; coning off wide parabolics for W yields hyperbolicity, but for nested parabolics $Q \lneq P \lneq W$, progress supported in Q can still be large in $\widehat{\Sigma}$ when Q and P are not wide. Coning off all proper subparabolics of P permits an inductive descent on rank detecting whether progress in the coned-off graph of P is genuinely a P phenomenon or is actually supported in a deeper parabolic. When P is irreducible affine it has no proper nonspherical parabolic subgroups, so its subsidiary coned-off graph has the same quasiisometry type as P. In particular, the higher rank affine parabolics are not hyperbolic, so their coned-off graphs cannot be part of an HHS structure.

1.3. **Connections to other results in the literature.** Caprace and Fujiwara [36, Proposition 4.5], building on work of Caprace and Haglund [37], characterized *rank-one* elements of W. Their result is essentially the case of Theorem 3.30 in which a group element acts loxodromically on Σ with a Morse axis. It is not clear how to prove our Theorem 3.30 directly from their result. A rank-one geodesic in a CAT(0) space is one that does not bound a half-plane [18]. For a periodic geodesic in a proper CAT(0) space this is equivalent to being strongly contracting [25, Theorem 5.4], but this fact relies on periodicity; arbitrary geodesics may encounter an irregular collection of non-hyperbolic regions, as opposed to a single half-plane.

Recent work of Ciobanu and Genevois [47, Theorem 5.1] showed that the Caprace-Fujiwara rank-one elements actually have strongly contracting axes in $(\Sigma^{(1)}, d_1)$. Even more recently, work of Druţu, Spriano,

[1]Specifically, it follows from [16, Corollary 7.11] that if (W, S) is a Coxeter system such that there is a $T \subset S$ with $T^\perp$ spherical and T is affine of type $\tilde{A}_2$, $\tilde{A}_n$ for $n \geq 4$, $\tilde{E}_6$, $\tilde{E}_7$, $\tilde{E}_8$, $\tilde{F}_4$, or $\tilde{G}_2$, then W is not a hierarchically hyperbolic group. It turns out, applying work of Hagen [65], that the irreducible higher rank affine Coxeter groups of types $\tilde{A}_3$, $\tilde{B}_n$, $\tilde{C}_n$, and $\tilde{D}_n$ actually are cocompactly cubulated, even though their actions on their Niblo-Reeves cube complexes are not cocompact.

and Zbinden [57, Corollary 1.13, Example G.3], showed that Morseness and strong contraction are equivalent for quasigeodesics in $(\Sigma^{(1)}, d_1)$. We extensively use the corresponding equivalence in (Σ, d_2), which follows from the CAT(0) geometry. However, the two conclusions are separate; strong contraction need not be preserved by quasiisometries [14]. This is also not an HHG fact; Rafi and Verberne constructed an example of a pseudo-Anosov element whose quasiaxis in a Cayley graph of its mapping class group is not strongly contracting [95].

Behrstock and Charney [19] classified rank-one elements in RAAGs. Using similar techniques, Charney and Sultan [46] gave a characterization of Morse geodesic rays in uniformly locally finite CAT(0) cube complexes that is very similar to (4) $\iff$ (3) of our Theorem 3.30, except that they require control on a chain of walls boundedly spaced along the ray. In the non-right-angled Coxeter group setting, this is too much to ask, so instead we weaken that condition to a lower bound on chain density. Section 3.1.2 is largely an adaptation of Charney and Sultan's construction to Davis complexes in a way that relates the Morse property to wide parabolics. The divergence argument in Section 3.1.2 is essentially the same as that of Cordes and Levcovitz [49], but we need to expose additional characterizations of Morseness in terms of the wall structure to interface with later sections. Section 3.1 does not really use the Coxeter structure in an essential way; it is a geometric argument about discrete wall systems in a CAT(0) space.

Hagen's contact graph for CAT(0) cube complexes [66] and Kim and Koberda's extension graph for RAAGs [74, 75] were curve graph counterparts in MCG-analogous structure theories for their respective groups, prior to the unification of MCGs, RAAGs, and other cubical groups as HHGs.

Outside the HHS paradigm, the free factor complex and free splitting complex provide curve graph surrogates for the outer automorphism group of a free group in MCG-inspired approaches to the study of its large-scale geometry [69, 68, 24, 70], and, recently, Kudlinska and Petyt [78] constructed a Morse recognizing hyperbolic space for free-by-cyclic groups by coning off maximal product subgroups.

1.4. **Plan of the paper.** Section 2 is a preliminary section collecting existing results on the structure of Coxeter groups, coarse geometry, non-geodesic spaces, and variations of the Morse property.

Section 3 works towards the characterization, Theorem 3.30, of the Morse property for a restricted class of quasigeodesic paths; specifically, paths for which the inverse image of their intersection with any wall is a connected set in the parameter space. Both combinatorial and CAT(0) geodesics have this property. Section 3.1 recalls the construction of bridges in CAT(0) spaces and uses them to characterize Morse quasigeodesics in terms of their interaction with the wall structure. In Section 3.2 we use grids, as in [38], to make the translation between conditions on sets of walls and the existence of wide parabolics. A main contribution of this section, Proposition 3.27, is an inductive argument showing that when a quasigeodesic has a sufficiently long interaction with a grid, then it has a long subsegment contained in a fixed radius neighborhood of some wide parabolic subcomplex, possibly different than the one associated with the original grid.

The proof of Theorem 3.30 in Section 3.3 is largely a formal combination of the results of the two preceding subsections. The reinterpretation of this result in terms of edge labels in Theorem 3.31 requires additional work using Coxeter-theoretic facts.

In Section 4 we construct the fine chain metric and work out its properties. Then in Section 4.3 we prove Σ with the fine chain metric is quasiisometric to the coned-off space. Roughly, the proof compares, quantitatively, the ways in which combinatorial geodesics fail to make progress in the metrics d_f and $\hat{d}$ during intervals that they spend in a wide parabolic subcomplex.

Section 5.1 builds out the dictionary of Table 1. These tools are used in Section 5.2 and Section 5.3 to prove acylindricity and the characterization of stability in terms of being purely loxodromic, respectively. These proofs are remarkably similar in structure to their MCG/HHS predecessors, [20, Theorem 14.3] and [23, Main Theorem], but both require significant Coxeter input to replace parts of the proofs where the relevant MCG/HHS machinery is missing.

2. Preliminaries

Let (W, S) be a finite rank Coxeter system, meaning W is a Coxeter group with a finite set S of fundamental generators. A Coxeter system (W, S) determines a *Coxeter diagram* that encodes a group presentation for W. The vertices of the Coxeter diagram are in bijection with S, whose elements are the generators of W, each of order 2. When s and t are distinct elements of S and st has order $m \in \{2, 3, \dots\} \cup \{\infty\}$ in W, then

Mapping class group MCG(S)	**Coxeter system** (W, S)
Teichmüller space $\mathcal{T}(S)$	Niblo-Reeves cube complex X_{NR}
Marking graph $\mathcal{M}(S)$	Davis complex $\Sigma = \Sigma(W, S)$
Curve graph $\mathcal{C}(S)$	Universal coned-off space $\widehat{\Sigma}$
Proper subsurface Z, its curve graph $\mathcal{C}(Z)$, $d_{\mathcal{M}(Z)}$, and $d_{\mathcal{C}(Z)}$	Pre-wide parabolic P, its coned-off parabolic graph $C(P)$, d_P, and δ_P
Reducible element preserving Z	Element of the normalizer $N_W(P)$ of P
Complexity of Z	Rank of P
Purely pseudo-Anosov	Purely $\widehat{\Sigma}$-loxodromic
Convex cocompact	Stable

TABLE 1. MCG/Coxeter dictionary

the Coxeter diagram has no edge between s and t when $m = 2$ and has an edge labelled m between s and t otherwise. (By convention, the label 3 is usually not written.)

A subset $T \subset S$ determines a *special subgroup* W_T of W generated by T that is isomorphic to the Coxeter group defined by the sub-Coxeter system involving only the generators in T. The special subgroups and their conjugates are called *parabolic subgroups*. If P is a parabolic subgroup conjugate to a special subgroup W_T then $|T|$ is the *rank* of P, and P is called *reducible* if there exists a nontrivial decomposition $T = T_0 \sqcup T_1$ such that there are no edges in the Coxeter diagram between vertices of T_0 and vertices of T_1. This implies $W_T = W_{T_0} \times W_{T_1}$. The parabolic is *irreducible* if it is not reducible.

Conceivably, a parabolic subgroup P could be conjugate to two different special subgroups W_T and W_U, but when this happens there is a conjugation in W taking the set T bijectively to the set U [52, Proposition 4.5.10]. Since the edge labels of the Coxeter diagram are determined by the orders of products of generators, this further implies that the subdiagram spanned by vertices of T is isomorphic as a Coxeter diagram to the subdiagram spanned by vertices of U. Thus, the rank and reducibility of P are well defined, independent of which special subgroup is taken as a representative of its conjugacy class.

The intersection of two parabolic subgroups is a parabolic subgroup of (W, S), and also of the induced Coxeter systems of the two parabolic subgroups [52, Lemma 5.3.6]. In particular, if $P \lneq Q$ are parabolic subgroups then $\mathrm{rank}(P) < \mathrm{rank}(Q)$.

A Coxeter group or one of its parabolic subgroups is called *spherical* when it is a finite group.

We use the same adjectives for $T \subset S$ as for the corresponding special subgroup W_T; ie, "T is spherical" means W_T is spherical.

Associated to a Coxeter system, there is a canonical decomposition of W as a product of its irreducible special subgroups, $W = \prod_i W_{T_i}$, where the T_i are the vertex sets of the connected components of the Coxeter diagram. For our purposes it will often be convenient to combine the spherical factors, so $W = W_{T_1} \times \cdots \times W_{T_n} \times W_U$, where the T_i are the nonspherical irreducible subsets of S and U is the union of the spherical irreducible subsets of S. Since parabolic subgroups inherit a Coxeter system from (W, S), we also get canonical product decompositions of a parabolic subgroup into irreducible parabolics/nonspherical irreducible parabolics plus one spherical factor.

The *Davis complex* $\Sigma = \Sigma(W, S)$ of the Coxeter system is a cell complex whose cells correspond to cosets of spherical special subgroups. Its n–skeleton is denoted $\Sigma^{(n)}$. It admits a piecewise Euclidean, locally finite, CAT(0) polyhedral structure, constructed as in [52, Section 7.3 and Section 12.1]. We normalize so that edges have length 1 (by taking $\mathbf{d} \equiv 1/2$ in the construction of [52, Section 12.1]). With this normalization, the geometry of Σ is completely determined by the Coxeter system (W, S).

For background on Coxeter groups, see [52, 29, 72]. For background on CAT(0) metric spaces, see [31, 52].

For the remainder of the paper we regard (W, S) and Σ as fixed, so terms referred to as "constants", e.g. in Section 2.2.3, are constants with respect to this fixed Coxeter system and choice of metric on Σ.

2.1. **Coxeter groups and walls.**

2.1.1. *Background on Coxeter groups.* A *reflection* in (W, S) is a conjugate of an element of S. A *wall* in Σ is the fixed set of a reflection. The unique reflection fixing wall M and exchanging the two components of $\Sigma \smallsetminus M$ is denoted $\mathfrak{r}_M$. Unlike in the right-angled case, general Coxeter groups can have conjugate generators, so the reflection $\mathfrak{r}_M$ is not necessarily conjugate to a unique element of S.

If M and M' are distinct walls and $x \in M \cap M'$, then in the cell σ of Σ containing x in its relative interior the walls restrict to distinct Euclidean hyperplanes in the Euclidean polytope σ, so their intersection is transverse. Hence, distinct walls in Σ are either disjoint or intersect transversely, and in the latter case we write $M \pitchfork M'$.

Lemma 2.1. *Point stabilizers in Σ are spherical parabolic subgroups.*

Proof. This is observed immediately following [52, Definition 5.2.6]. □

Lemma 2.2 ([52, Corollary D.2.9])**.** *Every finite subgroup of W is contained in a spherical parabolic subgroup.*

Lemma 2.3. *The group $G := \langle \mathfrak{r}_M, \mathfrak{r}_{M'} \rangle$ generated by reflections in distinct walls M and M' is dihedral. It is finite if and only if $M \pitchfork M'$.*

Proof. A group generated by a pair of distinct involutions is dihedral. Distinct walls M and M' are transverse if and only if they intersect. If $x \in M \cap M'$ then $G \leq \mathrm{Stab}(x)$, which is a spherical parabolic subgroup, by Lemma 2.1. Thus, G is a finite dihedral group. Conversely, if G is finite then by Lemma 2.2 it is contained in wW_Tw^{-1} for some $w \in W$ and spherical $T \subset S$. That parabolic subgroup stabilizes the cell σ with vertices wW_T. By construction, σ is a convex Euclidean polytope and wW_Tw^{-1} acts on it by Euclidean isometries preserving wW_T, so fixing the Euclidean barycenter x of the vertices. Thus, $x \in \mathrm{Fix}(\mathfrak{r}_M) \cap \mathrm{Fix}(\mathfrak{r}_{M'}) = M \cap M'$. □

A *reflection subgroup* of (W, S) is a subgroup of W that is generated by reflections.

Theorem 2.4 (Dyer [59]/Deodhar[54])**.** *Every reflection subgroup W' of (W, S) is a Coxeter group, and admits a canonical Coxeter system (W', S') for which the reflections are exactly the reflections of (W, S) contained in W'.*

Every wall M is convex and cuts Σ into two components. We say M *separates* A and B if A and B are contained in different components of $\Sigma \smallsetminus M$. The components of $\Sigma \smallsetminus M$ are the *open halfspaces* corresponding to M. A *closed halfspace* of M is the union of M with an open halfspace of M.

By construction, walls miss the vertices of Σ and intersect higher dimensional cells in a Euclidean codimension–1 hyperplane. This gives a natural piecewise Euclidean structure on a wall M from its intersections with the cells of Σ. Similarly, there are no tangential intersections between walls M and subcomplexes Υ of Σ: either they are disjoint or M cuts an edge of Υ, in which case we say M *cuts* Υ. In particular, the following are equivalent: $M \cap \Upsilon \neq \emptyset$, M cuts Υ, M cuts an edge of Υ, Υ contains vertices from both components of $\Sigma \smallsetminus M$.

The connected components of Σ minus the walls are the *open chambers*. Each open chamber contains a unique vertex of Σ, and every vertex is contained in a chamber. By choosing a base vertex $\mathbb{1} \in \Sigma$ we get a bijection between W and vertices of Σ by referring to the vertex $w.\mathbb{1}$ as w. The chamber containing $\mathbb{1}$ is the *fundamental chamber*, whose closure is a fundamental domain for $W \curvearrowright \Sigma$. This choice of basepoint identifies the Cayley graph of W with respect to S with the 1–skeleton of Σ.

It will sometimes be convenient to replace Σ by a subdivision in which walls of Σ are actual subcomplexes. The barycentric subdivision has this property, but that subdivision is finer than necessary. Instead we define an intermediate complex:

Definition 2.5. Let Σ' be the polyhedral subdivision of Σ obtained by subdividing each cell of Σ along its intersections with the walls of Σ. Equivalently, the cells of Σ' are the intersections of cells of Σ with closed chambers. Since finitely many walls meet each cell, Σ' is still a locally finite complex. It is the coarsest subdivision of Σ in which walls and their intersections are subcomplexes. Its 1–skeleton is the Hasse diagram of the face poset of Σ, i.e. one vertex for every cell of Σ and an edge connecting a cell-vertex with its codimension-1 face-vertices.

With respect to our choice of $\mathbb{1}$, the vertices of Σ corresponding to elements of a special subgroup W_T span a subcomplex, denoted Σ_T, isomorphic to the Davis complex of the Coxeter system (W_T, T). Furthermore,

$\Sigma_T^{(1)}$ is d_1–convex in $\Sigma^{(1)}$, [52, Lemma 4.3.3], and Σ_T is d_2–convex in Σ, [101, Proposition A.2]. Similarly, a coset of a special subgroup wW_T corresponds to a *parabolic subcomplex* $w\Sigma_T$, which is the full subcomplex of Σ spanned by vertices in the coset wW_T. These subcomplexes are closely related to the *residues* used in the buildings literature. The parabolic subcomplex $w\Sigma_T$ corresponds to the residue that is the union of chambers containing the vertices of $w\Sigma_T$. The parabolic subcomplex and the residue have the same stabilizer, which is the parabolic subgroup wW_Tw^{-1}. However, a parabolic subgroup may be the stabilizer of multiple parallel parabolic subcomplexes; we return to this point in Corollary 2.17.

Lemma 2.6. *If $a\Sigma_U \subset b\Sigma_T$ then* $\mathrm{Stab}(a\Sigma_U) \subset \mathrm{Stab}(b\Sigma_T)$.

Proof. Since edges of $a\Sigma_U$ are labelled by elements of U and edges of $b\Sigma_T$ are labelled by elements of T, we have $U \subset T$. Passing to vertex sets, $aW_U \subset bW_T$, so $a = bc$ for some $c \in W_T$, so $cW_Uc^{-1} \subset W_T$. Thus:

$$\mathrm{Stab}(a\Sigma_U) = aW_Ua^{-1} = bcW_Uc^{-1}b^{-1} \leq bW_Tb^{-1} = \mathrm{Stab}(b\Sigma_T) \qquad \square$$

Corollary 2.7. *If Υ is a parabolic subcomplex, a wall M cuts Υ if and only if $\mathfrak{r}_M \in \mathrm{Stab}(\Upsilon)$.*

Proof. If M cuts Υ then it cuts an edge of Υ, which is a parabolic subcomplex contained in Υ and stabilized by $\mathfrak{r}_M$. Lemma 2.6 gives $\mathfrak{r}_M \in \mathrm{Stab}(\Upsilon)$. If M does not cut Υ then it separates Υ and $\mathfrak{r}_M(\Upsilon)$, so $\mathfrak{r}_M \notin \mathrm{Stab}(\Upsilon)$. $\square$

An *affine* Coxeter group is one that is virtually a Euclidean reflection group. Coxeter classified the irreducible spherical and irreducible affine Coxeter groups [50], see also [52, Table 6.1 and Appendix C]. In particular, examination of [52, Table 6.1] shows that every proper subdiagram of an irreducible Euclidean diagram is a disjoint union of irreducible spherical diagrams, giving the well-known fact that an irreducible affine parabolic subgroup has no proper nonspherical subparabolics.

Definition 2.8. A Coxeter system is called *irreducible higher rank affine* if it is irreducible and affine of rank at least 3, in the sense that the fundamental generating set has cardinality at least 3. This corresponds to the geometric rank, the dimension of the Euclidean space on which it acts, being at least 2.

Definition 2.9. If $\mathcal{M}$ is a set of walls in Σ and $M_1,\dots,M_n$ are additional walls, let $W(\mathcal{M}, M_1, \dots, M_n) := \langle \mathfrak{r}_M \mid M \in \mathcal{M} \cup \bigcup_{i=1}^n \{M_i\} \rangle$ be the subgroup generated by reflections in those walls.

For any subset U of W there is a unique smallest parabolic subgroup $\mathrm{Pc}(U)$ containing it, known as its *parabolic closure*. In particular, with notation as above, $\mathrm{Pc}(\mathcal{M}, M_1, \dots, M_n) := \mathrm{Pc}(W(\mathcal{M}, M_1, \dots, M_n))$.

By Theorem 2.4, $W(\mathcal{M}, M_1, \dots, M_n)$ is a Coxeter group. $\mathrm{Pc}(\mathcal{M}, M_1, \dots, M_n)$ is also a Coxeter group, since it is a parabolic subgroup of (W, S).

Definition 2.10. A *chain* $\mathcal{K} = \{K_i\}_{i\in I}$ in Σ, indexed by $\emptyset \neq I \subset \mathbb{Z}$, is a collection of disjoint walls K_i, such that $i < j < k$ in I implies K_j separates K_i and K_k. The number of walls in the chain is denoted $|\mathcal{K}|$.

Definition 2.10 is a slight abuse of terminology. The complementary halfspaces of the walls $K \in \mathcal{K}$ can be labelled K^+ and K^- in such a way that for $i < j$ we have $K_i^- \subset K_j^-$ and $K_i^+ \supset K_j^+$. Then the set of halfspaces K_i^-, ordered consistently with the indices, gives a chain in the partially ordered set of halfspaces with respect to inclusion. Thus, it is really the nested sequence of halfspaces that is the "chain". Since the ordered index set is part of Definition 2.10, $\mathcal{K}$ contains the same information as the nested sequence of halfspaces K_i^-, so the terminology is justified.

We say a wall M or path γ is *transverse to a chain* $\mathcal{K}$ if it is transverse to every wall in the chain, and write $M \pitchfork \mathcal{K}$ or $\gamma \pitchfork \mathcal{K}$, respectively. We say *two chains $\mathcal{V}$ and $\mathcal{H}$ are transverse*, and write $\mathcal{V} \pitchfork \mathcal{H}$, if $V \pitchfork H$ for all $V \in \mathcal{V}$ and all $H \in \mathcal{H}$.

The *extremes* of a chain are the at most two walls of the chain that do not separate any of the other walls of the chain, i.e., the first and last walls, if they exist.

If $\mathcal{K} := \{K_i\}_{i\in I}$ is a chain and $\emptyset \neq J \subset I$ then $\{K_j\}_{j\in J}$ is a *subchain* of $\mathcal{K}$. It is a *convex subchain* if $j_0, j_1 \in J$ and $i \in I$ with $j_0 < i < j_1$ implies $i \in J$.

Lemma 2.11. *If $\mathcal{K} := \{K_i\}_{i\in I}$ is a chain then $W(\mathcal{K})$ and $\mathrm{Pc}(\mathcal{K})$ are irreducible and there exist $i_0, i_1 \in I$ such that $\mathrm{Pc}(\mathcal{K}) = \mathrm{Pc}(K_{i_0}, K_{i_1})$. If $|I| > 1$ then $W(\mathcal{K})$ is infinite.*

Proof. If $\mathcal{K}$ has only one wall then $W(\mathcal{K}) = \mathrm{Pc}(\mathcal{K}) \cong \mathbb{Z}/2\mathbb{Z}$ is irreducible. Otherwise, it contains disjoint walls. Reflections in disjoint walls generate an infinite dihedral group, by Lemma 2.3, so $W(\mathcal{K})$ is infinite

once $\mathcal{K}$ contains distinct walls, and no two reflections through distinct walls of $\mathcal{K}$ commute. Irreducibility of $W(\mathcal{K})$ and $\mathrm{Pc}(\mathcal{K})$ then follows from [38, Lemma 2.2 (ii)].

For the claim that the parabolic closure is determined by reflections in at most two of the walls, [34, Lemma 17] (see Lemma 2.21) implies that if $i < j < k$ are distinct indices in I then $\mathfrak{r}_{K_j} \in \mathrm{Pc}(\mathfrak{r}_{K_i}, \mathfrak{r}_{K_k})$, so for any finite subinterval $[i_0, i_1] \subset I$ we have $\mathrm{Pc}(\{K_i\}_{i_0 \le i \le i_1}) = \mathrm{Pc}(\{K_{i_0}, K_{i_1}\})$. Take an exhaustion of I by finite subintervals. This gives a nested sequence of parabolic closures of the corresponding convex subchains. A strict inclusion of parabolic subgroups requires a strict increase in rank, but the ranks of parabolic subgroups are bounded above by $|S|$, so the sequence of parabolic closures stabilizes at some finite stage. □

2.1.2. *Parabolic centralizers and normalizers.* There has been extensive work on describing normalizers and centralizers of parabolic subgroups of (W, S) [32, 71, 8, 33, 28, 10, 11, 87]. The case of spherical parabolics is complicated. The nonspherical irreducible case is much simpler, as we now briefly recall.

Lemma 2.12. *A reflection subgroup of a direct product of special subgroups splits as a direct product of reflection subgroups of the factors.*

Proof. Suppose (W, S) is a Coxeter system and r is a reflection in $W_{T_0} \times W_{T_1} \le W$ for some $T_0, T_1 \subset S$. Then $r = wsw^{-1}$ for some $w \in W_{T_0} \times W_{T_1}$ and $s \in T_0 \sqcup T_1$ [52, Lemma 4.2.3]. Suppose $w = uv$ for $u \in W_{T_0}$ and $v \in W_{T_1}$. If $s \in T_0$ then $r = uvsv^{-1}u^{-1} = usu^{-1} \in W_{T_0}$. If $s \in T_1$ then $r = uvsv^{-1}u^{-1} = vsv^{-1} \in W_{T_1}$. Thus, each reflection is in one factor or the other, and the factors commute, by hypothesis. □

Proposition 2.13. *Let (W, S) be a Coxeter system. Let $T \subset S$ be irreducible and nonspherical. Let $T^\perp := \{s \in S \smallsetminus T \mid \forall t \in T,\ st = ts\}$. Then:*

(1) If $wTw^{-1} \subset S$ for some $w \in W$ then $wTw^{-1} = T$.
(2) The normalizer of W_T in W is $N_W(W_T) = W_T \times W_{T^\perp}$.
(3) The centralizer of W_T in W is $Z_W(W_T) = W_{T^\perp}$.
(4) Every reflection in $N_W(W_T)$ is either in W_T or in $W_{T^\perp}$.

Proof. These results are well known, see [39, Lemma 3.3], [34, Lemma 22 and the preceding line], [35, Lemma 2.2], [49, Theorem 5.7]. They are usually attributed to Deodhar [53, Proposition 5.5], with further explanation in [77, Section 3.1]. Deodhar's result, and the text prior to it, generalizes some root system facts used by Howlett in the description of parabolic normalizers for finite Coxeter groups [71], which would later be used by Brink and Howlett [33] to describe parabolic normalizers for infinite Coxeter groups.

The proposition is the irreducible nonspherical special case of the parabolic normalizer theorem obtained from Howlett's argument, together with Deodhar's refinements of the elementary conjugating elements. The final assertion follows from Lemma 2.12. □

Lemma 2.14 ([52, Proposition 4.5.10])**.** *If $T, U \subset S$ and $wW_Tw^{-1} = W_U$ then there is an element $u \in W_U$ such that for $v := u^{-1}w$ we have $vTv^{-1} = U$.*

Corollary 2.15. *If P is an irreducible nonspherical parabolic subgroup of (W, S), then there is a unique $T \subset S$ such that P is conjugate to W_T.*

Proof. If P is conjugate to W_U and W_T then Lemma 2.14 says $vTv^{-1} = U$. Then since P is irreducible and nonspherical, Proposition 2.13 (1) says $T = U$. □

The following corollary of Proposition 2.13 establishes the definition of the *orthogonal factor* parabolic $P^\perp$ of an irreducible nonspherical parabolic P.

Corollary 2.16. *If P is an irreducible nonspherical parabolic subgroup of (W, S), then $N_W(P)$ is parabolic and there is a canonical parabolic subgroup $P^\perp$ such that $N_W(P) = P \times P^\perp$.*

Furthermore, if $R \subset N_W(P)$ is a set of reflections then $\langle R \rangle = \langle R \cap P \rangle \times \langle R \cap P^\perp \rangle$.

Proof. By definition, there exist an irreducible, nonspherical $T \subset S$ and $w \in W$ such that $wPw^{-1} = W_T$. Proposition 2.13 says $wN_W(P)w^{-1} = N_W(W_T) = W_T \times W_{T^\perp}$ and $Z_W(W_T) = W_{T^\perp}$. Take $P^\perp := Z_W(P) = w^{-1}W_{T^\perp}w$. Then $N_W(P) = P \times P^\perp$.

The direct product splitting of $\langle R \rangle$ follows from the claim of Proposition 2.13 that reflections in $N_W(W_T)$ are contained in either W_T or $W_{T^\perp}$. □

Remark. The construction of Corollary 2.16 is symmetric if $T^\perp$ is also irreducible and nonspherical, but that is not always the case. In general, $W_T \times W_{T^\perp}$ can be a proper subgroup of $N_W(W_{T^\perp})$. In the most extreme case, $T^\perp$ could be empty, with $W_{T^\perp}$ trivial and $N_W(W_{T^\perp}) = W$.

Corollary 2.17. *If P is an irreducible nonspherical parabolic subgroup then there is a unique parabolic subcomplex whose stabilizer is $N_W(P)$.*

Proof. Suppose $P = xW_Tx^{-1}$ for some $x \in W$ and $T \subset S$. Proposition 2.13 (2) and Corollary 2.16 imply that $N_W(P) = x(W_T \times W_{T^\perp})x^{-1} = xW_{T\sqcup T^\perp}x^{-1}$. The parabolic subcomplex $x\Sigma_{T\sqcup T^\perp}$ has stabilizer $N_W(P)$. Suppose $w\Sigma_U$ for some $w \in W$ and $U \subset S$ is another parabolic subcomplex with stabilizer equal to $N_W(P)$. Its stabilizer is wW_Uw^{-1}, so $W_U = w^{-1}xW_{T\sqcup T^\perp}x^{-1}w$. By Lemma 2.14, there exists $u \in W_U$ such that for $v := u^{-1}w^{-1}x$ we have $v(T \sqcup T^\perp)v^{-1} = U$. Let U' be the irreducible component of U such that $vTv^{-1} = U'$. Since T is irreducible and nonspherical, Proposition 2.13 (1) implies $T = U'$, so $v \in N_W(W_T) = W_{T\sqcup T^\perp}$, which implies $W_{T\sqcup T^\perp}v^{-1} = W_{T\sqcup T^\perp}$. Then:

$$wW_U = xv^{-1}u^{-1}W_U = xv^{-1}W_U = xv^{-1}(vW_{T\sqcup T^\perp}v^{-1}) = xW_{T\sqcup T^\perp}$$

Thus, $x\Sigma_{T\sqcup T^\perp}$ and $w\Sigma_U$ are the same parabolic subcomplex. □

2.1.3. *Caprace toolbox.* We state some results of Caprace that we will use.

Theorem 2.18 ([34, Theorem 8])**.** *There is a constant N such that the following holds. Suppose $\mathcal{K} = \{K_i\}_{0\le i\le n}$ is a chain and M and M' are walls that are each transverse to $\mathcal{K}$ and such that $\emptyset \neq M \cap M' \subset K_0$. If $n > N$ then $W(\mathcal{K} \cup \{M, M'\})$ is a Euclidean triangle group that is contained in a higher rank irreducible affine parabolic.*

Lemma 2.19 ([34, Lemma 11])**.** *Let $\mathcal{M}$ be a set of walls with $W(\mathcal{M}) \cong \mathcal{D}_\infty$. If a wall M is transverse to at least 8 walls of $\mathcal{M}$ then it is transverse to all of them and either $\mathfrak{r}_M$ centralizes $W(\mathcal{M})$ or $W(\mathcal{M} \cup \{M\})$ is a Euclidean triangle group.*

Proposition 2.20 ([34, Proposition 16])**.** *If R is a higher rank irreducible affine reflection subgroup then so is its parabolic closure.*

Lemma 2.21 ([34, Lemma 17])**.** *Let $\mathcal{K}$ be a chain containing at least 2 walls. Let M be a wall. If one of the following is true then $\mathfrak{r}_M \in \mathrm{Pc}(\mathcal{K})$.*

- *M separates two walls of $\mathcal{K}$.*
- *$W(\mathcal{K} \cup M) \cong \mathcal{D}_\infty$*
- *$W(\mathcal{K} \cup M)$ is a Euclidean triangle subgroup.*

2.2. **Metric geometry.**

2.2.1. *Coarse geometry.* Let (X, d) be a metric space. For $x \in X$ and $Y \subset X$, $d(x, Y) := \inf_{y\in Y} d(x, y)$. For $R \in \mathbb{R}$, the *R–neighborhood* of Y is $\mathcal{N}_R(Y) := \{x \in X \mid d(x, Y) < R\}$, and the *closed R–neighborhood* is $\bar{\mathcal{N}}_R(Y) := \{x \in X \mid d(x, Y) \le R\}$.

Subsets Y and Z of X are *coarsely equivalent* if they have finite Hausdorff distance, i.e. $\inf\{R \mid Y \subset \bar{\mathcal{N}}_R(Z)$ and $Z \subset \bar{\mathcal{N}}_R(Y)\}$ is finite.

For functions ϕ and ψ from X^n to $\mathbb{R}$, write $\phi \preceq \psi$ if there exist λ and ϵ such that for all $\underline{x} \in X^n$ we have $\phi(\underline{x}) \le \lambda\psi(\underline{x}) + \epsilon$. Write $\phi \overset{+}{\preceq} \psi$ if $\phi \preceq \psi$ with $\lambda = 1$. Write $\phi \asymp \psi$ if $\phi \preceq \psi$ and $\psi \preceq \phi$, and write $\phi \overset{+}{\asymp} \psi$ if $\phi \overset{+}{\preceq} \psi$ and $\psi \overset{+}{\preceq} \phi$.

A *quasiisometric embedding* is a map $\phi\colon X \to Y$ such that for $\phi^*(d_Y)(x, x') := d_Y(\phi(x), \phi(x'))$ we have $d_X \asymp \phi^*(d_Y)$ as maps on X^2. It is a *quasiisometry* if it is a quasiisometric embedding and $\phi(X)$ is coarsely equivalent to Y. It is a *biLipschitz embedding* if it is a quasiisometric embedding with 0 additive error.

A *path* is a continuous map from an interval of $\mathbb{R}$. A *geodesic* is an isometric embedding of an interval of $\mathbb{R}$, which is a segment, ray, etc according to the corresponding property of the domain. A *quasigeodesic* is a quasiisometric embedding of an interval of $\mathbb{R}$. A (λ, ϵ)–quasigeodesic is a quasigeodesic such that the multiplicative and additive errors of the quasiisometric embedding are λ and ϵ, respectively.

A quasigeodesic need not be a path, but there is a standard connect-the-dots trick, [31, III.H.1.11], that replaces a quasigeodesic in a geodesic metric space by a piecewise geodesic quasigeodesic path at bounded Hausdorff distance from the original, where the Hausdorff distance and the new quasigeodesic constants depend only on the original quasigeodesic constants.

2.2.2. *Distances.* We have several different measures of separation in Σ:

- $d_1(x,y)$ is the length of the shortest edge path between vertices x and y.
- $d_2(x,y)$ is the CAT(0) distance between x and y.
- $d_w(x,y) = |\mathcal{M}(x,y)|$ is the cardinality of the set $\mathcal{M}(x,y)$ of walls separating points x and y (which is exactly $d_1(x,y)$, when $x, y \in \Sigma^{(0)}$).
- $d_c(x,y)$ is the length of the longest chain of walls separating x and y.

Of these, only d_2 is a metric on Σ. The combinatorial metric d_1 extends to a geodesic metric on the 1–skeleton of Σ. The wall and chain measures d_w and d_c are metrics on the 0–skeleton. There are ways to extend them to metrics on all of Σ, but we will not use this fact. For us it will be enough to take coarse extensions, in the sense that if x and y are arbitrary points of Σ, then for $d \in \{d_1, d_w, d_c\}$ we let $d(x,y)$ be the minimum of $d(x',y')$ over vertices x' and y' such that x' is contained in a common closed chamber with x and y' is contained in a common closed chamber with y.

Niblo and Reeves [86] constructed a finite dimensional, locally finite CAT(0) cube complex X_{NR} corresponding to the Coxeter system (W,S). Its wall structure is isomorphic to that of Σ, and after choosing basepoints the orbit map gives a W–equivariant isometric embedding $(\Sigma^{(0)}, d_w) \to (X^{(0)}_{\text{NR}}, d_w)$. By the Helly property for walls in a CAT(0) cube complex, a collection of pairwise intersecting walls in X_{NR} has a common point of intersection, so the size of such a set is bounded by $\dim(X_{\text{NR}})$. Since the wall structures of Σ and X_{NR} are isomorphic, $\dim(X_{\text{NR}})$ is also the maximum cardinality of a set of pairwise intersecting walls in Σ, even though in Σ such a set need not have a common intersection point.

Lemma 2.22. *$d_2 \asymp d_1 \asymp d_w \asymp d_c$ as maps $\Sigma^2 \to \mathbb{R}$.*

Proof. There are only finitely many isometry types of cells of Σ, so every point is within uniformly bounded distance of a vertex. Moreover, any closest vertex to a point x is not separated from x by any wall. Thus, up to increasing the additive error afterwards, it suffices to prove the lemma on $\Sigma^{(0)}$.

W acts geometrically on Σ, freely and transitively on $\Sigma^{(0)}$, and $\Sigma^{(1)}$ is precisely the Cayley graph of W with respect to S, so d_1 and d_2 on $\Sigma^{(0)}$ are quasiisometric to the word metric on W with respect to S.

On $\Sigma^{(0)}$ $d_1 = d_w$; in one direction every edge is cut by a unique wall, so the number of walls separating vertices x and y is at most the length of a combinatorial geodesic between them. In the other direction, the Exchange Condition for reflection systems [52, Section 3.2] implies that a combinatorial geodesic is cut at most once by any given wall.

Since the wall structure of Σ is isomorphic to that of X_{NR}, the relation between d_w and d_c can be computed there. It is immediate from the definitions that $d_c \le d_w$. Conversely, for any w_1 and w_2 in W, consider the set of walls separating w_1 and w_2. If H is such a wall, let H^- be the halfspace with $w_1 \in H^-$ and $\partial H^- = H$, and let H^+ be its complementary halfspace. Consider the set of halfspaces H^- partially ordered by inclusion. Since they all contain w_1, two such halfspaces H_1^- and H_2^- are incomparable if and only if H_1 and H_2 cross. Therefore, antichains in the partially ordered set correspond to collections of pairwise crossing walls. Dilworth's Theorem says that in a partially ordered set the maximum size of an antichain is equal to the minimum size of a partition of the set into chains. At least one of these has at least average size, so there is some chain separating w_1 from w_2 containing at least $d_w(w_1,w_2)/\dim(X_{\text{NR}})$–many walls. Thus, $d_c \le d_w \le \dim(X_{\text{NR}}) d_c$. □

Lemma 2.22 enables the following little trick, which we will use frequently:

Lemma 2.23 (4–gon trick)**.** *Let x_0, x_1, y_0, and y_1 be four points. Let $\mathcal{V} := \{V_i\}_{i\in I}$ be a chain separating x_0 and y_0, oriented from x_0 to y_0, so that for all $i \in I$ we have $x_0 \in V_i^-$ and $y_0 \in V_i^+$. Let $I_x := \{i \in I \mid x_1 \in \bar{V}_i^+\}$, let $I_y := \{i \in I \mid y_1 \in \bar{V}_i^-\}$, let $I' := I \smallsetminus (I_x \cup I_y)$. Let $\mathcal{V}_x$, $\mathcal{V}_y$, and $\mathcal{V}'$ be the subchains of $\mathcal{V}$ with walls index sets I_x, I_y, and I', respectively. Then $I_x \subset I$ is an initial subinterval, $I_y \subset I$ is a terminal subinterval, and $\mathcal{V}'$, if nonempty, is a convex subchain of $\mathcal{V}$ separating x_1 from y_1. See Figure 1. Furthermore:*

$$|\mathcal{V}| - |\mathcal{V}'| \le d_c(x_0,x_1) + d_c(y_0,y_1) + 2 \asymp d_2(x_0,x_1) + d_2(y_0,y_1)$$

In particular, when $|\mathcal{V}|$ is much larger than $d_2(x_0,x_1)$ and $d_2(y_0,y_1)$, then $\mathcal{V}'$ is "most of" $\mathcal{V}$.

More generally, if $\mathcal{M}$ is any collection of walls separating x_0 and y_0 and $\mathcal{M}''$ is the subset of $\mathcal{M}$ separating x_1 and y_1 then:

$$|\mathcal{M}| - |\mathcal{M}''| \stackrel{+}{\preceq} d_w(x_0,x_1) + d_w(y_0,y_1)$$

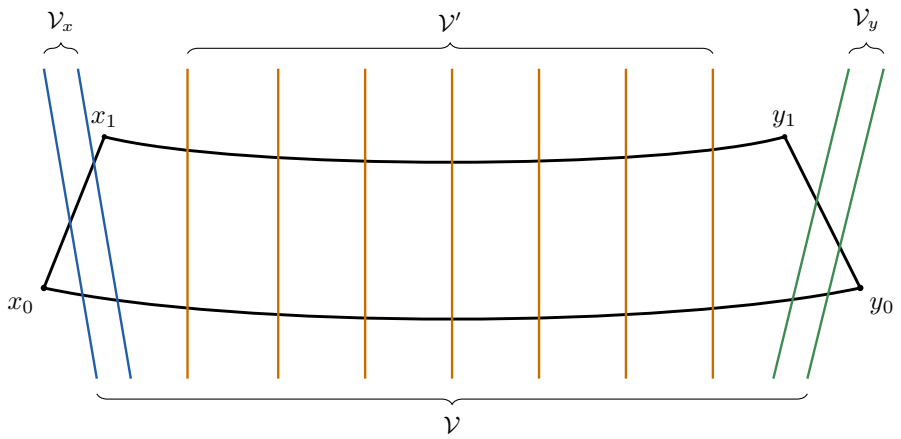


FIGURE 1. 4-gon trick: If $d_2(x_0, x_1)$ and $d_2(y_0, y_1)$ are small then most of the walls separating x_0 and y_0 also separate x_1 and y_1.

Proof. If $i < j$ are indices in I with $j \in I_x$ then $x_1 \in \bar{V}_j^+ \subset \bar{V}_i^+$, so $i \in I_x$. Thus, I_x is an initial subinterval of I. A similar argument shows I_y is a terminal subinterval of I. If $I' \neq \emptyset$, then it is the middle subinterval of I separating I_x and I_y. For $i \in I'$, the condition $i \notin I_x$ implies $x_1 \in V_i^-$, and the condition $i \notin I_y$ implies $y_1 \in V_i^+$, so V_i separates x_1 and y_1.

Define $I'_x := \{i \in I \mid x_1 \in V_i^+\}$. Since $\mathcal{V}$ is a chain, its walls are disjoint, so x_1 belongs to at most one of its walls, so I'_x is all except possibly the last index of I_x. Then $\{V_i\}_{i \in I'_x}$ is a chain separating x_0 and x_1, giving $d_c(x_0, x_1) \geq |I'_x| \geq |I_x| - 1$. A similar estimate holds for y_0 and y_1, so:

$$\begin{aligned} |\mathcal{V}| - |\mathcal{V}'| &= |I| - |I'| \\ &\leq |I_x| + |I_y| \\ &\leq d_c(x_0, x_1) + d_c(y_0, y_1) + 2 \end{aligned}$$

The final coarse equivalence comes from Lemma 2.22.

For the more general d_w version, the walls of Σ are countable, so we can number them and make all the same index definitions as in the previous case, just without any geometric significance to the indexing. The corresponding $\mathcal{M}'$ is a subset of $\mathcal{M}''$, so it suffices to coarsely bound $|\mathcal{M}| - |\mathcal{M}'|$. For that, a similar computation to the previous case applies, because the maximal number of pairwise intersecting walls is $\dim X_{\mathrm{NR}}$, so $|I_x| - |I'_x|$ and $|I_y| - |I'_y|$ are still uniformly bounded. □

We can extend Lemma 2.22 to a point-parabolic subcomplex distance comparison:

Lemma 2.24. *The following functions on $W = \Sigma^{(0)}$ are coarsely equivalent, with constants independent of $T \subset S$:*

$$w \mapsto \begin{cases} d_1(w, \Sigma_T) \\ d_2(w, \Sigma_T) \\ d_w(w, \Sigma_T) \\ d_c(w, \Sigma_T) \end{cases}$$

Proof. Let $w \in W = \Sigma^{(0)}$. By [52, Lemma 4.3.3] there is a unique decomposition $w = vu$ where $v \in W_T$ and u is the unique minimal length representative of $W_T w$. Thus, v is the closest vertex of W_T to w. Take a d_1–geodesic γ from w to v. A wall M separating w from v cuts the interior of γ. Split γ as a concatenation $\gamma' + \gamma''$ of nontrivial subpaths meeting at $M \cap \gamma$. Then $\gamma' + \mathfrak{r}_M(\gamma'')$ tightens to an edge path from w to $\mathfrak{r}_M(v)$ of length at most $|\gamma| - 1$, with v and $\mathfrak{r}_M(v)$ on opposite sides of M. If M separates w from v but does not separate w from Σ_T then it cuts Σ_T. However, the reflections whose walls cut Σ_T are precisely the reflections in W_T, which stabilize Σ_T. This gives a contradiction, because then $\mathfrak{r}_M(v)$ is a vertex of Σ_T that is closer to w than v. Thus, the walls separating w from Σ_T are precisely the walls separating w from v. Now apply Lemma 2.22 for distances between w and v. This accounts for the wall-based distances in the statement.

The CAT(0) distance follows because every point in Σ_T is within uniformly bounded distance of a vertex, so $d_2(w, \Sigma_T) \stackrel{+}{\asymp} d_2(w, \Sigma_T^{(0)})$. □

We will not use the following result, but it must be mentioned that in addition to point-point and point-parabolic subcomplex coarse equivalences $d_1 \asymp d_2 \asymp d_c$, there are also point-wall and wall-wall versions.

Theorem 2.25 (Strong Parallel Walls Theorem, [34, Theorem C] and [41, Theorem 2.3])**.** *Let $\Sigma = \Sigma(W, S)$ be the Davis complex of a finite rank Coxeter system. Let M and M' be disjoint walls in Σ, and let x be a point in Σ.*

(1) $d_c(M, x) \asymp d_1(M, x) \asymp d_2(M, x)$
(2) $d_c(M, M') \asymp d_1(M, M') \asymp d_2(M, M')$

The constants of the equivalences are independent of M, M', and x.

We remark that the references only state that these distances are related by some function, but it is straightforward to check that the proof actually produces a linear function.

In contrast to Lemma 2.24 and Theorem 2.25, distances to arbitrary CAT(0)-convex sets are not approximated by wall distances. Here are two explicit examples:

Example 2.26. Consider the standard square tiling of $\mathbb{R}^2$.

First, let L be a line that is not vertical or horizontal. It is CAT(0)–convex, but it crosses every vertical and horizontal line, so the intersection of halfspaces containing it is an empty intersection, hence is the whole plane. There are points of the plane arbitrarily far from L, but they are not separated from L by any wall.

Second, for each $n \in \mathbb{N}$, consider $v := (n, n)$ and the closed ball B of radius n about $(0, 0)$. The intersection of halfspaces containing B is the square bounded by vertical and horizontal lines at distance $n + 1/2$ from $(0, 0)$. This set contains v, so no wall separates v and B, but $d_2(v, B) = (\sqrt{2} - 1)n$.

2.2.3. *Constants.* We fix some constants of the Coxeter system (W, S) and the choice of Σ.

Let Δ be the maximal d_1–diameter of a spherical special subgroup of (W, S).

Let $\omega(r)$ be the maximum number of walls that intersect any closed d_2–ball of radius r in Σ. This is well-defined, by local finiteness of the wall structure and cocompactness of $W \curvearrowright \Sigma$.

Let $\ell := \max\{N + 2, 8\}$, where N is the constant of Theorem 2.18 (and 8 is the constant of Lemma 2.19).

Take $\mathfrak{R}$ large enough that for all $x \in \Sigma$ and Υ a parabolic subcomplex, Lemma 2.24 implies:

$$d_2(x, \Upsilon) \geq \mathfrak{R} \implies d_c(x, \Upsilon) \geq 4\ell - 1$$

Next we define functions ρ_0 and $\mathfrak{L}$ that take as input quasigeodesic parameters and carry information about chain density along a quasigeodesic. Suppose that Lemma 2.22 gives Λ and E such that for all $x, y \in \Sigma$:

$$d_2(x, y)/\Lambda - E \leq d_c(x, y) \leq \Lambda d_2(x, y) + E$$

Given $\lambda \geq 1$ and $\epsilon \geq 0$, define $\rho_0(\lambda, \epsilon) := \frac{1}{2\lambda\Lambda}$ and $\mathfrak{L}(\lambda, \epsilon) := 2\lambda(\Lambda E + \epsilon)$.

Observe that if $\gamma\colon I \to \Sigma$ is a (λ, ϵ)–quasigeodesic and $[x, y] \subset I$ then:

$$|y - x| \geq \mathfrak{L}(\lambda, \epsilon) \implies \frac{d_c(\gamma(x), \gamma(y))}{|y - x|} \geq \rho_0(\lambda, \epsilon)$$

So if $|y - x|$ is large enough then a chain realizing $d_c(\gamma(x), \gamma(y))$ has length at least $|y - x|\rho_0(\lambda, \epsilon)$. In particular, there do exist chains separating $\gamma(x)$ and $\gamma(y)$ whose *density*, in terms of number of walls per unit length in the parameter space, is at least $\rho_0(\lambda, \epsilon)$.

Remark. All of these constants can be effectively bounded. The point is that there are only finitely many conjugacy classes of spherical subgroups in W, and they can be enumerated in terms of the known types of irreducible spherical Coxeter diagrams appearing as full subgraphs in the Coxeter diagram for (W, S). Caprace's constant from Theorem 2.18 and a bound on $\dim(X_{\mathbb{NR}})$ [86, Lemma 3] are expressed in terms of the values of the Tits form on subsystems of the canonical root system corresponding to spherical subgroups. Since the Tits form has an explicit closed form, these quantities are computable. Likewise, the cells of the Davis complex are determined by the orthogonal representations of the spherical subgroups, which can be explicitly computed, so there are only finitely many isometry types of cells in Σ and they are isometric to convex Euclidean polytopes that are described by linear algebra. Effective bounds on the distance relations in Lemma 2.22 and Lemma 2.24 can be extracted from the geometry of cells of Σ and the bound on $\dim(X_{\mathbb{NR}})$.

2.3. **Morse and hyperbolic geometry.**

2.3.1. *Variations of the Morse property.* Let Z be a subset of a metric space X. We define several properties that describe quantitatively how Z sits inside X in a manner similar to a quasiconvex subspace of a hyperbolic space. These properties and equivalences between them were first stated for the case of Z being a geodesic or quasigeodesic [25, 9, 100, 99], see also the survey [42].

Definition 2.27. A subset $Z \subset X$ is *Morse* or has the *Morse property* if it is quasigeodesically quasiconvex, in the sense that there is a real valued function $\mu(\lambda, \epsilon) := \sup_\gamma \sup_{x\in\gamma} d(x, Z)$, where the first supremum is taken over (λ, ϵ)–quasigeodesic segments γ with both endpoints on Z. A function bounding μ from above is known as a *Morse gauge* for Z; the given μ is the *optimal Morse gauge*.

The content of the definition is whether or not μ defines a real valued function; if Z is not Morse then there is some input pair (λ, ϵ) on which μ takes value ∞.

The next definitions make sense using a *set valued* "closest points projection":

$$\pi_Z \colon X \to 2^Z : x \mapsto \{z \in Z \mid d(x, z) = d(x, Z)\}$$

In fact, in more general situations an "almost closest points projection" is good enough; the key point is that point images must be nonempty. In the settings that we apply this concept, we have proper X and closed Z, so point images of π_Z are nonempty. A priori they could have arbitrarily large diameters. If Z is convex and X is CAT(0) then point images of π_Z are single points, and, in fact, $\pi_Z \colon X \to Z$ is 1–Lipschitz.

Definition 2.28. A closed subset Z of a proper geodesic metric space X is *χ–strongly contracting* if $\chi \geq \sup_{x,y} \operatorname{diam}(\pi_Z(x) \cup \pi_Z(y))$, where the supremum is taken over points x and y such that $d(x, y) \leq d(x, Z)$ and the *diameter* of $Y \subset X$ is $\operatorname{diam}(Y) := \sup_{y,y'\in Y} d(y, y')$.

Definition 2.29 (Divergence)**.** For a closed subset Z of a proper geodesic metric space X and parameters $0 < M \leq 1$ and $N > 2$, define the *optimal divergence gauge* $\delta(r; M, N) := \inf_{x,y} \bar{d}_{Mr}(x, y)$, where the infimum is taken over all pairs x and y at distance r from Z with $d(x, y) \geq Nr$. The modified distance $\bar{d}_{Mr}(x, y)$ denotes the infimal length of a path in X connecting x to y and staying outside the Mr–neighborhood of Z, or ∞ if no such path exists. Any lower bound on $\delta(r; M, N)$ is a *divergence gauge* for Z with respect to parameters M and N.

This notion of divergence appears in [42]. It is the same, up to a suitable equivalence relation on functions, to other familiar versions of divergence, and the equivalence class of the function does not depend on the choices of parameters M and N [42, Proposition 8.4]. The equivalence relation distinguishes functions of different polynomial degrees, so phrases such as "Z has at least quadratic divergence" make sense, and are established by showing there is some valid choice of M and N such that Z has a divergence gauge with respect to M and N that is bounded below by a quadratic function of r.

By convention, $\sup \emptyset = -\infty$ and $\inf \emptyset = \infty$, so if for some r there are no suitable pairs (x, y) in Definition 2.29 or no suitable paths between them, then $\delta(r; M, N) = \infty$. This is the expected behavior: every closed bounded set certainly has some Morse/contraction/divergence gauge that can be defined in terms of its diameter. For instance, for a geodesic segment of length $L > 0$ in the Euclidean plane, the boundary of its r–neighborhood has diameter $L + 2r$ and length $2\pi r + 2L$, so any two points on it can be connected by a subsegment of the boundary of length at most $\pi r + L$, but for $r > L$ there is no pair of points on the boundary at distance at least $3r$ from each other. The optimal divergence gauge $\delta(r; 1, 3)$ is between $3r$ and $\pi r + L$ while $r \leq L$, but then is ∞ for $r > L$. The segment admits a quadratic divergence gauge $q_L(r) := 3r^2/L$. The behavior of $\delta(r; 1, 3)$ is Euclidean-like on scales at most L, while on large scales the bounded segment begins to look "point-like", hence Morse in a trivial way. The bound q_L degenerates as $L \to \infty$. In contrast, divergence gauges for geodesic segments in the hyperbolic plane can be bounded uniformly, independent of the segment diameters.

Definition 2.30 (BGIP)**.** A closed subset Z of a proper geodesic metric space has the *Bounded Geodesic Image Property* if there exists C such that for every geodesic γ, if $\gamma \cap \mathcal{N}_C(Z) = \emptyset$ then $\operatorname{diam}(\pi_Z(\gamma)) \leq C$.

Definition 2.31. A closed subset Z of a proper geodesic metric space is *strongly constricting* if there exists C such that for every geodesic segment γ with endpoints x and y, if $\operatorname{diam}(\pi_Z(x) \cup \pi_Z(y)) > C$ then γ passes within distance C of both $\pi_Z(x)$ and $\pi_Z(y)$.

Proposition 2.32 ([43], [102, Proposition 3.1], [42, Proposition 8.15, Corollary 3.6])**.** *When X is proper and CAT(0), Morse, strongly contracting, and at least quadratic divergence are equivalent. Moreover, there are effective quantitative bounds between the relevant gauges.*

Proposition 2.33 ([15, Proposition 2.9])**.** *Strong contraction, strong constriction, and the Bounded Geodesic Image Property are equivalent, with effective bounds between the relevant gauges.*

We close this subsection with some contraction robustness lemmas.

Lemma 2.34. *Let X be a proper geodesic metric space. Given χ, λ, and ϵ, there exists D such that for every χ–strongly contracting (λ, ϵ)–quasigeodesic path $\gamma\colon I \to X$ and for every $s, t \in I$ the Hausdorff distance between $\gamma([s,t])$ and any geodesic between $\gamma(s)$ and $\gamma(t)$ is at most D.*

Proof. Pick a geodesic $\alpha\colon [0, L] \to X$ between $\gamma(s)$ and $\gamma(t)$. By [15, Lemma 2.14], there is a D' depending on χ, λ, and ϵ such that $\gamma([s,t]) \subset \bar{\mathcal{N}}_{D'}(\alpha)$.

Conversely, suppose Z is a connected with $\{\alpha(0), \alpha(L)\} \subset Z \subset \bar{\mathcal{N}}_{D'}(\alpha)$. Define the Gromov product coordinate of $x \in X$ along α by:

$$f(x) := \left(d(\alpha(0), \alpha(L)) + d(\alpha(0), x) - d(\alpha(L), x)\right)/2$$

The function $f\colon X \to [0, L]$ is 1–Lipschitz and for $p \in [0, L]$ satisfies $f \circ \alpha(p) = p$. It follows that $f(Z) = [0, L]$. In particular, for every $p \in [0, L]$ there exists $z \in Z$ such that $f \circ \alpha(p) = f(z)$. By hypothesis, there exists $q \in [0, L]$ with $d(\alpha(q), z) \leq D'$. We bound the difference between the two "projections" of z:

$$|p - q| = |f(\alpha(p)) - f(\alpha(q))| = |f(z) - f(\alpha(q))| \leq D'$$

Then the triangle inequality gives $d(\alpha(p), z) \leq d(\alpha(p), \alpha(q)) + d(\alpha(q), z) \leq 2D'$. Since $p \in [0, L]$ was arbitrary, $\alpha \subset \bar{\mathcal{N}}_{2D'}(Z)$. Applying this argument with $D := 2D'$ and $Z := \gamma([s,t])$ completes the lemma. □

Lemma 2.35. *Given χ, λ, and ϵ, there exists χ', λ', ϵ', and D such that if $\gamma\colon I \to X$ is a χ–strongly contracting, (λ, ϵ)–quasigeodesic in a proper geodesic metric space X, then:*

- *There is a continuous, χ'–strongly contracting (λ', ϵ')–quasigeodesic at bounded Hausdorff distance from γ. Moreover, if γ is compact or a closed ray we can choose the continuous version to have the same endpoints/basepoint.*
- *Every subsegment of γ is χ'–strongly contracting.*
- *Every geodesic segment with both endpoints on γ is χ'–strongly contracting and is D–Hausdorff equivalent to any subsegment of γ with the same endpoints.*
- *Every geodesic ray based on γ and asymptotic to γ is χ'–strongly contracting and is D–Hausdorff equivalent to any subray of γ with which it is asymptotic and shares a basepoint.*
- *Every biinfinite geodesic asymptotic to γ is χ'–strongly contracting and D–Hausdorff equivalent to γ.*

Proof. The first item follows by using the connect-the-dots trick to get a continuous quasigeodesic that is coarsely equivalent to γ, with Hausdorff distance bounded in terms of λ and ϵ. By [13, Lemma 6.3], strong contraction is preserved by coarse equivalence, so the new quasigeodesic is strongly contracting, with contraction constant determined by that of γ and the Hausdorff distance between them.

Take D_0 to be the constant provided by Lemma 2.34. Using properness to take limits of geodesic segments in the ray and biinfinite cases, there is a geodesic α of the same type as γ and at Hausdorff distance at most D_0 from γ. By [13, Lemma 6.3], α is strongly contracting, with contraction constant determined by χ and D_0. By [45, Lemma 3.6], subsegments of α are uniformly strongly contracting, with contraction constant depending only on that of α.

Fix $s < t$ in I and suppose β is a geodesic from $\gamma(s)$ to $\gamma(t)$. Choose p and q in α within distance D_0 of $\gamma(s)$ and $\gamma(t)$, respectively. The subsegment of α from p to q is strongly contracting, as are the geodesic segments of length at most D_0 joining $\gamma(s)$ to p and joining q to $\gamma(t)$. The triangle lemma [45, Lemma 3.8] says that if two sides of a geodesic (almost) triangle are strongly contracting, then so is the third side, with contraction bound determined by those of the first two sides. One application of the triangle lemma gives that a geodesic from $\gamma(s)$ to q is strongly contracting, and a second application gives that β is strongly contracting, with contraction constant determined by that of α and the quantity D_0. Lemma 2.34 gives that β and $\gamma([s,t])$ are at Hausdorff distance at most D_0, so another application of [13, Lemma 6.3] transfers strong contraction of β to strong contraction of $\gamma([s,t])$. This proves the second and third items.

Uniform strong contraction of compact subsegments of quasigeodesic transfers to strong contraction, with the same constant, of the whole quasigeodesic. This is an easy point-by-point argument using a long enough compact subsegment to control each projection. Thus, the previous argument also implies that any subray γ' of γ is uniformly strongly contracting. Let β be a geodesic ray with the same basepoint and asymptotic to γ'. Then strong contraction of γ' forces β and γ' to be at uniformly bounded Hausdorff distance, [45, Lemma 4.4], and then [13, Lemma 6.3] transfers strong contraction of γ' to β.

Finally, suppose that γ is biinfinite and β is some biinfinite geodesic asymptotic to γ. Properness gives geodesic rays α^+ and α^- based at $\gamma(0)$ and asymptotic to $\gamma([0,\infty))$ and $\gamma([0,-\infty))$, respectively. By the preceding paragraph, α^+ and α^- are uniformly strongly contracting and uniformly Hausdorff equivalent to the corresponding subray of γ. The rays α^+ and α^- together with β form an ideal geodesic triangle with two strongly contracting sides, so the truncation argument in the proof of the triangle lemma gives a uniform Hausdorff distance bound between β and $\alpha^+ \cup \alpha^-$. Hence, β is uniformly Hausdorff equivalent to γ, so [13, Lemma 6.3] makes β uniformly strongly contracting.

Take χ' and D to be the maxima of the strong contraction and Hausdorff distance bounds from the various cases of this argument. □

2.3.2. *Working in non-geodesic spaces.* To work in the setting of hyperbolic spaces that are not necessarily geodesic, Blachère, Haïssinsky, and Mathieu [27] introduced the following notion:

Definition 2.36. A map $\gamma\colon I \to X$ from a subinterval of $\mathbb{R}$ to a metric space X is a *(parameterized)* (λ,ϵ,τ)*–quasiruler* if it is a (λ,ϵ)–quasigeodesic and the triangle inequality is almost degenerate, in the sense that for all $r<s<t$ in I:

$$d(\gamma(r),\gamma(s)) + d(\gamma(s),\gamma(t)) - \tau \leq d(\gamma(r),\gamma(t)) \leq d(\gamma(r),\gamma(s)) + d(\gamma(s),\gamma(t))$$

The space X is *quasiruled* if there exist λ, ϵ, and τ such that every pair of points in X is the pair of endpoints of a (λ,ϵ,τ)–quasiruler.

Abbott and Incerti-Medici [4] make the following refinement:

Definition 2.37. A map $\gamma\colon I \to X$ from a subinterval of $\mathbb{R}$ to a metric space X is an *unparameterized* τ*–quasiruler* if it satisfies two conditions:
bounded jumps: $\forall t \in I$, $\limsup_{s\in I,\, |t-s|\to 0} d(\gamma(s),\gamma(t)) \leq \tau$
almost degeneracy: $\forall r<s<t\in I$, $d(\gamma(r),\gamma(s)) + d(\gamma(s),\gamma(t)) - \tau \leq d(\gamma(r),\gamma(t))$

There is another definition in the literature that is related:

Definition 2.38. A $(1,\epsilon)$–quasigeodesic is also known as an ϵ*–rough geodesic*, and a space for which there exists ϵ such that every pair of points is connected by an ϵ–rough geodesic is a *rough geodesic space*.

Abbott and Incerti-Medici prove [4, Lemma A.4] that given τ there exist λ and ϵ such that every unparameterized τ–quasiruler satisfying an additional metric properness condition can be reparameterized to make it a (λ,ϵ,τ)–quasiruler. They also prove [4, Lemma A.6] that quasiruled spaces are roughly geodesic, although the rough geodesics need not be the reparameterizations of the quasirulers. In Section 4 we introduce a non-geodesic metric on Σ and $X_{\mathbb{NR}}$ such that combinatorial geodesics in $X_{\mathbb{NR}}$ can be reparameterized to be rough geodesic quasirulers, see Corollary 4.5. Getting the same reparameterization to be both roughly geodesic and a uniform quasiruler, in this particular setting, is the reason for proving Corollary 4.5 instead of quoting Abbott and Incerti-Medici.

2.4. **Parabolic subcomplexes.**

2.4.1. *Wide groups.* A group is *constricted* if all of its asymptotic cones have cut points or *unconstricted* otherwise [56]. A group is *wide* if none of its asymptotic cones have cut points [55], which is a strictly stronger condition than being unconstricted. For finitely generated groups, the existence of a biinfinite Morse geodesic implies that the group is constricted.

For Coxeter groups there is a nice wide/constricted dichotomy visible from the Coxeter diagram:

Proposition 2.39. *Let $W = P_1 \times \cdots \times P_n \times K$ be the canonical product decomposition of (W,S) where the P_i are special subgroups corresponding to nonspherical connected components of the Coxeter diagram, and K is the maximal spherical special subgroup. Suppose $W \neq K$. Then W is wide if and only if either $n \geq 2$ or $n = 1$ and P_1 is higher rank affine.*

Proof. If $n \geq 2$ or $n = 1$ and P_1 is higher rank affine then W is virtually a product of infinite groups. Asymptotic cones of products are products of asymptotic cones, so any asymptotic cone of a product of two infinte groups has no cut points, so the group is wide. Otherwise, Caprace and Fujiwara [36, Proposition 4.5] say that W contains a rank-one element (equivalently, a Morse element), so it is constricted. □

Definition 2.40 (cf [49, Definition 3.1] and the definition $\mathbb{T}_0$ of [21, Section A.1])**.** A *wide parabolic subgroup* P of (W, S) is a nonspherical parabolic that is wide as a Coxeter group, in the sense of Proposition 2.39, which occurs exactly if either $P = B \times C$, where B and C are nonspherical parabolics of (W, S), or $P = A \times K$, where A is an irreducible higher rank affine parabolic of (W, S) and K is a spherical parabolic of (W, S).

A *wide parabolic subcomplex* is a parabolic subcomplex whose stabilizer is a wide parabolic subgroup.

2.4.2. *Gate projections to parabolic subcomplexes.*

Lemma 2.41. *Given a parabolic subcomplex $\Upsilon := w\Sigma_T$ and a vertex $x \in \Sigma^{(0)} = W$ there exists a unique d_1–closest vertex $\mathfrak{p}_\Upsilon(x)$ of Υ to x. Furthermore, for every vertex $y \in \Upsilon^{(0)}$:*

$$d_1(x, y) = d_1(x, \mathfrak{p}_\Upsilon(x)) + d_1(\mathfrak{p}_\Upsilon(x), y)$$

The map $\mathfrak{p}_\Upsilon \colon \Sigma^{(0)} \to \Upsilon^{(0)}$ is the gate map *or* gate projection to Υ*. It is 1–Lipschitz.*

Proof. By [52, Lemma 4.3.1], there is a unique shortest element v of $W_\emptyset x^{-1} w W_T$, and there exists $z \in W_T$ such that $x^{-1}y = vz$ with $|x^{-1}y|_S = |v|_S + |z|_S$. Take $\mathfrak{p}_\Upsilon(x) := xv$. Then $d_1(x, y) = |x^{-1}y|_S = |v|_S + |z|_S = d_1(x, \mathfrak{p}_\Upsilon(x)) + d_1(\mathfrak{p}_\Upsilon(x), y)$. Uniqueness follows immediately.

The fact that $\mathfrak{p}_\Upsilon$ is 1–Lipschitz is a standard, easy computation. □

Corollary 2.42. *If $U \subset T \subset S$ then $\mathfrak{p}_{\Sigma_U} = \mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}$.*

Proof.

$$\begin{aligned}
&d_1(x, \mathfrak{p}_{\Sigma_U}(x)) \\
&\quad = d_1(x, \mathfrak{p}_{\Sigma_T}(x)) + d_1(\mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U}(x)) \\
&\quad = d_1(x, \mathfrak{p}_{\Sigma_T}(x)) + d_1(\mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x)) + d_1(\mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U}(x)) \\
&\quad \geq d_1(x, \mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x)) + d_1(\mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U}(x)) \\
&\quad \geq d_1(x, \mathfrak{p}_{\Sigma_U}(x)) + d_1(\mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U}(x))
\end{aligned}$$

Thus, $d_1(\mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x), \mathfrak{p}_{\Sigma_U}(x)) = 0$. □

Corollary 2.43. *A wall M separates x from $\mathfrak{p}_\Upsilon(x)$ if and only if M separates x from Υ.*

Proof. A wall that separates x from all of Υ certainly separates x from $\mathfrak{p}_\Upsilon(x) \in \Upsilon$. For the converse, suppose M is a wall that separates x from $y := \mathfrak{p}_\Upsilon(x)$ but not from Υ, so there is a vertex $z \in \Upsilon$ such that x and z are on one side of M, and y is on the other. Then the concatenation of the d_1–geodesic from x to y with the d_1–geodesic from y to z crosses M twice, so it is not a combinatorial geodesic, so $d_1(x, z) < d_1(x, y) + d_1(y, z)$. This contradicts the gate property $d_1(x, z) = d_1(x, y) + d_1(y, z)$. □

The following interaction between gate maps and the group action follows directly from the definition.

Corollary 2.44. *For $w \in W$, Υ a parabolic subcomplex, and $x \in \Sigma^{(0)}$:*

$$\mathfrak{p}_{w\Upsilon}(x) = w\mathfrak{p}_\Upsilon(w^{-1}x)$$

Lemma 2.45. *For any $U \subset S$ and $g \in W_{U^\perp}$, for all $x \in \Sigma^{(0)}$:*

$$\mathfrak{p}_{\Sigma_U}(gx) = \mathfrak{p}_{\Sigma_U}(x)$$

Proof. Let $T := U \sqcup U^\perp$. Write $\mathfrak{p}_{\Sigma_T}(x) = ab$ for $a \in W_U$ and $b \in W_{U^\perp}$. Using that $g\Sigma_T = \Sigma_T$ and Corollary 2.44, $\mathfrak{p}_{\Sigma_T}(gx) = \mathfrak{p}_{g\Sigma_T}(gx) = g\mathfrak{p}_{\Sigma_T}(g^{-1}gx) = g\mathfrak{p}_{\Sigma_T}(x)$. This plus Corollary 2.42 gives:

$$\mathfrak{p}_{\Sigma_U}(g\mathfrak{p}_{\Sigma_T}(x)) = \mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(gx) = \mathfrak{p}_{\Sigma_U}(gx)$$

Thus, $\mathfrak{p}_{\Sigma_U}(gx) = \mathfrak{p}_{\Sigma_U}(gab) = \mathfrak{p}_{\Sigma_U}(agb)$. The latter is agb times the shortest element of $(agb)^{-1}W_U = b^{-1}g^{-1}W_U$. Since $b^{-1}g^{-1} \in W_{U^\perp}$, there is no cancellation between $b^{-1}g^{-1}$ and words in W_U, so the

shortest element of $b^{-1}g^{-1}W_U$ is $b^{-1}g^{-1}$, which gives $\mathfrak{p}_{\Sigma_U}(agb) = agbb^{-1}g^{-1} = a$. The same argument gives $\mathfrak{p}_{\Sigma_U}(ab) = a$. Thus:

$$\mathfrak{p}_{\Sigma_U}(x) = \mathfrak{p}_{\Sigma_U} \circ \mathfrak{p}_{\Sigma_T}(x) = \mathfrak{p}_{\Sigma_U}(ab) = a = \mathfrak{p}_{\Sigma_U}(gab) = \mathfrak{p}_{\Sigma_U}(gx) \qquad \square$$

Lemma 2.46. *For $i \in \{0,1\}$, let $\Upsilon_i := w_i\Sigma_{T_i}$ be parabolic subcomplexes. Let u be the unique shortest element of $W_{T_0}w_0^{-1}w_1W_{T_1}$. Take $v_0 \in W_{T_0}$ and $v_1 \in W_{T_1}$ such that $w_0^{-1}w_1 = v_0uv_1^{-1}$. Let $U_0 := \{s \in T_0 \mid u^{-1}su \in T_1\}$ and $U_1 := \{s \in T_1 \mid usu^{-1} \in T_0\}$. Then $\mathfrak{p}_{\Upsilon_0}(\Upsilon_1) = w_0v_0W_{U_0}$ is the vertex set of a parabolic subcomplex $w_0v_0\Sigma_{U_0}$. Furthermore, right-multiplication by u defines an isomorphism $w_0v_0\Sigma_{U_0} \to w_1v_1\Sigma_{U_1}$.*

Proof. The element u exists by [52, Lemma 4.3.1], and $d_1(\Upsilon_0, \Upsilon_1) = |u|_S$. For every vertex $w_0v_0x \in w_0v_0W_{U_0} \subset \Upsilon_0^{(0)}$ we have $d_1(w_0v_0x, w_0v_0xu) = |u|_S$ and $w_0v_0xu = w_0v_0uu^{-1}xu \in w_0v_0uW_{U_1} = w_1v_1W_{U_1} \subset \Upsilon_1^{(0)}$. Thus $w_0v_0W_{U_0} \subset \mathfrak{p}_{\Upsilon_0}(\Upsilon_1)$.

Conversely, suppose $z \in W_{T_0}$ such that $w_0v_0z \in \mathfrak{p}_{\Upsilon_0}(\Upsilon_1)$. Then there is some $y \in W_{T_1}$ such that $w_0v_0z = \mathfrak{p}_{\Upsilon_0}(w_1v_1y)$. By construction, $\mathfrak{p}_{\Upsilon_0}(w_1v_1y)$ is w_1v_1yx, where x is the unique shortest element of $(w_1v_1y)^{-1}w_0v_0W_{T_0} = y^{-1}u^{-1}W_{T_0}$. Let $a := x^{-1}y^{-1}u^{-1} \in W_{T_0}$. We have $w_0v_0z = w_1v_1 \cdot yx = w_0v_0u \cdot u^{-1}a^{-1}$, so $z = a^{-1}$ and $x = y^{-1}u^{-1}z$. Since u^{-1} is the shortest element of $W_{T_1}u^{-1}W_{T_0} \ni x$, there are $b \in W_{T_0}$ and $c \in W_{T_1}$ such that $x = cu^{-1}b$ and $|x|_S = |c|_S + |u^{-1}|_S + |b|_S$. But since x is shortest in $y^{-1}u^{-1}W_{T_0} \subset W_{T_1}u^{-1}W_{T_0}$, the element b is trivial. Thus, $x = cu^{-1}$, which gives $W_{T_0} \ni z = uycu^{-1} \in uW_{T_1}u^{-1}$. By the remark following [52, Lemma 5.3.6], $uW_{T_1}u^{-1} \cap W_{T_0} = W_{U_0}$, so $z \in W_{U_0}$. Thus, $w_0v_0W_{U_0} \supset \mathfrak{p}_{\Upsilon_0}(\Upsilon_1)$.

An edge in the full subcomplex spanned by $\mathfrak{p}_{\Upsilon_0}(\Upsilon_1)$ is of the form $w_0v_0p - w_0v_0pq$ for some $p \in W_{U_0}$ and $q \in U_0$. Right-multiplication by u sends these vertices to $w_0v_0pu = w_0v_0uu^{-1}pu = w_1v_1u^{-1}pu$ and $w_0v_0pqu = w_1v_1u^{-1}pu \cdot u^{-1}qu$, respectively. These are two elements of $w_1v_1W_{U_1}$ that differ by $u^{-1}qu \in U_1$, so they are adjacent vertices of $w_1v_1W_{U_1}$. Thus, right-multiplication by u gives a bijection from $w_0v_0W_{U_0}$ to $w_1v_1W_{U_1}$ that preserves adjacency, so extends to an isomorphism of the full subcomplexes spanned by those vertex sets. $\square$

Corollary 2.47. $\mathrm{Stab}(\mathfrak{p}_{\Upsilon_0}(\Upsilon_1)) = \mathrm{Stab}(\mathfrak{p}_{\Upsilon_1}(\Upsilon_0))$

The next result is a partial converse of Corollary 2.47.

Lemma 2.48. *Let P be a nonspherical irreducible parabolic subgroup of (W,S). Let $T \subset S$ be the unique subset such that P is conjugate to W_T. Suppose $\Upsilon_0 = w_0\Sigma_{T_0}$ and $\Upsilon_1 = w_1\Sigma_{T_1}$ are parabolic subcomplexes with $P = \mathrm{Stab}(\Upsilon_0) = \mathrm{Stab}(\Upsilon_1)$. Then $T = T_0 = T_1$ and the unique shortest element u of $W_Tw_0^{-1}w_1W_T$ belongs to $W_{T^\perp}$ and satisfies:*

- $\mathfrak{p}_{\Upsilon_1}(\Upsilon_0) = \Upsilon_0^{(0)}u = \Upsilon_1^{(0)}$.
- *Left-multiplication by $w_0uw_0^{-1} \in P^\perp$ and right-multiplication by $u \in W_T^\perp$ induce the same isomorphism $\Upsilon_0 \to \Upsilon_1$.*

Proof. The set T exists by Corollary 2.15, and Proposition 2.13 (1) further implies $T = T_0 = T_1$. Proposition 2.13 (2) gives $w_0^{-1}w_1 \in N_W(W_T) = W_T \times W_{T^\perp}$, so we can write $w_0^{-1}w_1 = vu$ where $v \in W_T$ and $u \in W_{T^\perp}$ is the shortest element of $W_Tw_0^{-1}w_1W_T$. The projection formula in Lemma 2.46 gives $U_0 = U_1 = T$, and that right-multiplication by u is the gate map $\mathfrak{p}_{\Upsilon_1}(\Upsilon_0) = \Upsilon_0^{(0)}u = \Upsilon_1^{(0)}$. Finally, for any $x \in W_T$ we have $w_0uw_0^{-1} \cdot w_0x = w_0ux = w_0x \cdot u$, so on $\Upsilon_0^{(0)}$ left-multiplication by $w_0uw_0^{-1}$, right-multiplication by $u \in W_{T^\perp} = W_T^\perp$, and $\mathfrak{p}_{\Upsilon_1}$ all define the same map. $\square$

Lemma 2.49. *If $x \in \Sigma^{(0)}$ and Υ is a parabolic subcomplex then $\mathfrak{p}_\Upsilon(x)$ is the unique d_2–closest vertex of Υ to x. Furthermore, there exists a closed chamber containing both $\pi_\Upsilon(x)$ and $\mathfrak{p}_\Upsilon(x)$.*

Proof. Let $y \in \Upsilon^{(0)} \smallsetminus \{\mathfrak{p}_\Upsilon(x)\}$. Let z be the first vertex after y on a d_1–geodesic from y to $\mathfrak{p}_\Upsilon(x)$. Since Υ is d_1–convex, $z \in \Upsilon$. Let M be the wall dual to the edge between z and y. Then M cuts Υ, so by Corollary 2.43 it does not separate x and $\mathfrak{p}_\Upsilon(x)$. Thus, vertices x, $\mathfrak{p}_\Upsilon(x)$, and z are in one open halfspace M^- of M and y is in the other. Let p be the point at which the unique d_2–geodesic from x to y crosses M. The reflection through M is an isometry that fixes p and exchanges y and z, so:

$$d_2(x,y) = d_2(x,p) + d_2(p,y) = d_2(x,p) + d_2(p,z) \geq d_2(x,z)$$

In fact, $d_2(x,p)+d_2(p,z)>d_2(x,z)$, because x and z are contained in the d_2–convex open halfspace M^- and $p\in M$ is not, so p does not lie on the unique d_2–geodesic from x to z. Iterating along the d_1–geodesic, conclude $d_2(x,\mathfrak{p}_\Upsilon(x))<d_2(x,y)$.

The proof of the further statement is similar: suppose M is a wall with $\mathfrak{p}_\Upsilon(x)\in M^-$ and $\pi_\Upsilon(x)\in M^+$. Then M cuts Υ, so M does not separate x and $\mathfrak{p}_\Upsilon(x)$, so $x\in M^-$. Using the reflect and straighten trick as in the previous argument, this produces a point $\mathfrak{r}_M(\pi_\Upsilon(x))$ of Υ that is strictly closer to x than $\pi_\Upsilon(x)$ is. That is a contradiction, so no wall separates $\mathfrak{p}_\Upsilon(x)$ and $\pi_\Upsilon(x)$. □

3. Characterizing the Morse property for quasigeodesics

Section 3.1 reviews the concept of a "bridge" between convex subsets of a CAT(0) space, and then characterizes the Morse property for quasigeodesic paths in terms of dense transverse chains, Theorem 3.7. Section 3.2 describes "grids" and uses them to produce wide parabolics. Theorem 3.23 in that subsection relates a quasigeodesic's interactions with wide parabolic subcomplexes and dense transverse chains. Theorem 3.30 in Section 3.3 gives the main result of this section characterizing the Morse property for certain quasigeodesics.

The guiding intuition is that a quasigeodesic should be Morse if it avoids the wide parabolic subcomplexes. For a precise characterization, we need to make "avoids" quantitative and *uniform*. If γ is a quasigeodesic path with a long subsegment γ' contained in the R–neighborhood of a wide parabolic subcomplex Υ, then Lemma 2.23 implies that for any long chain $\mathcal{V}$ separating the endpoints of γ' there is a convex subchain $\mathcal{V}'$ of $\mathcal{V}$ that also cuts Υ, with $|\mathcal{V}|-|\mathcal{V}'|$ bounded in terms of R. One might hope in the other direction that if a long chain $\mathcal{V}$ is transverse to both γ and a wide parabolic subcomplex Υ, then, maybe after throwing away some walls from the ends of $\mathcal{V}$, the subsegment γ' between the extremes of the remaining chain would be contained in a uniform neighborhood of Υ. That is not true, see Example 3.1. The main novelty of this section is an argument, whose heart is Proposition 3.27, that we can make a trade: there is a convex subchain $\mathcal{V}'$ of $\mathcal{V}$, consisting of a definite fraction of its walls, such that γ is closer in the wall geometry to the wide parabolic subcomplex Υ' stabilized by $N_W(\mathrm{Pc}(\mathcal{V}'))$ than it was to Υ. An induction on rank shows that after uniformly finitely many such trades we get a subsegment of γ contained in a uniform neighborhood of the resulting wide parabolic subcomplex.

Example 3.1. Take the Coxeter diagram with vertices $S:=\{a,b,c,x,y,z\}$ and edges labelled ∞ for ab, ac, bc, xy, xz, yz, and cz (all other generator pairs commute). Let $U:=\{a,b\}$ and $T:=\{a,b,c\}$. Then $U^\perp=\{x,y,z\}$ and $T^\perp=\{x,y\}$, and both $W_T\times W_{T^\perp}$ and $W_U\times W_{U^\perp}$ are wide.

Let $\Upsilon:=\Sigma_{T\sqcup T^\perp}$ and $\Upsilon':=\Sigma_{U\sqcup U^\perp}$. Let $\gamma_0\subset\Sigma_U\subset\Sigma_T$ be a long geodesic segment, let $\mathcal{V}$ be a chain of U–walls separating its endpoints, and consider $\gamma_n:=(zx)^n\gamma_0$. The product structure on Υ' shows the γ_n are all geodesic segments contained in Υ' that are all cut by $\mathcal{V}$. However, they get arbitrarily far from Υ as $n\to\infty$: the retraction $W\twoheadrightarrow W_{\{x,z\}}$ killing $\{a,b,c,y\}$ is 1–Lipschitz and sends $\gamma_n^{(0)}$ to $(zx)^n$ while sending $\Upsilon^{(0)}$ onto $\{1,x\}$.

Thus, Υ is simply the wrong wide parabolic subcomplex, despite being cut by the same chain $\mathcal{V}$ that cuts γ_n. We trade shrinking T to U for expanding $T^\perp$ to $U^\perp$, thereby exchanging Υ for a different wide parabolic subcomplex Υ' closer to, and in this example actually containing, γ_n.

3.1. Bridges and divergence.

Section 3.1.1 recalls the CAT(0) bridge machinery of Bestvina, Kleiner, and Sageev [26], and specializes those results to walls in Davis complexes. Section 3.1.2 states and proves the main result of this subsection, Theorem 3.7, which characterizes the Morse property for quasigeodesic paths in terms of dense transverse chains.

3.1.1. *CAT(0) bridges.*

Lemma 3.2 (CAT(0) bridge lemma (cf [26, Lemma 2.3]))**.** *Let X be a complete CAT(0) space. Let A and B be nonempty closed convex subsets of X such that $d_2(A,B)$ is realized. Define $A_0:=\{a\in A\mid d_2(a,B)=d_2(A,B)\}$, $B_0:=\{b\in B\mid d_2(b,A)=d_2(A,B)\}$, and*

$$\mathrm{Bridge}(A,B):=\bigcup_{a\in A_0}[a,\pi_B(a)]$$

(1) A_0 and B_0 are closed and convex.
(2) $\pi_B|_{A_0}\colon A_0\to B_0$ and $\pi_A|_{B_0}\colon B_0\to A_0$ are inverse isometries.
(3) $\mathrm{Bridge}(A,B)$ is closed and convex and is isometric to a product $A_0\times[0,d_2(A,B)]$.

(4) Suppose there exists $\epsilon > 0$ such that:

$$(\prec) \qquad a \in A,\ d_2(a, A_0) = 1 \implies d_2(a, B) \geq d_2(A, B) + \epsilon$$

Then for every $a \in A$ with $d_2(a, A_0) \geq 1$ we have:

$$d_2(a, B) \geq\ d_2(A, B) + \epsilon d_2(a, A_0)$$

(5) If X is a locally finite complex with a cocompact isometry group then for every D there exists ϵ such that if A and B are convex subcomplexes with $d_2(A, B) \leq D$ then ϵ is a flaring bound *for A and B satisfying $(\prec)$.*

As an example, consider two lines in $\mathbb{E}^2$ intersecting at angle $0 < \theta < \pi/2$. Their bridge is the single point of intersection. Projection of one line to the other is surjective, so projections between convex sets need not be coarsely equivalent to the bridge (unlike in the cubical case). The optimal flaring bound satisfying $(\prec)$ is $\epsilon = \sin(\theta)$, so it depends on the pair of convex sets.

Lemma 3.2 is just [26, Lemma 2.3] stated slightly differently, with the additional hypothesis that X is complete, and with the final item of [26, Lemma 2.3] split into two separate statements.

Corollary 3.3. *Walls in Σ flare: walls are convex, distances between walls are realized, and for all D there exists ϵ such that if M and M' are walls of Σ with $d_2(M, M') \leq D$ then ϵ is a flaring bound for M and M' satisfying $(\prec)$.*

Proof. There is a subdivision Σ' of Σ, as in Definition 2.5, in which the walls of Σ are subcomplexes. Walls are convex in Σ, so they are also convex in Σ', since the underlying metric space is the same. Since Σ' is locally finite and W acts cocompactly, distances between convex subcomplexes are realized. Thus, walls satisfy the hypotheses of Lemma 3.2 and the additional hypotheses of Lemma 3.2 (5). □

Lemma 3.4. *For all D there exists $k > 0$ such that for all walls M, M' of Σ with $d_2(M, M') \leq D$, all paths γ from M to M', and all R we have:*

$$\gamma \not\subset \mathcal{N}_R(\mathrm{Bridge}(M, M')) \implies |\gamma| \geq kR - 2$$

Proof. Let $D_0 := d_2(M, M') \leq D$. By Corollary 3.3, given D there exists a flaring bound ϵ for M and M'. Define $k := \frac{\epsilon}{1+\epsilon}$.

Let $M_0 := \{m \in M \mid d_2(m, M') = d_2(M, M')\}$ and $M_0' := \{m \in M' \mid d_2(m, M) = d_2(M, M')\}$. Suppose there exists $x \in \gamma$ at least R–far from the bridge. Let $m := \pi_M(x)$ and $m' := \pi_{M'}(x)$.

$$(1) \qquad |\gamma| \geq d_2(x, m) + d_2(x, m') \geq d_2(m, m') \geq d_2(m, M')$$

If $d_2(x, m) \geq R$ or $d_2(x, m') \geq R$ then $|\gamma| \geq R > kR - 2$.

If $d_2(m, M_0) < 1$ and $d_2(m', M_0') < 1$ then $|\gamma| > 2(R - 1) > kR - 2$.

Up to swapping the roles of M and M', it remains to consider the case that $d_2(m, M_0) \geq 1$ and $d_2(x, m) < R$. In this case Lemma 3.2 gives:

$$d_2(m, M') \geq D_0 + \epsilon d_2(m, M_0) \geq D_0 + \epsilon(d_2(x, M_0) - d_2(x, m))$$

Since $M_0 \subset \mathrm{Bridge}(M, M')$, we have $d_2(x, M_0) \geq d_2(x, \mathrm{Bridge}(M, M')) \geq R$, so:

$$(2) \qquad d_2(m, M') \geq D_0 + \epsilon(R - d_2(x, m))$$

Combining equations (1) and (2):

$$\begin{aligned} |\gamma| &\geq \max\{D_0 + \epsilon(R - d_2(x, m)), d_2(x, m)\} \\ &\geq \frac{\epsilon}{1+\epsilon} R + \frac{D_0}{1+\epsilon} \\ &> kR - 2 \end{aligned}$$

□

Corollary 3.5. *For all D there exists $k > 0$, as in Lemma 3.4, such that for all walls M, M' of Σ with $d_2(M, M') \leq D$, all (λ, ϵ)–quasigeodesics $\gamma\colon [0, L] \to \Sigma$ from M to M', and all R we have:*

$$\gamma \not\subset \mathcal{N}_R(\mathrm{Bridge}(M, M')) \implies L \geq (k/\lambda)R - (2/\lambda)(\epsilon + 1)$$

Proof. Suppose $t \in [0, L]$ is such that $\gamma(t) \notin \mathcal{N}_R(\mathrm{Bridge}(M, M'))$. By Lemma 3.4, the length of the concatenation of the geodesic from $\gamma(0)$ to $\gamma(t)$ with the geodesic from $\gamma(t)$ to $\gamma(L)$ has length at least $kR - 2$, so $\lambda t + \epsilon + \lambda(L - t) + \epsilon \geq kR - 2$, which implies $L \geq (k/\lambda)R - (2/\lambda)(\epsilon + 1)$. □

Lemma 3.6. *For all C and D there exists B such that for all walls M, M' of Σ such that $d_2(M, M') \leq D$ and such that the longest chain transverse to both M and M' contains at most C–many walls, the diameter of* $\mathrm{Bridge}(M, M')$ *is at most B.*

Proof. Let $D_0 := d_2(M, M')$ and $M_0 := \{m \in M \mid d_2(m, M') = D_0\}$. Since $\mathrm{Bridge}(M, M')$ is isometric to a product $M_0 \times [0, D_0]$ and $D_0 \leq D$, it suffices to bound the diameter of M_0. For any $x, y \in M_0$, consider the geodesic 4–gon whose sides are $[x, y]$, $[y, \pi_{M'}(y)]$, $[\pi_{M'}(y), \pi_{M'}(x)]$, and $[\pi_{M'}(x), x]$. Let $\mathcal{K}$ be a chain realizing $d_c(x, y)$ and $\mathcal{K}'$ the subchain of $\mathcal{K}$ consisting of walls separating $\pi_{M'}(x)$ and $\pi_{M'}(y)$. By hypothesis $|\mathcal{K}'| \leq C$, so Lemma 2.22 and Lemma 2.23 give:

$$d_2(x, y) \asymp d_c(x, y) \overset{+}{\asymp} d_c(x, y) - C \leq |\mathcal{K}| - |\mathcal{K}'| \preceq 2D_0 \qquad \square$$

Remark. The converse of Lemma 3.6 is not true: having a small bridge does not limit the number of common crossers. This is different than in the right-angled case. For example, the Davis complex of the 3,3,3 triangle group is the regular hexagonal tiling of the plane. The walls are three families of parallel lines. Any line M from the first family crosses any line M' from the second family, so their bridge is the single point $M \cap M'$, but the entire infinite third family consists of lines that cross both M and M'.

3.1.2. *Wall characterization of the Morse property.*

Theorem 3.7. *For all λ and ϵ the following are equivalent for all (λ, ϵ)–quasigeodesic paths $\gamma\colon I \to \Sigma$. Moreover, there are explicit bounds between the constraints of each item that are independent of γ.*

- *(a) There exist $K > 2$, r_0, and a quadratic polynomial $q(r)$ with positive leading coefficient such that γ has divergence gauge $\delta(r; 1, K)$ satisfying $\delta(r; 1, K) \geq q(r)$ for all $r \geq r_0$.*
- *(b) There exists μ such that γ is μ–Morse.*
- *(c) There exists χ such that γ is χ–strongly contracting.*
- *(d) There exist C and L_0 such that for every compact subinterval $I' \subset I$ of length at least L_0 there exists a chain $\mathcal{V} := \{V_i\}_{1 \leq i \leq n}$ separating the endpoints of $\gamma(I')$ such that for every pair V_i, V_{i+1} of consecutive walls of $\mathcal{V}$ there are at most C–many walls transverse to $\{V_i, V_{i+1}\}$ and such that for all $0 \leq i \leq n$ we have $|t_{i+1} - t_i| \leq 7L_0$, where $I' = [t_0, t_{n+1}]$ and $t_i := \min\{t \in I' \mid \gamma(t) \in V_i\}$.*
- *(e) There exist C and $0 < \rho$ such that there exists L_1 such that for every compact subinterval $I' \subset I$ of length at least L_1 there exists a chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ such that for every pair V, V' of consecutive walls of $\mathcal{V}$ the longest chain $\mathcal{H}$ transverse to $\{V, V'\}$ has $|\mathcal{H}| \leq C$.*

The proof of the theorem will follow from some auxiliary results. Lemma 3.8 and Lemma 3.9 use strong contraction to produce walls that are "almost orthogonal to" γ. Proposition 3.10 assembles these walls into a chain with good properties. Proposition 3.11 derives quadratic divergence from such a chain.

Lemma 3.8. *For all χ there exists χ' such that for all λ and ϵ there exists χ'' such that for every χ–strongly contracting (λ, ϵ)-quasigeodesic path $\gamma\colon I \to \Sigma$, and for every wall M such that* $\mathrm{diam}(\pi_\gamma(M)) > 2\chi'$*, there is a subinterval $J_M \subset I$ such that:*

$$\gamma^{-1}(\pi_\gamma(M)) \subset J_M \quad \textit{and} \quad \gamma(J_M) \subset \bar{\mathcal{N}}_{\chi''}(M)$$

Proof. By Proposition 2.33, strong contraction is quantitatively equivalent to strong constriction, so there exists χ' depending only on χ such that if $[x, y] \subset \Sigma$ is a geodesic, $\mathrm{diam}(\pi_\gamma(x) \cup \pi_\gamma(y)) > \chi'$ implies $[x, y]$ passes within distance χ' of $\pi_\gamma(x)$ and $\pi_\gamma(y)$. Suppose M is a wall such that $\mathrm{diam}(\pi_\gamma(M)) > 2\chi'$. For every $x_0 \in \pi_\gamma(M)$ there exists $x_1 \in \pi_\gamma(M)$ with $d_2(x_0, x_1) > \chi'$, else $\mathrm{diam}(\pi_\gamma(M))$ would be at most $2\chi'$. Choose $m_0, m_1 \in M$ such that $x_i \in \pi_\gamma(m_i)$. Since M is convex, it contains the geodesic $[m_0, m_1]$, which, by the result above, passes within distance χ' of $\pi_\gamma(m_0)$ and $\pi_\gamma(m_1)$. Since $\mathrm{diam}(\pi_\gamma(m_0)) \leq \chi$, the geodesic $[m_0, m_1] \subset M$ passes within distance $\chi + \chi'$ of x_0. Since x_0 was an arbitrary point of $\pi_\gamma(M)$, this gives $\pi_\gamma(M) \subset \bar{\mathcal{N}}_{\chi+\chi'}(M)$.

Take J_M to be the smallest closed subinterval of I containing $\gamma^{-1}(\pi_\gamma(M))$. Suppose I'' is a maximal open subinterval of J_M with $\gamma(I'') \subset \bar{\mathcal{N}}^c_{\chi+\chi'}(M)$. Convexity of $d_2(\cdot, M)$ implies the geodesic between the endpoints of $\gamma(\bar{I}'')$ is contained in $\bar{\mathcal{N}}_{\chi+\chi'}(M)$. Lemma 2.34 provides D depending on χ, λ, and ϵ such that $\gamma(\bar{I}'')$ and the geodesic between its endpoints have Hausdorff distance at most D. Take $\chi'' := \chi + \chi' + D$. $\square$

Lemma 3.9. *For all χ, λ, and ϵ there exist L_0 such that for every χ–strongly contracting (λ, ϵ)–quasigeodesic path segment $\gamma\colon I \to \Sigma$, for every subinterval $I' \subset I$ of length L_0 there exists a wall M that separates the*

endpoints of $\gamma(I')$, separates the endpoints of $\gamma(I)$, and admits a subinterval $J_M \subset I$ such that $\gamma^{-1}(\pi_\gamma(M)) \subset J_M$ and $|J_M| \leq L_0$.

Proof. Take χ' and χ'' of Lemma 3.8. Let ω be the wall growth function of Section 2.2.3. Lemma 2.34 provides D depending on χ, λ, and ϵ such that $\gamma(I)$ and the geodesic α between its endpoints have Hausdorff distance at most D. From the construction of χ'' in the proof of Lemma 3.8, we have $\chi'' \geq D$. By Lemma 2.22, $d_w \asymp d_2$, and γ gives coarse equivalence between $|\cdot|$ on its domain and d_2 on its image, so we can choose $L_0 \geq \lambda(2\chi' + \epsilon)$ large enough that $|x-y| \geq L_0$ implies $d_w(\gamma(x), \gamma(y)) > 2\omega(\chi'')$.

Suppose $[x,y] = I' \subset I = [a,b]$, with $|I'| = L_0$. Consider $M \in \mathcal{M}(\gamma(x),\gamma(y)) \smallsetminus \mathcal{M}(\gamma(a),\gamma(b))$, the walls that separate endpoints of $\gamma(I')$ but fail to separate endpoints of $\gamma(I)$. Since M fails to separate the endpoints of $\gamma(I)$, which are the endpoints of α, by convexity of halfspaces there is a closed halfspace $\bar{M}^+$ of M containing α. Since M separates $\gamma(x)$ and $\gamma(y)$, we may assume $\gamma(x) \in M^-$ and $\gamma(y) \in M^+$. Then $d_2(\gamma(x), M) = d_2(\gamma(x), \bar{M}^+) \leq d_2(\gamma(x), \alpha) \leq D$. Thus, every wall that separates the endpoints of $\gamma(I')$ but fails to separate the endpoints of $\gamma(I)$ passes within distance $D \leq \chi''$ of one of the endpoints of $\gamma(I')$.

For each wall $M \in \mathcal{M}(\gamma(x),\gamma(y))$, let J_M be the smallest closed interval of I such that $\gamma^{-1}(\pi_\gamma(M)) \subset J_M$. Notice that since γ is continuous, for every $M \in \mathcal{M}(\gamma(x),\gamma(y))$ there exists $t \in (x,y)$ such that $\gamma(t) \in M$, and J_M contains this time t, so every J_M has nonempty intersection with $[x,y]$. Let $\mathcal{M}^+ \subset \mathcal{M}(\gamma(x),\gamma(y))$ be the walls for which $|J_M| > L_0$. For each $M \in \mathcal{M}^+$, the interval J_M contains a point of $[x,y]$ and is strictly longer than $[x,y]$, so it contains at least one of the endpoints, x or y. Since J_M is smallest, each endpoint of J_M is either in, or is an accumulation point of, $\gamma^{-1}(\pi_\gamma(M))$, so $\operatorname{diam}(\pi_\gamma(M))$ is at least the d_2–distance between the endpoints of $\gamma(J_M)$, which is at least $|J_M|/\lambda - \epsilon > L_0/\lambda - \epsilon \geq 2\chi'$. Lemma 3.8 says $\gamma(J_M) \subset \bar{\mathcal{N}}_{\chi''}(M)$. Since J_M contains at least one of x or y, the wall M passes within distance χ'' of either $\gamma(x)$ or $\gamma(y)$.

A wall of $\mathcal{M}(\gamma(x),\gamma(y))$ can fail to satisfy the lemma either by failing to separate the endpoints of $\gamma(I)$, or by belonging to $\mathcal{M}^+$. We have shown in both cases that such a wall passes within distance χ'' of an endpoint of $\gamma(I')$. There are only at most $2\omega(\chi'')$–many walls passing within distance χ'' of one of the two endpoints of $\gamma(I')$, but the choice of L_0 implies that $|\mathcal{M}(\gamma(x),\gamma(y))| = d_w(\gamma(x),\gamma(y)) > 2\omega(\chi'')$, so there are walls of $\mathcal{M}(\gamma(x),\gamma(y))$ that separate the endpoints of $\gamma(I)$ and have $|J_M| \leq L_0$. □

Proposition 3.10. *For all χ, λ, and ϵ there exist C and L_0 such that for every χ–strongly contracting (λ,ϵ)–quasigeodesic path $\gamma\colon I \to \Sigma$, with I a compact interval of length at least L_0, there exists a chain $\mathcal{V}$ separating the endpoints of $\gamma(I)$ such that for every pair V, V' of consecutive walls of $\mathcal{V}$ the number of walls transverse to $\{V, V'\}$ is at most C, and such that $|t_{i+1} - t_i| \leq 7L_0$, where the t_i are defined as follows. If $\mathcal{V} = \{V_i\}_{1 \leq i \leq n}$, then let $t_0 := \min I$, $t_{n+1} := \max I$, and for each $1 \leq i \leq n$, let $t_i := \min \gamma^{-1}(\gamma(I) \cap V_i)$.*

Proof. Given χ, λ, and ϵ take χ', χ'', L_0, and ω as in Lemma 3.9, and take $C := \omega(\chi'')$. By shifting the parameterization of γ, if necessary, we may assume $(0, L_0) \subset I = [a,b]$. Let $s_i := iL_0$ for $i \in \mathbb{Z}$. If $(s_i, s_{i+1}) \subset I$ then by Lemma 3.9 there exists a wall $V_i \in \mathcal{M}(\gamma(s_i),\gamma(s_{i+1})) \cap \mathcal{M}(\gamma(a),\gamma(b))$ and an interval $J_i \subset I$ of length at most L_0 such that $\gamma^{-1}(\pi_\gamma(V_i)) \subset J_i$. In particular, J_i contains every time t such that $\gamma(t) \in V_i$, since at such times $\pi_\gamma(\gamma(t)) = \gamma(t)$. Since J_i has length at most L_0 and $J_i \cap (s_i, s_{i+1}) \neq \emptyset$, we have $J_i \subset ((i-1)L_0, (i+2)L_0) \cap I$. In particular, if $j \geq i+3$ and $(s_i, s_{j+1}) \subset I$ then J_i is disjoint from $J_j \subset ((j-1)L_0, (j+2)L_0)$, which implies that V_i and V_j are disjoint, since a point in $V_i \cap V_j$ would project via π_γ to a point in $\pi_\gamma(V_i) \cap \pi_\gamma(V_j)$, but $\gamma^{-1}(\pi_\gamma(V_i) \cap \pi_\gamma(V_j)) \subset J_i \cap J_j = \emptyset$.

Suppose for $i+1 < j < k-1$ in $\mathbb{Z}$ that $(s_i, s_{k+1}) \subset I$ and M is a wall transverse to both V_i and V_k. For $z \in M \cap V_i$, we have $\pi_\gamma(z) \subset \pi_\gamma(M) \cap \pi_\gamma(V_i)$, so $\gamma^{-1}(\pi_\gamma(M))$ contains a point in J_i. Similarly, $\gamma^{-1}(\pi_\gamma(M))$ contains a point in J_k. Now, $J_i \subset ((i-1)L_0, (i+2)L_0)$ and $J_k \subset ((k-1)L_0, (k+2)L_0)$ so $[jL_0, (j+1)L_0]$ separates J_i and J_k in I. It follows that $\operatorname{diam}(\pi_\gamma(M)) > L_0/\lambda - \epsilon > 2\chi'$, so Lemma 3.8 implies $\gamma([jL_0, (j+1)L_0]) \subset \bar{\mathcal{N}}_{\chi''}(M)$. Thus, every point of $\gamma([jL_0, (j+1)L_0])$ is χ''–close to every wall that is transverse to both V_i and V_k. The constant C is the maximum number of walls intersecting any χ''–ball, so there are at most C–many walls transverse to both V_i and V_k when $k \geq i+4$ and $(s_i, s_{k+1}) \subset I$.

Take $I' := \{i \in 4\mathbb{Z} \mid (iL_0, (i+1)L_0) \subset I\}$. Then $\mathcal{V} := \{V_i\}_{i \in I'}$ contains at least one wall, V_0, since $(0, L_0) \subset I$. Since wall indices in $\mathcal{V}$ differ by at least 4, the argument above gives that they are disjoint and there are at most C walls transverse to any distinct pair of them. Furthermore, since all the walls were chosen to be in $\mathcal{M}(\gamma(a),\gamma(b))$, there are no facing triples in $\mathcal{V}$, so it is a chain.

Finally, we bound the differences in first crossing times between consecutive walls of $\mathcal{V}$. There is some k such that the i-th wall of $\mathcal{V}$ is $V_{4(i+k)}$, so $t_i \in J_{4(i+k)} \subset ((4(i+k)-1)L_0, (4(i+k)+2)L_0)$. Then $|t_{i+1} - t_i| \leq (4(i+1+k)+2)L_0 - (4(i+k)-1)L_0 = 7L_0$. The first wall is in $J_{4(1+k)} \subset ((4(1+k)-1)L_0, (4(1+k)+2)L_0)$.

The fact that it is the first wall means $(4kL_0, (4k+1)L_0) \not\subset I$, so the initial endpoint of I is greater than $4kL_0$, so $|t_1 - t_0| \leq (4k+6)L_0 - 4kL_0 = 6L_0 \leq 7L_0$. The symmetric argument at the terminal endpoint of I gives $|t_{n+1} - t_n| \leq 6L_0 \leq 7L_0$. □

Proposition 3.11. *Given λ, ϵ, C, $0 < \rho$, and L, suppose $\gamma\colon I \to \Sigma$ is a compact (λ, ϵ)–quasigeodesic path with the property that for every compact subinterval $I' \subset I$ of length at least L there exists a chain $\mathcal{V}$ that separates the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ such that for every pair of consecutive walls V, V' of $\mathcal{V}$, every chain $\mathcal{H} \pitchfork \{V, V'\}$ has $|\mathcal{H}| \leq C$. Then there are $K = K(\lambda, \rho)$, $r_0 = r_0(\lambda, \epsilon, \rho, C, L)$, and a quadratic polynomial $q(r)$ with positive leading coefficient whose coefficients depend on λ, ϵ, ρ, and C, such that γ has divergence gauge $\delta(r; 1, K)$ satisfying $\delta(r; 1, K) \geq q(r)$ for $r \geq r_0$.*

Proof. Suppose λ_0 and ϵ_0 are constants such that Lemma 2.22 implies that for all $x, y \in \Sigma$:

$$d_2(x,y)/\lambda_0 - \epsilon_0 \leq d_c(x,y) \leq \lambda_0 d_2(x,y) + \epsilon_0$$

Take $K := 2 + 2\lambda(1 + 2\lambda_0)/\rho$. Consider $r \geq r_0$ where r_0 is a constant that is a maximum of three quantities appearing in the course of the proof.

Suppose there exist points x and y such that $d_2(x, \gamma) = d_2(y, \gamma) = r$ and $d_2(x, y) \geq Kr$, so that $d_2(\pi_\gamma(x), \pi_\gamma(y)) \geq (K-2)r$. Suppose there exists a path α from x to y in $\mathcal{N}_r^c(\gamma)$.

Pick $x' \in \pi_\gamma(x)$ and $y' \in \pi_\gamma(y)$ and let I' be the smallest closed interval whose endpoints map to x' and y'. This implies $|I'| \geq ((K-2)r - \epsilon)/\lambda$, which is at least L once $r_0 \geq (\lambda L + \epsilon)/(K-2)$, so there exists a chain $\mathcal{V} := \{V_i\}_{1 \leq i \leq n}$ as in the statement.

For each i, let t_i be the first time in I' such that $\gamma(t_i) \in \bar{V}_i^+$. Since $\bar{V}_{i+1}^+ \subset V_i^+$, the first crossing times $t_i < t_{i+1}$ are ordered compatibly with the index.

We have $n-1$ intervals $[t_i, t_{i+1}]$ with $\sum_{i=1}^{n-1} |t_{i+1} - t_i| \leq |I'|$ and $n = |\mathcal{V}| \geq \rho|I'|$, so fewer than $|I'|/(2/\rho)$ many of the intervals have length strictly greater than $2/\rho$.

By Lemma 2.23 the convex subchain $\mathcal{V}'$ of $\mathcal{V}$ consisting of the walls that separate x and y satisfies:

$$|\mathcal{V}| - |\mathcal{V}'| \leq d_c(x, x') + 1 + d_c(y, y') + 1 \leq 2(\lambda_0 d_2(x, x') + \epsilon_0 + 1) = 2(\lambda_0 r + \epsilon_0 + 1)$$

See Figure 2.

The number of intervals $[t_i, t_{i+1}]$ such that both V_i and V_{i+1} are in $\mathcal{V}'$ is $|\mathcal{V}'| - 1$, and:

$$|\mathcal{V}'| - 1 \geq |\mathcal{V}| - 2(\lambda_0 r + \epsilon_0 + 1) - 1 \geq \rho|I'| - 2\lambda_0 r - (2\epsilon_0 + 3)$$

Fewer than $\rho|I'|/2$ of those are long, so the number of intervals with length at most $2/\rho$ such that V_i and V_{i+1} are in $\mathcal{V}'$ is at least:

$$\begin{aligned} \rho|I'| - 2\lambda_0 r - (2\epsilon_0 + 3) - \rho|I'|/2 &= \rho|I'|/2 - 2\lambda_0 r - (2\epsilon_0 + 3) \\ &\geq \frac{\rho}{2}\frac{(K-2)r - \epsilon}{\lambda} - 2\lambda_0 r - (2\epsilon_0 + 3) \\ &= r - \left(\frac{\rho\epsilon}{2\lambda} + 2\epsilon_0 + 3\right) \end{aligned} \tag{3}$$

Assume $r_0 \geq \frac{\rho\epsilon}{2\lambda} + 2\epsilon_0 + 4$, so that (3) gives at least one good interval.

Let V_i and V_{i+1} be a pair of consecutive walls of $\mathcal{V}'$ such that $0 < t_{i+1} - t_i \leq 2/\rho$. This implies $d_2(V_i, V_{i+1}) \leq 2\lambda/\rho + \epsilon$. Corollary 3.5 says the length of a quasigeodesic with fixed constants connecting V_i and V_{i+1} not contained in $\mathcal{N}_R(\mathrm{Bridge}(V_i, V_{i+1}))$ has parameter interval length growing linearly in R, with constants depending on the quasigeodesic constants and $d_2(V_i, V_{i+1})$ but not on V_i and V_{i+1}. Since $t_{i+1} - t_i \leq 2/\rho$, the corresponding segment of γ is contained in a uniform neighborhood of $\mathrm{Bridge}(V_i, V_{i+1})$. Lemma 3.6 says $\mathrm{diam}(\mathrm{Bridge}(V_i, V_{i+1}))$ is bounded in terms of $2\lambda/\rho + \epsilon$ and C. Thus, $\mathrm{Bridge}(V_i, V_{i+1})$ and $\gamma([t_i, t_{i+1}])$ are coarsely equivalent, with constants depending on ρ, λ, ϵ, and C.

On the other hand, since α is required to stay far from γ, this implies α stays outside $\mathcal{N}_R(\mathrm{Bridge}(V_i, V_{i+1}))$ for some $R \stackrel{+}{\asymp} r$, so Lemma 3.4 says the length of the subsegment of α from its first encounter with V_i to its first encounter with V_{i+1} is bounded below by a linear function of r, with positive slope, whose constants depend on $d_2(V_i, V_{i+1})$, which is bounded above by $2\lambda/\rho + \epsilon$. Assume r_0 is large enough that this linear function takes positive values for all $r \geq r_0$. The bound from (3) says the number of such subsegments of α is linear in r with slope 1. Those subsegments of α have disjoint interiors, so the total length of α is bounded below by a quadratic polynomial $q(r)$ with positive leading coefficient whose coefficients depend on ρ, λ, ϵ, and C.

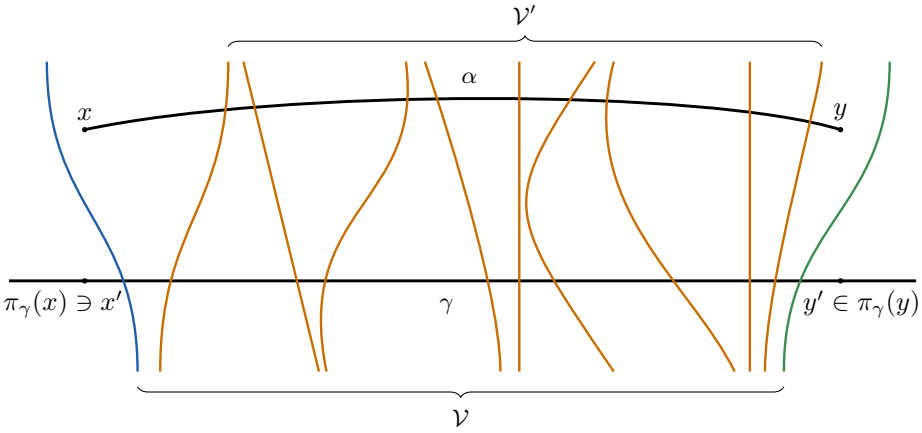


FIGURE 2. Density control on the chain separating $x', y' \in \gamma$ forces linearly many of the interwall subsegments of γ to be short and the corresponding interwall subsegments of α to be long.

We have shown that $\delta(r; 1, K) \geq q(r)$ when $r \geq r_0$ as desired. Notice that if for some r there is no such pair of points (x, y) or there is no such path α, then $\delta(r; 1, K) = \infty$, so it is still true that $\delta(r; 1, K) \geq q(r)$. □

Proof of Theorem 3.7. The equivalence of (a), (b), and (c) is Proposition 2.32.

Suppose (c) holds. By Lemma 2.35, there exists χ' depending only on χ, λ, and ϵ, such that every compact subsegment of γ is χ'–strongly contracting. Proposition 3.10 then implies (d).

Assuming (d), set $\rho := 1/14L_0$ and $L_1 := 14L_0$. Let I' be a compact subinterval of I of length at least L_1, take $\mathcal{V} := \{V_i\}_{1 \leq i \leq n}$ to be the chain supplied by (d), and define t_0 and t_{n+1} to be the endpoints of I'. Since the subsegments given by (d) have length at most $7L_0$:

$$|\mathcal{V}|/|I'| = n/|I'| \geq \frac{\frac{|I'|}{7L_0} - 1}{|I'|} \geq \frac{1}{7L_0} - \frac{1}{14L_0} = \frac{1}{14L_0} = \rho$$

This gives (e).

Finally, the proof (e) $\implies$ (a) amounts to showing that uniform quadratic divergence on all compact subintervals passes to the whole quasigeodesic. Suppose (e) holds with constant C, ρ, and L_1. Take K, r_0, and q as provided by Proposition 3.11 with respect to λ, ϵ, C, ρ, and L_1. Given $r \geq r_0$ and points x and y at d_2–distance r from γ with $d_2(x, y) \geq Kr$, take $x' \in \pi_\gamma(x)$ and $y' \in \pi_\gamma(y)$, and consider a compact subinterval $J \subset I$ containing a point from $\gamma^{-1}(x')$ and a point from $\gamma^{-1}(y')$. Then $d_2(x, \gamma(J)) = d_2(y, \gamma(J)) = r$. Proposition 3.11 says that every path from x to y that stays outside $\mathcal{N}_r(\gamma(J))$ has length bounded below by $q(r)$, for the quadratic function q. The paths from x to y that stay outside $\mathcal{N}_r(\gamma(I))$ are a subset of those that stay outside $\mathcal{N}_r(\gamma(J))$, so the same lower length bound applies. This shows that q is a quadratic lower bound on the divergence of $\gamma(I)$. □

3.2. **Grids and wide parabolics.** Section 3.2.1 describes grids and uses them to produce wide parabolics. Section 3.2.2 states and proves the main result of this subsection, Theorem 3.23, which relates a quasigeodesic's interactions with wide parabolic subcomplexes and dense transverse chains.

3.2.1. *Grids.*

Definition 3.12. A *grid* is a pair $(\mathcal{V}, \mathcal{H})$ (vertical/horizontal) of transverse chains that each contain at least two walls.

Definition 3.13. Let ℓ be as in Section 2.2.3. If $\mathcal{V}$ is a chain, define its *interior* $\mathring{\mathcal{V}}$ to be all the walls V of $\mathcal{V}$ such that there are at least $\ell - 1$ walls of $\mathcal{V}$ that come before V in the chain and at least $\ell - 1$ walls that come after V.

The goal of this subsection is to prove[2] the following proposition, which is essentially the Grid Lemma attributed to Caprace and Przytycki by Caprace and Marquis [38, Lemma 2.8].

Proposition 3.14. *Let $(\mathcal{V},\mathcal{H})$ be a grid in Σ such that $|\mathcal{V}|\geq 2\ell-1$ and $|\mathcal{H}|\geq 2\ell$. Then $\mathrm{Pc}(\mathcal{V})\times\mathrm{Pc}(\mathcal{V})^{\perp}$ is a wide parabolic and either $\mathrm{Pc}(\mathcal{V})=\mathrm{Pc}(\mathcal{H})$ is irreducible higher rank affine or $\mathrm{Pc}(\mathring{\mathcal{H}})\leq\mathrm{Pc}(\mathcal{V})^{\perp}$.*

Lemma 3.15 (cf [35, Lemma 2.3][3])**.** *Let R_1 and R_2 be finite sets of reflections in (W,S) such that $\langle R_1\rangle$ and $\langle R_2\rangle$ are irreducible, and such that $r_1r_2=r_2r_1$ for all $r_1\in R_1$ and $r_2\in R_2$. Suppose $\langle R_1\rangle$ is infinite. Then one of the following is true:*

- $\mathrm{Pc}(R_1\cup R_2)=\mathrm{Pc}(R_1)\times\mathrm{Pc}(R_2)$
- $R_2\leq\mathrm{Pc}(R_1)$

If, in addition, $\langle R_2\rangle$ is infinite, then $R_2\leq\mathrm{Pc}(R_1)$ implies $\mathrm{Pc}(R_1)=\mathrm{Pc}(R_2)$ is higher rank affine.

Example 3.16. Consider the Coxeter system (W,S) defined by the Coxeter diagram $a\overset{4}{-}b\overset{4}{-}c\overset{4}{-}d$, which is not affine. Consider the finite sets of reflections $R_1:=\{bab,c,d\}$ and $R_2:=\{a\}$. Then $aca=c$, $ada=d$ and $a(bab)a=(abab)a=(baba)a=bab$, so R_1 and R_2 commute. In fact, from Allcock's description of reflection centralizers [11], one finds that $\langle R_1\rangle$ is the entire reflection part of $Z_W(\langle R_2\rangle)$. The group $\mathrm{Pc}(R_2)\cong\mathbb{Z}/2\mathbb{Z}$ is irreducible. The group $\langle R_1\rangle$ is infinite and irreducible; to see this, consider $P:=\langle a,b,c\rangle$, which is a Euclidean triangle parabolic subgroup of W. In its usual $\mathrm{Isom}(\mathbb{E}^2)$ representation the elements $bab,c\in R_1\cap P$ are reflections through parallel lines, so $\langle R_1\rangle$ contains an infinite dihedral subgroup. This also shows that $\langle R_1\rangle$ is irreducible, since c commutes with neither bab nor d. Furthermore, P is 2–spherical and rank 3, so every proper parabolic subgroup is finite, so $\mathrm{Pc}(\langle bab,c\rangle)=P$. But $d\in R_1\leq\mathrm{Pc}(R_1)\geq\mathrm{Pc}(\langle bab,c\rangle)$, so $\mathrm{Pc}(R_1)=W$. Thus, $\mathrm{Pc}(R_2)\lneq\mathrm{Pc}(R_1)$ and $\mathrm{Pc}(R_1)$ is not affine. This contradicts both conclusions in the $R_2\leq\mathrm{Pc}(R_1)$ branch of [35, Lemma 2.3].

Lemma 3.17. *Let $(\mathcal{V},\mathcal{H})$ be a grid in Σ. Suppose one of the following is true:*

- *$W(\mathcal{V})$ and $W(\mathcal{H})$ commute.*
- *$W(\mathcal{V})$ is dihedral and $\mathcal{V}$ contains at least 8 walls.*

Then one of the following is true:

- $\mathrm{Pc}(\mathcal{H})\leq\mathrm{Pc}(\mathcal{V})^{\perp}$
- $\mathrm{Pc}(\mathcal{H})=\mathrm{Pc}(\mathcal{V})$ *is an irreducible higher rank affine parabolic.*

Proof. By Lemma 2.11, $\mathrm{Pc}(\mathcal{V})$ and $\mathrm{Pc}(\mathcal{H})$ are irreducible and nonspherical. Furthermore, since it does not change $\mathrm{Pc}(\mathcal{V})$ and $\mathrm{Pc}(\mathcal{H})$, we may assume $\mathcal{V}$ and $\mathcal{H}$ are finite chains, so that $W(\mathcal{V})$ and $W(\mathcal{H})$ are finitely generated reflection groups.

By Lemma 3.15 we can partition the set $R:=\{\mathfrak{r}_H\mid H\in\mathcal{H}\}$ of reflections through walls of $\mathcal{H}$ into disjoint subsets:

- $R_0:=\{\mathfrak{r}_H\mid H\in\mathcal{H},\,\mathfrak{r}_H\notin Z_W(W(\mathcal{V}))\}$
- $R_1:=\{\mathfrak{r}_H\mid H\in\mathcal{H},\,\mathfrak{r}_H\in Z_W(W(\mathcal{V}))\cap\mathrm{Pc}(\mathcal{V})\}$
- $R_2:=\{\mathfrak{r}_H\mid H\in\mathcal{H},\,\mathfrak{r}_H\in\mathrm{Pc}(\mathcal{V})^{\perp}\}$

Furthermore, if $|R_1|\geq 2$ then $\mathrm{Pc}(R_1)=\mathrm{Pc}(\mathcal{V})$ is higher rank affine.

Reflections in distinct walls of $\mathcal{H}$ do not commute, so at least one of $R\cap\mathrm{Pc}(\mathcal{V})$ or R_2 is empty.

$W(\mathcal{V})$ and $W(\mathcal{H})$ commute if and only if $R_0=\emptyset$, in which case we are done, since then either $R=R_1$ or $R=R_2$. The former gives $|R_1|\geq 2$, so $\mathrm{Pc}(\mathcal{H})=\mathrm{Pc}(R_1)=\mathrm{Pc}(\mathcal{V})$ is higher rank affine. The latter gives $\mathrm{Pc}(\mathcal{H})=\mathrm{Pc}(R_2)\leq\mathrm{Pc}(\mathcal{V})^{\perp}$.

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[2]The proof of the Grid Lemma presented in [38] has a gap. In our language, that proof supposes that there is a grid $(\mathcal{V},\mathcal{H})$ with $\mathcal{V}=\{V_i\}_{0\leq i\leq k}$, and supposes there is *any* $0\leq i_0\leq k$ such that there is an $H_j\in\mathring{\mathcal{H}}$ such that $\mathfrak{r}_{V_{i_0}}$ and $\mathfrak{r}_{H_j}$ do not commute. It then applies [38, Lemma 2.9], which is Theorem 2.18, to $\{V_i\}_{0\leq i\leq i_0}$. However, Theorem 2.18 requires the chain to be sufficiently long, which is not true if i_0 is small. Proposition 3.20 below argues that given the noncommuting pair of walls there are four possible configurations to which one might try to apply Theorem 2.18, and at least one of them satisfies that theorem's hypotheses. This fills the gap.

[3]Lemma 3.15 is a corrected version of [35, Lemma 2.3] stating what is actually proven there. The original omits the hypothesis that $\langle R_2\rangle$ is infinite in the final sentence, but still concludes $\mathrm{Pc}(R_1)=\mathrm{Pc}(R_2)$ is higher rank affine. This is an error, as demonstrated by Example 3.16. The specific error in the proof of [35, Lemma 2.3] is the application of [35, Lemma 2.2] in the case $R_2\leq\mathrm{Pc}(R_1)$ without explicitly requiring $\langle R_2\rangle$ to be nonspherical.

If $R_0 \neq \emptyset$ then, by hypothesis, $W(\mathcal{V})$ is dihedral and $\mathcal{V}$ contains at least 8 walls. For $\mathfrak{r}_H \in R_0$, Lemma 2.19 says $W(H, \mathcal{V})$ is a Euclidean triangle group, so Lemma 2.21 gives $\mathfrak{r}_H \in \mathrm{Pc}(\mathcal{V})$, and Proposition 2.20 says $\mathrm{Pc}(\mathcal{V})$ is higher rank affine. Since $\emptyset \neq R_0 \subset R \cap \mathrm{Pc}(\mathcal{V})$, $R_2 = \emptyset$, so $R = R_0 \cup R_1$, which gives $\mathrm{Pc}(\mathcal{H}) \leq \mathrm{Pc}(\mathcal{V})$. Since irreducible affine parabolics do not contain proper nonspherical parabolic subgroups and $\mathrm{Pc}(\mathcal{H})$ is a nonspherical parabolic, $\mathrm{Pc}(\mathcal{H}) = \mathrm{Pc}(\mathcal{V})$. □

Lemma 3.18. *Let $(\mathcal{V}, \mathcal{H})$ be a grid in Σ, with $\mathcal{V} = \{V_i\}_{i\in I}$ and $\mathcal{H} = \{H_j\}_{j\in J}$. Suppose, for some $i_0 \in I$ and $j_0 \in J$ there is a wall M that contains $V_{i_0} \cap H_{j_0}$. Then M is transverse to at least two of the following chains:*

$$\mathcal{V}^- := \{V_i\}_{\{i\in I | i\leq i_0\}} \qquad \mathcal{V}^+ := \{V_i\}_{\{i\in I | i\geq i_0\}}$$
$$\mathcal{H}^- := \{H_j\}_{\{j\in J | j\leq j_0\}} \qquad \mathcal{H}^+ := \{H_j\}_{\{j\in J | j\geq j_0\}}$$

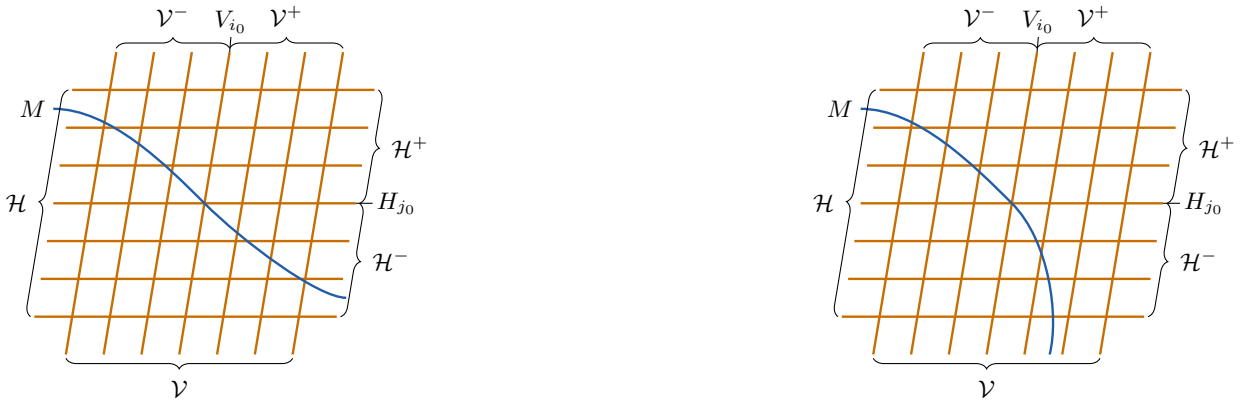


FIGURE 3. Two possibilities for a wall M containing $V_{i_0} \cap H_{j_0}$, separating $V_{i_0}^+ \cap H_{j_0}^+$ from $V_{i_0}^- \cap H_{j_0}^-$ and $M \not\pitchfork \mathcal{H}^+$. We have $M \pitchfork \mathcal{V}^-$ and at least one of $M \pitchfork \mathcal{V}^+$ or $M \pitchfork \mathcal{H}^-$.

Proof. If M is equal to V_{i_0} then it is transverse to all of $\mathcal{H}$, so it is transverse to both of $\mathcal{H}^-$ and $\mathcal{H}^+$. If M is equal to H_{j_0} then it is transverse to both $\mathcal{V}^-$ and $\mathcal{V}^+$. Now assume M is not one of V_{i_0} or H_{j_0}. Then $V_{i_0} \cup H_{j_0}$ divides Σ into four regions, $V_{i_0}^- \cap H_{j_0}^-$, $V_{i_0}^+ \cap H_{j_0}^-$, $V_{i_0}^+ \cap H_{j_0}^+$, and $V_{i_0}^- \cap H_{j_0}^+$. The wall M meets two of these, which are opposites, either $V_{i_0}^- \cap H_{j_0}^-$ and $V_{i_0}^+ \cap H_{j_0}^+$ or $V_{i_0}^- \cap H_{j_0}^+$ and $V_{i_0}^+ \cap H_{j_0}^-$, and separates the other two. Suppose, without loss of generality, that M separates $V_{i_0}^- \cap H_{j_0}^-$ from $V_{i_0}^+ \cap H_{j_0}^+$.

Suppose i_0 and j_0 are not extremes. Suppose M is not transverse to $\mathcal{H}^+$. Then there is some first $j_1 > j_0$ in J such that M is disjoint from H_{j_1}. For any $i < i_0$ there is a path from $V_{i_0}^- \cap H_{j_0}^-$ to $V_{i_0}^+ \cap H_{j_0}^+$ by travelling through V_i and H_{j_1}. Specifically, choose a point $a \in V_i \subset V_{i_0}^-$ in the interior of $H_{j_0}^-$, which exists, since $V_i \pitchfork H_{j_0}$, a point $b \in V_i \cap H_{j_1}$, which exists, since $V_i \pitchfork H_{j_1}$, and a point $c \in H_{j_1} \subset H_{j_0}^+$ in the interior of $V_{i_0}^+$, which exists, since $V_{i_0} \pitchfork H_{j_1}$. By convexity of walls, $[a, b] \subset V_i \subset V_{i_0}^-$ and $[b, c] \subset H_{j_1} \subset H_{j_0}^+$. Now, since M separates $V_{i_0}^- \cap H_{j_0}^-$ from $V_{i_0}^+ \cap H_{j_0}^+$, it cuts either $[a, b]$ or $[b, c]$, but if M is disjoint from $H_{j_1} \supset [b, c]$ then it must be that M cuts $[a, b] \subset V_i$, so $M \pitchfork V_i$. Since this was true for every $i < i_0$, we conclude $M \not\pitchfork \mathcal{H}^+$ implies $M \pitchfork \mathcal{V}^-$.

An analogous argument yields that M is transverse to at least one of $\mathcal{H}^-$ or $\mathcal{V}^+$.

When one or both of i_0 and j_0 are extremes then the argument is similar, with the added observation that if, say, i_0 is the least element of I, then $\mathcal{V}^-$ is the single wall V_{i_0}, which is transverse to M by hypothesis. □

Lemma 3.19. *Let M and M' be distinct walls such that $\mathfrak{r}_M$ and $\mathfrak{r}_{M'}$ do not commute. Let $M'' := \mathfrak{r}_M(M')$. Then M'' is distinct from M and M', and $M \cap M' = M \cap M'' = M' \cap M''$.*

Proof. If $M = M'' = \mathfrak{r}_M(M')$ then $M = \mathfrak{r}_M(M) = \mathfrak{r}_M^2(M') = M'$. If $M' = M'' = \mathfrak{r}_M(M')$ then since $\mathfrak{r}_{M''} = \mathfrak{r}_M \mathfrak{r}_{M'} \mathfrak{r}_M$, we get that $\mathfrak{r}_M$ and $\mathfrak{r}_{M'}$ commute. Both are contradictions.

The first intersection equality is direct: $M \cap M' \subset M$, so it is fixed pointwise by $\mathfrak{r}_M$, but $\mathfrak{r}_M(M \cap M') = M \cap \mathfrak{r}_M(M') = M \cap M''$.

For the equality $M \cap M' = M' \cap M''$, the main step is to show $\mathfrak{r}_M \in \mathrm{Pc}(M', M'')$. In the canonical representation of (W, S) on $\mathbb{R}^{|S|}$, halfspaces of walls in Σ correspond to roots in $\mathbb{R}^{|S|}$, so ($\pm$–pairs of) roots correspond to reflections, see [72, Section 5]. It is well-known, cf [33, Proposition 3.3], or [94, Lemma 3.1] for a short proof, that if a root α is in the linear span of roots $\beta_1, \ldots, \beta_n$, then the corresponding reflections satisfy $\mathfrak{r}_\alpha \in \mathrm{Pc}(\mathfrak{r}_{\beta_1}, \ldots, \mathfrak{r}_{\beta_n})$. Since M, M', and M'' are distinct, the corresponding roots α_M, $\alpha_{M'}$, and $\alpha_{M''}$ are pairwise non-colinear, but since $M'' = \mathfrak{r}_M(M')$, the reflection formula for roots gives that $\alpha_{M''}$ is in the 2–dimensional subspace spanned by α_M and $\alpha_{M'}$. But then $\alpha_{M'}$ and $\alpha_{M''}$ also span that 2–dimensional subspace, so α_M is in the span of $\alpha_{M'}$ and $\alpha_{M''}$, so $\mathfrak{r}_M \in \mathrm{Pc}(M', M'')$.

To finish the lemma, if $x \in M' \cap M''$ then $\mathrm{Stab}(x)$ is a spherical parabolic subgroup that contains $\mathfrak{r}_{M'}$ and $\mathfrak{r}_{M''}$, so it contains $\mathrm{Pc}(M', M'')$, so it contains $\mathfrak{r}_M$. But $\mathfrak{r}_M(x) = x$ if and only if $x \in M$, so $M' \cap M'' \subset M$. Thus, $M' \cap M'' = M' \cap M'' \cap M = M' \cap M' \cap M = M' \cap M$. □

Proposition 3.20. *Let $(\mathcal{V}, \mathcal{H})$ be a grid in Σ such that $\mathcal{V}$ and $\mathcal{H}$ are finite and each contain at least $(2\ell - 1)$–many walls. If $W(\mathring{\mathcal{H}}) \not\subset Z_W(W(\mathcal{V}))$ or $W(\mathring{\mathcal{V}}) \not\subset Z_W(W(\mathcal{H}))$ then $\mathrm{Pc}(\mathcal{V}) = \mathrm{Pc}(\mathcal{H})$ is irreducible higher rank affine.*

Proof. The bound on the chain size implies $\mathring{\mathcal{V}}$ and $\mathring{\mathcal{H}}$ are nonempty. The cases are symmetric, so suppose $W(\mathring{\mathcal{H}}) \not\subset Z_W(W(\mathcal{V}))$. For $\mathcal{V} = \{V_i\}_{i \in I}$ and $\mathcal{H} = \{H_j\}_{j \in J}$ there are $i_0 \in I$ and $j_0 \in J$ such that $\mathfrak{r}_{V_{i_0}}$ does not commute with $\mathfrak{r}_{H_{j_0}}$, and, furthermore, $H_{j_0} \in \mathring{\mathcal{H}}$. Since V_{i_0} and H_{j_0} are intersecting walls whose reflections do not commute, Lemma 3.19 says that $\mathfrak{r}_{V_{i_0}}(H_{j_0})$ is distinct from V_{i_0} and H_{j_0} and $\emptyset \neq V_{i_0} \cap H_{j_0} = V_{i_0} \cap \mathfrak{r}_{V_{i_0}}(H_{j_0}) = H_{j_0} \cap \mathfrak{r}_{V_{i_0}}(H_{j_0})$. As described by Lemma 3.18, there are four subchains $\mathcal{V}^+$, $\mathcal{V}^-$, $\mathcal{H}^+$, and $\mathcal{H}^-$ with respect to i_0 and j_0, and $\mathfrak{r}_{V_{i_0}}(H_{j_0})$ is transverse to at least two of them. Since j_0 is an interior index, both of $\mathcal{H}^+$ and $\mathcal{H}^-$ contain at least ℓ–many walls, and at least one of $\mathcal{V}^+$ or $\mathcal{V}^-$ does, since $|\mathcal{V}^+| + |\mathcal{V}^-| = |\mathcal{V}| + 1 \geq 2\ell$. Thus, there is at least one of $\mathcal{V}^+$, $\mathcal{V}^-$, $\mathcal{H}^+$, or $\mathcal{H}^-$ that contains at least ℓ–many walls and is transverse to $\mathfrak{r}_{V_{i_0}}(H_{j_0})$.

Suppose that $\mathcal{V}^+$ is a chain transverse to $\mathfrak{r}_{V_{i_0}}(H_{j_0})$ with at least ℓ–many walls. Then Theorem 2.18 for $\mu := \mathfrak{r}_{V_{i_0}}(H_{j_0})$ and $\mu' := H_{j_0}$ and the chain $\mathcal{V}^+$ gives that $W(H_{j_0}, \mathfrak{r}_{V_{i_0}}(H_{j_0}), \mathcal{V}^+) = W(H_{j_0}, \mathcal{V}^+)$ is a Euclidean triangle group that is contained in an irreducible affine parabolic $\mathcal{A}$, which has rank at least 3 since it contains the triangle group. Then Lemma 2.21 says $\mathfrak{r}_{H_{j_0}} \in \mathrm{Pc}(\mathcal{V}^+) \leq \mathcal{A}$. Irreducible affine Coxeter groups have no proper nonspherical parabolics, so since $\mathrm{Pc}(\mathcal{V}^+)$ is nonspherical and parabolic in $\mathcal{A}$, it equals $\mathcal{A}$. Furthermore, a chain in an irreducible affine Coxeter group consists of a family of parallel hyperplanes, so it generates an infinite dihedral reflection subgroup. Since $\mathfrak{r}_{H_{j_0}} \in \mathrm{Pc}(\mathcal{V}^+)$, we do not have $\mathrm{Pc}(\mathcal{H}) \leq \mathrm{Pc}(\mathcal{V}^+)^\perp$, so Lemma 3.17 implies $\mathrm{Pc}(\mathcal{H}) = \mathrm{Pc}(\mathcal{V}^+) = \mathcal{A}$, with $W(\mathcal{H}) \cong \mathcal{D}_\infty \cong W(\mathcal{V}^+)$.

Now consider V_i with $i < i_0$. It is transverse to $\mathcal{H}$, and $|\mathcal{H}| \geq 2\ell - 1 \geq 8$, so by Lemma 2.19 and Lemma 2.21 either $\mathfrak{r}_{V_i} \in \mathcal{A}$ or $\mathfrak{r}_{V_i}$ centralizes $W(\mathcal{H})$. In the latter case, apply Lemma 3.15 with $R_1 = \{\mathfrak{r}_H \mid H \in \mathcal{H}\}$, which is finite, since $\mathcal{H}$ is finite, and $R_2 = \{\mathfrak{r}_{V_i}\}$. The first option of Lemma 3.15 implies $\mathfrak{r}_{V_i}$ centralizes $\mathrm{Pc}(\mathcal{H}) = \mathrm{Pc}(\mathcal{V}^+)$, which is not true, since V_i is disjoint from the walls of $\mathcal{V}^+$. Thus, the second option of Lemma 3.15 holds, so $\mathfrak{r}_{V_i} \in \mathrm{Pc}(\mathcal{H}) = \mathcal{A}$. Thus, $\mathcal{A} = \mathrm{Pc}(\mathcal{V}) = \mathrm{Pc}(\mathcal{H})$ is irreducible higher rank affine.

The proof in the case that $\mathcal{V}^-$ is the long chain transverse to $\mathfrak{r}_{V_{i_0}}(H_{j_0})$ is exactly the same. If instead we use one of the chains $\mathcal{H}^+$ or $\mathcal{H}^-$ the only change is to swap the roles of $\mathcal{V}$ and $\mathcal{H}$ and take $\mu' := V_{i_0}$ in the application of Theorem 2.18. □

Corollary 3.21. *Let $(\mathcal{V}, \mathcal{H})$ be a grid in Σ such that $2\ell - 1 \leq |\mathcal{V}| < \infty$ and $2\ell \leq |\mathcal{H}| < \infty$. If $W(\mathcal{V}) \ncong \mathcal{D}_\infty$ then $\mathrm{Pc}(\mathring{\mathcal{H}}) \leq \mathrm{Pc}(\mathcal{V})^\perp$.*

Proof. If $\mathrm{Pc}(\mathcal{V})$ is affine then $W(\mathcal{V})$ is dihedral, contrary to hypothesis, so Proposition 3.20 forces $W(\mathring{\mathcal{H}})$ to centralize $W(\mathcal{V})$. Apply Lemma 3.15 to $W(\mathcal{V})$ and $W(\mathring{\mathcal{H}})$, both of which are nonspherical since $|\mathcal{V}| \geq 2\ell - 1 > 2$ and $|\mathring{\mathcal{H}}| \geq 2$, and irreducible by Lemma 2.11. Again, since $\mathrm{Pc}(\mathcal{V})$ cannot be affine, the remaining alternative is $\mathrm{Pc}(\mathring{\mathcal{H}}) \leq \mathrm{Pc}(\mathcal{V})^\perp$. □

Proof of Proposition 3.14. By Lemma 2.11, if $\mathcal{V}$ is infinite we can pass to a finite subchain of length at least $2\ell - 1$ and long enough that its parabolic closure is all of $\mathrm{Pc}(\mathcal{V})$. Thus, we may assume $\mathcal{V}$ is finite.

If $\mathcal{H}$ is infinite then every wall H in its interior is in the interior of some finite subchain of length at least 2ℓ and long enough that its parabolic closure is all of $\mathrm{Pc}(\mathcal{H})$. Proving the proposition for such finite horizontal chains implies either $\mathrm{Pc}(\mathcal{H}) = \mathrm{Pc}(\mathcal{V})$ is higher rank irreducible affine, and we are done, or not, in which case

$\mathfrak{r}_H \in \mathrm{Pc}(\mathcal{V})^\perp$. Since the latter is true for every $H \in \mathring{\mathcal{H}}$, we have $\mathrm{Pc}(\mathring{\mathcal{H}}) \leq \mathrm{Pc}(\mathcal{V})^\perp$. Thus, it suffices to prove the proposition for finite chains $\mathcal{V}$ and $\mathcal{H}$.

The cardinality lower bounds give that $\mathrm{Pc}(\mathcal{V})$ and $\mathrm{Pc}(\mathring{\mathcal{H}})$ are irreducible and nonspherical, by Lemma 2.11. If $W(\mathcal{V}) \ncong \mathcal{D}_\infty$ then Corollary 3.21 says $\mathrm{Pc}(\mathring{\mathcal{H}}) \leq \mathrm{Pc}(\mathcal{V})^\perp$. If $W(\mathcal{V}) \cong \mathcal{D}_\infty$ then Lemma 3.17 says either $\mathrm{Pc}(\mathcal{H})$ is contained in $\mathrm{Pc}(\mathcal{V})^\perp$ or $\mathrm{Pc}(\mathcal{V}) = \mathrm{Pc}(\mathcal{H})$ is irreducible higher rank affine. Thus, either $\mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ is a product of two infinite factors, or the first factor is irreducible higher rank affine, so in either case $\mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ is wide. □

3.2.2. *Avoiding wide parabolic subcomplexes.* Recall we defined constants $\mathfrak{R}$ and ℓ and functions $\rho_0(\lambda, \epsilon)$ and $\mathfrak{L}(\lambda, \epsilon)$ in Section 2.2.3.

Definition 3.22. A path $\gamma\colon I \to \Sigma$ has *connected wall preimages* if for every wall M the set $\gamma^{-1}(\gamma(I) \cap M)$ is a connected subset of I (possibly empty).

Theorem 3.23. *For all λ and ϵ, we have the implications (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) $\implies$ (v) for all (λ, ϵ)–quasigeodesics $\gamma\colon I \to \Sigma$. If γ is also continuous with connected wall preimages then (v) $\implies$ (i). Moreover, there are explicit bounds between the constraints of each item that depend on λ and ϵ, but are independent of γ.*

- *(i) For all L there exists a wide parabolic subcomplex Υ and compact subinterval $I' \subset I$ of length L such that $\gamma(I') \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$.*
- *(ii) There exists R such that for all L there exists a wide parabolic subcomplex Υ and compact subinterval $I' \subset I$ of length L such that $\gamma(I') \subset \bar{\mathcal{N}}_R(\Upsilon)$.*
- *(iii) For all $0 < \rho$ there exists L_0 such that for all $L_1 \geq L_0$ there exists a compact subinterval $I' \subset I$ of length at least L_1 such that every chain $\mathcal{V}$ that separates the endpoints of $\gamma(I')$ and has $|\mathcal{V}|/|I'| \geq \rho$ contains a pair of consecutive walls mutually transverse to an infinite chain.*
- *(iv) For all $0 < \rho$ there exists L_0 such that for all $L_1 \geq L_0$ there exists a compact subinterval $I' \subset I$ of length at least L_1 such that every chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ contains a pair of consecutive walls mutually transverse to a chain of length 2ℓ.*
- *(v) For all L' there exists a grid $(\mathcal{V}, \mathcal{H})$ with chain lengths $|\mathcal{V}| \geq L'$ and $|\mathcal{H}| = 2\ell$, and such that there is a compact subinterval $I' \subset I$ such that $\mathcal{V}$ separates the endpoints of $\gamma(I')$.*

The remainder of the subsection proves the theorem in steps. The implication (i) $\implies$ (ii) follows by choosing $R = \mathfrak{R}$. Lemma 3.24 gives (ii) $\implies$ (iii). The implication (iii) $\implies$ (iv) follows by passing to a subchain. Lemma 3.25 gives (iv) $\implies$ (v). The proof for (v) $\implies$ (i) is broken into three pieces: Lemma 3.26 is the inductive step for Proposition 3.27, which is then applied in Lemma 3.28.

We will see in the proof of Theorem 3.30 that Theorem 3.23 characterizes when γ fails to be Morse. Notice that the items of Theorem 3.23 are impossible to satisfy if I is bounded, because each of them asserts the existence of arbitrarily long subintervals with some property. This is the expected behavior; bounded quasigeodesic segments are always Morse.

Lemma 3.24. *Suppose $\gamma\colon I \to \Sigma$ is a (λ, ϵ)–quasigeodesic such that there exists R such that for all L there exists a wide parabolic subcomplex Υ and a compact subinterval $I' \subset I$ of length L such that $\gamma(I') \subset \bar{\mathcal{N}}_R(\Upsilon)$. Then for all $0 < \rho$ there exists L_0 such that for all $L_1 \geq L_0$ there exists a compact subinterval $I' \subset I$ of length at least L_1 such that every chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ contains a pair of consecutive walls mutually transverse to a biinfinite chain.*

Proof. Fix R as in the hypothesis and take any $0 < \rho$. By Lemma 2.22 there exists $L_0 \geq \mathfrak{L}(\lambda, \epsilon)$ such that for all $x, y \in \Sigma$:

$$d_2(x, y) \leq R \implies d_c(x, y) \leq \rho L_0/4 - 2$$

By hypothesis, given $L_1 \geq L_0$ there exists a wide parabolic subcomplex Υ and compact $I' \subset I$ of length at least L_1 such that $\gamma(I') \subset \bar{\mathcal{N}}_R(\Upsilon)$. Suppose that $\mathcal{V}$ is a chain separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$.

Let $I' = [z_0, z_1]$. Apply the 4–gon trick, Lemma 2.23, to the four points $\gamma(z_0)$, $\pi_\Upsilon(\gamma(z_0))$, $\pi_\Upsilon(\gamma(z_1))$, and $\gamma(z_1)$ to find a convex subchain $\mathcal{V}'$ of $\mathcal{V}$ separating $\pi_\Upsilon(\gamma(z_0))$ and $\pi_\Upsilon(\gamma(z_1))$, with:

$$|\mathcal{V}| - |\mathcal{V}'| \leq d_c(\gamma(z_0), \pi_\Upsilon(\gamma(z_0))) + d_c(\gamma(z_1), \pi_\Upsilon(\gamma(z_1))) + 2 \leq 2(\rho L_0/4 - 2) + 2$$

But $|\mathcal{V}| \geq \rho L_1$, so $|\mathcal{V}'| \geq |\mathcal{V}|/2 + 2$. In particular, there is at least one consecutive pair V,V' of walls of $\mathcal{V}$ that separate $\pi_\Upsilon(\gamma(z_0))$ and $\pi_\Upsilon(\gamma(z_1))$, so cut Υ.

If Υ is a product of two infinite factors then up to translation by W we may assume $\Upsilon = \Sigma_{T \sqcup T'}$ where T and T' are subsets of S with no edges connecting them in the Coxeter diagram. A wall M cutting Υ has reflection $\mathfrak{r}_M \in W_T \times W_{T'}$. Two walls M and M' cutting Υ give a reflection subgroup $\langle \mathfrak{r}_M, \mathfrak{r}_{M'} \rangle$ that is finite if and only if $M \pitchfork M'$. Lemma 2.12 says $\langle \mathfrak{r}_M, \mathfrak{r}_{M'} \rangle$ splits as a direct product of reflection subgroups of the two factors. If $\mathfrak{r}_M$ and $\mathfrak{r}_{M'}$ live in opposite factors then $\langle \mathfrak{r}_M, \mathfrak{r}_{M'} \rangle \cong (\mathbb{Z}/2\mathbb{Z})^2$. If they live in the same factor, say W_T, then $\langle \mathfrak{r}_M, \mathfrak{r}_{M'} \rangle$ is a subgroup of W_T, so M and M' cross in Σ if and only if they cross in Σ_T. Thus, walls from $\mathcal{V}$ that cut Υ, being disjoint in Σ, either all have reflections in W_T or all have reflections in $W_{T'}$. Without loss of generality, assume the former. By hypothesis $W_{T'}$ is infinite, so there exist walls H and H' of Σ such that $\langle \mathfrak{r}_H, \mathfrak{r}_{H'} \rangle \cong \mathcal{D}_\infty \leq W_{T'}$. Then the $\langle \mathfrak{r}_H, \mathfrak{r}_{H'} \rangle$ orbits of H and H' give a chain in Σ consisting of walls with reflections in $W_{T'}$. These are transverse to $\{V, V'\}$, since every reflection in W_T commutes with every reflection in $W_{T'}$.

The other option is that up to W–translation $\Upsilon = \Sigma_{T \sqcup T'}$ where T is irreducible higher rank affine and T' is spherical. Walls of $\mathcal{V}$ are disjoint, so those that cut Υ have reflections in W_T, since in a spherical parabolic subcomplex all walls cross. Since W_T is affine, Σ_T is a polyhedral tiling of $\mathbb{E}^n$ for some $n > 1$, and walls of Σ_T are affine codimension–1 hyperplanes that come in finitely many parallel families, with every wall from one family crossing every wall from every other family. Since the walls of $\mathcal{V}$ are disjoint in Σ, their intersections with Σ_T all belong to the same parallel family. Any one of the other parallel families gives a chain transverse to $\Sigma_T \cap \mathcal{V}$, so taking the corresponding walls of Σ gives an infinite chain transverse to $\{V, V'\}$. □

Lemma 3.25. *Suppose $\gamma\colon I \to \Sigma$ is a (λ, ϵ)–quasigeodesic such that for all $0 < \rho$ there exists L_0 such that for all $L_1 \geq L_0$ there exists a compact subinterval $I' \subset I$ of length at least L_1 such that every chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ contains a pair of consecutive walls mutually transverse to a chain of length 2ℓ. Then for all L' there exists a grid $(\mathcal{V}, \mathcal{H})$ with $\mathcal{V}$ separating the endpoints of a compact subsegment of γ, $|\mathcal{V}| \geq L'$, and $|\mathcal{H}| = 2\ell$.*

Proof. Assume $L' \geq 2$, and set $\rho := \rho_0(\lambda, \epsilon)/(L' - 1)$. Take $L_1 := \max\{\mathfrak{L}(\lambda, \epsilon), L_0(\rho)\}$. The definitions of $\mathfrak{L}(\lambda, \epsilon)$ and $\rho_0(\lambda, \epsilon)$ imply that for any compact subinterval $I' \subset I$ of length at least L_1, there does exist a chain $\mathcal{V} = \{V_i\}_{i_0 \leq i \leq i_1}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho_0(\lambda, \epsilon)$. Consider the subchain $\{V_{j_k}\}_{0 \leq k \leq (i_1 - i_0)/(L'-1)}$ where $j_k = i_0 + (L' - 1)k$, which contains at least $\rho|I'|$–many walls, so, by hypothesis, contains a pair of consecutive walls V and V', transverse to a chain $\mathcal{H}$ of length 2ℓ. Let $\mathcal{V}'$ be the subchain of $\mathcal{V}$ consisting of all walls from V to V'. By construction, $|\mathcal{V}'| = L'$, and the two extremes are transverse to $\mathcal{H}$, so $\mathcal{V}' \pitchfork \mathcal{H}$. □

In the following, when $\gamma\colon I \to \Sigma$ is a compact path with connected wall preimages and $\mathcal{K} = \{K_i\}_{i_0 \leq i \leq i_1}$ is a chain separating the endpoints of $\gamma(I)$, we assume that the parameterization of γ and the chain order are compatible. The *subsegment of γ between the extremes of $\mathcal{K}$* means $\gamma(J)$ for $J := [\max \gamma^{-1}(\gamma(I) \cap K_{i_0}), \min \gamma^{-1}(\gamma(I) \cap K_{i_1})]$. Note that all but the extreme walls of $\mathcal{K}$ separate the endpoints of $\gamma(J)$.

Lemma 3.26. *Let $\gamma\colon I \to \Sigma$ be a compact path with connected wall preimages. Let $\mathcal{V}$ be a chain that separates the endpoints of $\gamma(I)$ and has $|\mathcal{V}| \geq 2\ell$. Let Υ be the unique parabolic subcomplex with stabilizer $\mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ (recall Corollary 2.17). Suppose there is a $t \in I$ such that $d_c(\gamma(t), \Upsilon) \geq 4\ell - 1$. Define $\mathcal{V}^-$ to be the set of walls $V \in \mathcal{V}$ for which $\max \gamma^{-1}(\gamma(I) \cap V) < t$, and define $\mathcal{V}^+$ to be the set of walls $V \in \mathcal{V}$ for which $\min \gamma^{-1}(\gamma(I) \cap V) > t$. Then $\mathcal{V}^-$ and $\mathcal{V}^+$ are disjoint convex subchains of $\mathcal{V}$, at least one of which is transverse to a chain $\mathcal{H}$ with $|\mathcal{H}| \geq 2\ell$ and $\mathfrak{r}_H \notin \mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ for all $H \in \mathcal{H}$.*

Proof. By hypothesis, there is a chain $\mathcal{K}$ with $|\mathcal{K}| \geq 4\ell - 1$ separating $x := \gamma(t)$ from Υ. Every wall $K \in \mathcal{K}$ is disjoint from $\{x\} \cup \Upsilon$. Every reflection in $\mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ cuts Υ, so $\mathfrak{r}_K \notin \mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ for all $K \in \mathcal{K}$.

If $I = [a, b]$, let $\gamma^- = \gamma([a, t])$, and let $\gamma^+ = \gamma([t, b])$.

Since γ has connected wall preimages and $\mathcal{V}$ is a chain separating the endpoints of γ, V comes before V' in $\mathcal{V}$ if and only if $\gamma^{-1}(\gamma(I) \cap V)$ is a nonempty closed interval that occurs in I before the nonempty closed interval $\gamma^{-1}(\gamma(I) \cap V')$. It follows that $\mathcal{V}^-$ is an initial subchain of $\mathcal{V}$, $\mathcal{V}^+$ is a terminal subchain of $\mathcal{V}$, they are disjoint, and they account for all but at most one wall of $\mathcal{V}$.

For every $V \in \mathcal{V}^-$ there is a path δ_V from Υ to x that travels from a point in $V \cap \Upsilon$ through V to $V \cap \gamma \subset \gamma^-$, and then along γ^- to x. The same is true for $V \in \mathcal{V}^+$, with γ^+ replacing γ^-. Since $K \in \mathcal{K}$

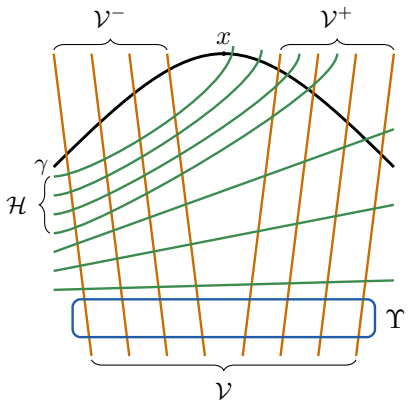


FIGURE 4. Inductive step: A point $x \in \gamma$ between the extremes of $\mathcal{V}$ and far from Υ gives a splitting of $\mathcal{V}$ into subchains $\mathcal{V}^+$ and $\mathcal{V}^-$ such that one of them is part of a grid with a chain $\mathcal{H}$ consisting of "most" of the walls separating x from Υ. Only at most $2\ell - 1$ of the walls separating x from Υ can be transverse to all of $\mathcal{V}$ without cutting Υ.

separates x from Υ, it cuts every such path. If K does not meet γ^- then for all $V \in \mathcal{V}^-$ it must be that K cuts the part of δ_V contained in V, so $K \pitchfork \mathcal{V}^-$. Similarly, $K \cap \gamma^+ = \emptyset$ implies $K \pitchfork \mathcal{V}^+$. At least one of these is true, since the fact that $\gamma^{-1}(K \cap \gamma(I))$ is connected and does not contain t implies that $\gamma^{-1}(K \cap \gamma(I))$ does not contain points from both $[a,t]$ and $[t,b]$. Moreover, suppose some wall $K \in \mathcal{K}$ meets γ^+. If there is a wall $K' \in \mathcal{K}$ separating K from x then it must cut the subsegment of γ^+ between x and $K \cap \gamma^+$, so K' meets γ^+ and does not meet γ^-. Conversely, suppose there is a wall $K' \in \mathcal{K}$ that is separated from x by K. Since K meets γ^+ it does not meet γ^-, so K' also does not meet γ^-, else we would have a path from K' to x through γ^- avoiding K. Thus, if $\mathcal{K}$ is ordered from Υ toward x, then $\mathcal{K}$ splits into an initial subchain $\mathcal{K}'$ consisting of walls disjoint from γ and a terminal subchain $\mathcal{K}''$ consisting of walls that either all meet γ^+ or all meet γ^-.

Suppose $|\mathcal{K}'| \geq 2\ell$. Since $\mathcal{K}'$ is disjoint from γ it is transverse to $\mathcal{V}$. Apply Proposition 3.14 to $(\mathcal{V}, \mathcal{K}')$. Then for $K \in \mathring{\mathcal{K}}'$ we have $\mathfrak{r}_K \in \mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$, which is a contradiction. Thus, $|\mathcal{K}'| \leq 2\ell - 1$, which implies $|\mathcal{K}''| \geq 2\ell$. Take $\mathcal{H} := \mathcal{K}''$. If the walls of $\mathcal{K}''$ meet γ^+ then $\mathcal{H} \pitchfork \mathcal{V}^-$, and if the walls of $\mathcal{K}''$ meet γ^- then $\mathcal{H} \pitchfork \mathcal{V}^+$. □

Proposition 3.27. *Let $C := 3^{|S|-3}$. For every compact path $\gamma\colon I \to \Sigma$ with connected wall preimages, for every grid $(\mathcal{V}, \mathcal{H})$ with $\mathcal{V}$ separating the endpoints of γ and chain lengths $|\mathcal{V}| \geq C(2\ell - 1)$ and $|\mathcal{H}| \geq 2\ell$, there exists a wide parabolic subcomplex Υ and a convex subchain $\mathcal{V}'$ of $\mathcal{V}$ such that $|\mathcal{V}'| \geq |\mathcal{V}|/C$ and the subsegment of γ between the extremes of $\mathcal{V}'$ is contained in $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$.*

Proof. Suppose there is a grid $(\mathcal{V}, \mathcal{H})$ satisfying the hypotheses. Note that if such a grid exists then by Proposition 3.14 either both $\mathrm{Pc}(\mathcal{V})$ and $\mathrm{Pc}(\mathcal{V})^\perp$ are nonspherical, in which case $|S| \geq 4$, or $\mathrm{Pc}(\mathcal{V})$ is irreducible higher rank affine, in which case $|S| \geq 3$ with equality only if $W = \mathrm{Pc}(\mathcal{V})$ is a Euclidean triangle group. If W is a Euclidean triangle group we are finished by taking $\Upsilon = \Sigma$. Thus, we may suppose $|S| > 3$.

The proof is by induction on rank: we guess a wide product parabolic subcomplex and subsegment of γ, and if the lemma is not satisfied we change the guesses in such a way that the rank of the first factor of the parabolic decreases.

Let $\mathcal{V}_0 := \mathcal{V} = \{V_i\}_{i_0 \leq i \leq i_1}$ and $\mathcal{H}_0 := \mathcal{H}$. Let $j_0 := i_0$, $j_1 := i_0 + \lfloor \frac{i_1 - i_0}{3} \rfloor$, $j_2 := j_1 + \lfloor \frac{i_1 - i_0}{3} \rfloor$, and $j_3 := i_1$. Let $\mathcal{V}_{01} := \{V_j\}_{j_0 \leq j \leq j_1}$, $\mathcal{V}_{02} := \{V_j\}_{j_1 \leq j \leq j_2}$, and $\mathcal{V}_{03} := \{V_j\}_{j_2 \leq j \leq j_3}$; each of these is a convex subchain of $\mathcal{V}_0$ containing at least $\frac{|\mathcal{V}_0|}{3}$-many walls. Define Υ_0 to be the unique parabolic subcomplex with stabilizer $\mathrm{Pc}(\mathcal{V}_0) \times \mathrm{Pc}(\mathcal{V}_0)^\perp$, which is wide by Proposition 3.14 applied to $(\mathcal{V}, \mathcal{H})$. Consider the subsegment γ' of γ between the extremes of $\mathcal{V}_{02}$. If $\gamma' \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon_0)$ then we are done, since $|\mathcal{V}_{02}| \geq |\mathcal{V}|/3 \geq |\mathcal{V}|/C$. Otherwise, let $x = \gamma'(t)$ be a point with t in the interior of the domain of γ' and $d_2(x, \Upsilon_0) > \mathfrak{R}$. Define $\mathcal{V}_0^-$ to be the convex subchain of $\mathcal{V}$ consisting of walls V with $\max \gamma^{-1}(\gamma(I) \cap V) < t$, and define $\mathcal{V}_0^+$ to be the convex subchain of $\mathcal{V}$ consisting of walls V with $\min \gamma^{-1}(\gamma(I) \cap V) > t$.

Apply Lemma 3.26 to produce a chain $\mathcal{H}_1'$ with $|\mathcal{H}_1'| \geq 2\ell$ that is transverse to either $\mathcal{V}_0^-$ or $\mathcal{V}_0^+$. Let $\mathcal{V}_1$ be whichever of $\mathcal{V}_0^-$ or $\mathcal{V}_0^+$ is transverse to $\mathcal{H}_1'$. Lemma 3.26 also says that reflections in walls of $\mathcal{H}_1'$ are not in $\mathrm{Pc}(\mathcal{V}_0) \times \mathrm{Pc}(\mathcal{V}_0)^\perp$. Observe that $\mathcal{V}_{01} \subset \mathcal{V}_0^-$ and $\mathcal{V}_{03} \subset \mathcal{V}_0^+$, so in either case $|\mathcal{V}_1| \geq |\mathcal{V}_0|/3 \geq 3^{|S|-4}(2\ell - 1)$.

If $W(\mathcal{V}_0) \cong \mathcal{D}_\infty$ we get a contradiction, because $\mathcal{V}_1$ contains more than 8 walls of $\mathcal{V}_0$, so Lemma 2.19 upgrades $\mathcal{H}_1' \pitchfork \mathcal{V}_1$ to $\mathcal{H}_1' \pitchfork \mathcal{V}_0$, but then Lemma 3.17 says $\mathrm{Pc}(\mathcal{H}_1')$ is either contained in $\mathrm{Pc}(\mathcal{V}_0)$ or $\mathrm{Pc}(\mathcal{V}_0)^\perp$. Thus, if $W(\mathcal{V}_0) \cong \mathcal{D}_\infty$ then we must have had $\gamma' \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon_0)$, and we are done.

Otherwise, Proposition 3.14 says $\mathrm{Pc}(\mathcal{V}_1) \times \mathrm{Pc}(\mathcal{V}_1)^\perp$ is wide and contains reflections through all walls of $\mathcal{H}_1 := \mathring{\mathcal{H}}_1'$. Moreover, if $W(\mathcal{V}_1) \ncong \mathcal{D}_\infty$ then $\mathrm{Pc}(\mathcal{V}_1)$ is not higher rank irreducible affine, so $\mathrm{Pc}(\mathcal{H}_1) \leq \mathrm{Pc}(\mathcal{V}_1)^\perp$. Now, $\mathcal{V}_1 \subset \mathcal{V}_0$ implies $\mathrm{Pc}(\mathcal{V}_1) \leq \mathrm{Pc}(\mathcal{V}_0)$, which implies $\mathrm{Pc}(\mathcal{V}_0)^\perp \leq \mathrm{Pc}(\mathcal{V}_1)^\perp$. In the $W(\mathcal{V}_1) \ncong \mathcal{D}_\infty$ case these cannot be equalities, because in that case we know that for all $H \in \mathcal{H}_1$ we have $\mathfrak{r}_H \in \mathrm{Pc}(\mathcal{V}_1)^\perp \smallsetminus \mathrm{Pc}(\mathcal{V}_0)^\perp$. Thus $W(\mathcal{V}_1) \ncong \mathcal{D}_\infty \implies \mathrm{Pc}(\mathcal{V}_1) \lneq \mathrm{Pc}(\mathcal{V}_0)$.

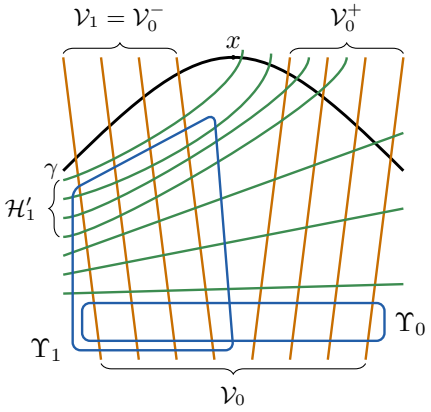


FIGURE 5. Using the induction step to produce a new candidate wide parabolic subcomplex Υ_1 corresponding to $\mathcal{V}_1 \subset \mathcal{V}_0$.

Define Υ_1 to be the unique parabolic subcomplex with stabilizer $\mathrm{Pc}(\mathcal{V}_1) \times \mathrm{Pc}(\mathcal{V}_1)^\perp$, and repeat the construction. See Figure 5. Either the induction will stop with the segment of γ corresponding to the middle third of $\mathcal{V}_1$ contained in $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon_1)$, which is guaranteed to be the case if $W(\mathcal{V}_1) \cong \mathcal{D}_\infty$, or $\mathrm{Pc}(\mathcal{V}_1) \lneq \mathrm{Pc}(\mathcal{V}_0)$, in which case we produce a further $\mathcal{V}_2$, $\mathcal{H}_2$, Υ_2, and repeat.

The ranks of parabolics in a strictly decreasing sequence are strictly decreasing, so this iteration terminates in at most $|S|$ steps. In fact, it terminates in at most $|S| - 4$ steps, since either at some point in the procedure we reach an affine $\mathrm{Pc}(\mathcal{V}_i)$, or at each step we have both $\mathrm{Pc}(\mathcal{V}_i)$ and $\mathrm{Pc}(\mathcal{V}_i)^\perp$ nonspherical. In the latter case, $\mathrm{Pc}(\mathcal{V}_0)^\perp$ has rank at least 2, which means $\mathrm{Pc}(\mathcal{V}_0)$ has rank at most $|S| - 2$, so the rank of $\mathrm{Pc}(\mathcal{V}_i)$ is at least 2 and at most $|S| - 2 - i$, whence $i \leq |S| - 4$. Thus, in $0 \leq i \leq |S| - 4$ steps we produce a wide parabolic subcomplex Υ_i such that $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon_i)$ contains the subsegment of γ between the extremes of $\mathcal{V}_{i2}$, where:

$$|\mathcal{V}_{i2}| \geq \frac{|\mathcal{V}_i|}{3} \geq \frac{|\mathcal{V}|}{3^{i+1}} \geq \frac{|\mathcal{V}|}{3^{|S|-3}} = \frac{|\mathcal{V}|}{C}$$ □

Lemma 3.28. *Suppose $\gamma\colon I \to \Sigma$ is a (λ, ϵ)–quasigeodesic path with connected wall preimages, such that for all L' there exists a grid $(\mathcal{V}, \mathcal{H})$ with $\mathcal{V}$ separating the endpoints of a compact subsegment of γ and chain lengths $|\mathcal{V}| \geq L'$ and $|\mathcal{H}| = 2\ell$. Then for all L there exists a wide parabolic subcomplex Υ and subinterval $I' \subset I$ of length at least L such that $\gamma(I') \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$.*

Proof. Fix arbitrary L. Let C be the constant of Proposition 3.27. By Lemma 2.22, we can choose L' sufficiently large that for all $x, y \in \Sigma$:

$$d_c(x, y) \geq L'/C - 2 \implies d_2(x, y) \geq \lambda L + \epsilon$$

Set $L'' := \max\{C(2\ell - 1), L'\}$. Take a grid $(\mathcal{V}, \mathcal{H})$ as hypothesized, with $|\mathcal{V}| \geq L''$ and pass to a compact subsegment of γ whose endpoints are separated by $\mathcal{V}$. Since $|\mathcal{V}| \geq C(2\ell - 1)$, we can apply Proposition 3.27 to $(\mathcal{V}, \mathcal{H})$ to get a convex subchain $\mathcal{V}'$ of $\mathcal{V}$ and a wide parabolic subcomplex Υ such that the subsegment $\gamma(I')$ of γ between the extremes of $\mathcal{V}'$ is contained in $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$. All but the extreme walls of $\mathcal{V}'$ separate the

endpoints of $\gamma(I')$. The number of such walls is at least $|\mathcal{V}|/C - 2 \geq L'/C - 2$, so the d_2–distance between the endpoints of $\gamma(I')$ is at least $\lambda L + \epsilon$. Since γ is a (λ, ϵ)–quasigeodesic, $|I'| \geq L$. □

3.3. **The characterization theorems.** We can now characterize certain Morse quasigeodesics, specifically, Morse quasigeodesic paths with connected wall preimages. This family includes both the d_1– and d_2–geodesics: d_1–geodesics are uniformly d_2–quasigeodesic, by Lemma 2.22, and every wall preimage is empty or a single point. For a d_2–geodesic, connected wall preimages follows from convexity of walls.

Definition 3.29. A chain is *fine* if for every pair of consecutive walls V and V' in the chain, every chain $\mathcal{H}$ transverse to both V and V' has $|\mathcal{H}| < 2\ell$, where ℓ is the constant defined in Section 2.2.3.

Theorem 3.30 (Characterization of the Morse property)**.** *Let $\Sigma = \Sigma(W, S)$ be the Davis complex of a finite rank Coxeter system. There are constants $\mathfrak{R}$ and ℓ depending only on (W, S), such that for all λ and ϵ, we have implications (1) $\implies$ (2) $\implies$ (3) $\implies$ (4) $\implies$ (5) for all (λ, ϵ)–quasigeodesic paths $\gamma\colon I \to \Sigma$. If γ additionally has connected wall preimages then (5) $\implies$ (1). Moreover, there are effective bounds between the constraints of each implication that depend on λ and ϵ but are independent of γ.*

1. *There is an upper bound on the lengths of chains $\mathcal{V}$ that separate the endpoints of a compact subsegment of γ and for which there exists a grid $(\mathcal{V}, \mathcal{H})$ with $|\mathcal{H}| = 2\ell$.*
2. *There exists $0 < \rho$ such that there exists L such that for every compact subinterval $I' \subset I$ of length at least L there exists a fine chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$.*
3. *There exist C and $0 < \rho$ such that there exists L such that for every compact subinterval $I' \subset I$ of length at least L there exists a chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with $|\mathcal{V}|/|I'| \geq \rho$ such that for every pair V, V' of consecutive walls of $\mathcal{V}$, every chain $\mathcal{H} \pitchfork \{V, V'\}$ has $|\mathcal{H}| \leq C$.*
4. *There exists μ such that γ is μ–Morse.*
5. *There is an upper bound on the lengths of subintervals $I' \subset I$ for which there exists a wide parabolic subcomplex Υ with $\gamma(I') \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$.*

Proof. Recall that Theorem 3.7 characterized the Morse property in terms of equivalent items (a)-(e), while Theorem 3.23 gave items (i)-(v) describing the interaction of the quasigeodesic with wide parabolics. Compared to Items (1)-(5), we have: (1)= ¬(v), (2)= ¬(iv), (3)=(e), (4)=(b), and (5)= ¬(i).

Logically, the implication (iv) $\implies$ (v) in Theorem 3.23 gives (1) $\implies$ (2), but we also need to demonstrate effective quantitative bounds. We argue as in the proof of (iv) $\implies$ (v), which was Lemma 3.25: Suppose that (1) gives a positive integral upper bound B, and define $\rho := \rho_0(\lambda, \epsilon)/B > 0$. For every parameter subinterval I' of length at least $\mathfrak{L}(\lambda, \epsilon)$, there exists a chain separating the endpoints of $\gamma(I')$ of density at least $\rho_0(\lambda, \epsilon)$. Take such a chain $\mathcal{V}$, and pass to the subchain $\mathcal{V}'$ consisting of every B-th wall, which has density at least ρ. If $\mathcal{V}'$ is not fine then it has consecutive walls V and V' transverse to a chain $\mathcal{H}$ of length 2ℓ. Then the convex subchain $\mathcal{V}''$ of $\mathcal{V}$ from V to V' makes a grid with $\mathcal{H}$. This is a contradiction, since $|\mathcal{V}''| = B + 1 > B$, so $\mathcal{V}'$ is fine. Thus, (1) with bound B implies (2) with $\rho = \rho_0(\lambda, \epsilon)/B$ and $L = \mathfrak{L}(\lambda, \epsilon)$.

The implication (2) $\implies$ (3) is immediate from the definition of fine chain, by taking $C = 2\ell - 1$.

Theorem 3.7 (e) $\implies$ (b) gives (3) $\implies$ (4).

Suppose (4). We have (4)=(b) $\implies$ (e)=(3), by Theorem 3.7, so there are C, ρ, and L depending on μ, λ, and ϵ that satisfy (3). With this ρ, the proof of Lemma 3.24 constructs L_0 depending on ρ, $R = \mathfrak{R}$, λ, and ϵ such that for any subinterval I' of length at least L_0 if $\gamma(I')$ is contained in the closed $\mathfrak{R}$–neighborhood of a wide parabolic subcomplex Υ then every chain of density at least ρ and separating the endpoints of $\gamma(I')$ contains a consecutive pair of walls mutually transverse to an infinite chain. Now if $|I'| \geq \max\{L, L_0\}$ and there exists a wide parabolic subcomplex Υ with $\gamma(I') \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$, this contradicts (3). Thus, $\max\{L, L_0\}$ is a bound satisfying (5).

Now suppose γ also has connected wall preimages and suppose (5) holds with bound A. Lemma 3.28 takes a given L and produces L' such that if $(\mathcal{V}, \mathcal{H})$ is a grid with $|\mathcal{V}| \geq L'$ and $|\mathcal{H}| = 2\ell$, with $\mathcal{V}$ separating the endpoints of a compact subsegment of γ, then there is a smaller compact subsegment $\gamma(I')$ contained in the closed $\mathfrak{R}$–neighborhood of a wide parabolic subcomplex, with $|I'| \geq L$. Take $L := A + 1$, and take the corresponding L'. This L exceeds the upper bound of (5), so there is no grid $(\mathcal{V}, \mathcal{H})$ with $|\mathcal{V}| \geq L'$ and $|\mathcal{H}| = 2\ell$ that separates the endpoints of a compact subsegment of γ. Thus, (1) holds with bound L'. □

Theorem 3.30 applies to combinatorial geodesics, but via the identification of $\Sigma^{(1)}$ with the Cayley graph of W with respect to S, each edge of $\Sigma^{(1)}$ is labelled with one of the generators. For combinatorial geodesics we

can give an additional characterization of the Morse property in terms of labels on subsegments. Furthermore, we can tighten the geometric condition of Theorem 3.30 of being contained in a fixed neighborhood of a wide parabolic subcomplex to actually being contained in a wide parabolic subcomplex:

Theorem 3.31 (Characterization of the Morse property for combinatorial geodesics)**.** *Let $\Sigma = \Sigma(W, S)$ be the Davis complex of a finite rank Coxeter system. There are effective quantitative equivalences between the following items, uniformly over all combinatorial geodesics γ in Σ.*

1. *There exists μ such that γ is μ–Morse.*
2. *There is an upper bound on the lengths of subsegments γ' of γ for which there exists a wide parabolic subcomplex Υ with $\gamma' \subset \Upsilon$.*
3. *There is an upper bound on the lengths of subsegments γ' of γ for which there exists $T \subset S$ such that W_T is a wide special subgroup and all edges of γ' have label in T.*

Proof. The implications (1) $\implies$ (2) and (3) $\implies$ (2) are easy, and (2) $\implies$ (3) follows from a similar argument to the second step of the proof of $\neg(1) \implies \neg(3)$. We prove $\neg(1) \implies \neg(3)$. There are two parts to the proof. The first is to go from a combinatorial geodesic having a long subsegment contained in the $\mathfrak{R}$–neighborhood of a wide parabolic subcomplex $\Upsilon = w\Sigma_T$ to having a long subsegment such that every edge e_i is dual to a wall that cuts Υ. The second part is to argue that the edge labels of the e_i are all contained in a common wide subset of S. The complication in this step is that in non-right-angled Coxeter groups the edges dual to a fixed wall might not all have the same label; the labels might only be conjugate generators. This means the subset of S containing the labels of the edges e_i might not be T—it might be some subset of S related to T by a conjugation.

For the first step, we start with a counting argument. Suppose that $\mathcal{M}$ is the set of walls dual to edges of a combinatorial geodesic segment α, and suppose $\mathcal{M}'$ is some subset of those walls. Partition α into a concatenation of subpaths that are either single edges dual to a wall in $\mathcal{M} \smallsetminus \mathcal{M}'$ or a maximal subpath such that every edge is dual to a wall in $\mathcal{M}'$. Suppose that the lengths of the latter are bounded above by N. There are at most $|\mathcal{M}| - |\mathcal{M}'| + 1$ such subsegments, so $|\mathcal{M}| = |\alpha| \leq N(|\mathcal{M}| - |\mathcal{M}'| + 1) + |\mathcal{M}| - |\mathcal{M}'|$. Thus:

$$(\dagger) \qquad \frac{|\mathcal{M}'|}{|\mathcal{M}| - |\mathcal{M}'| + 1} \leq N$$

Suppose $\gamma\colon I \to \Sigma$ is a combinatorial geodesic. Then γ is a (λ, ϵ)–quasigeodesic path with connected wall preimages. Furthermore, γ is injective and parameterized by arc length, and the quasigeodesic constants are uniform over all combinatorial geodesics.

Suppose $\neg(1)$, so γ is not Morse. It follows from Theorem 3.30 that for every L there exists a wide parabolic subcomplex Υ and subinterval $I' \subset I$ of length L such that $\gamma' := \gamma(I')$ is an edge path contained in $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$. Let x and y be the end vertices of γ', and consider the set of all walls $\mathcal{M}$ separating x and y, of which there are exactly L–many, since they are the walls cutting the edges of γ'. Let $\mathcal{M}'$ be the subset of $\mathcal{M}$ consisting of walls that cut Υ. A d_w–version of the 4–gon trick, Lemma 2.23, gives:

$$|\mathcal{M}| - |\mathcal{M}'| \preceq d_w(x, \mathfrak{p}_\Upsilon(x)) + d_w(y, \mathfrak{p}_\Upsilon(y)) \preceq \mathfrak{R}$$

Since the right-hand side is a fixed constant, this gives $|\mathcal{M}'| \overset{+}{\asymp} |\mathcal{M}| = L$, so:

$$(\ddagger) \qquad \frac{|\mathcal{M}'|}{|\mathcal{M}| - |\mathcal{M}'| + 1} \succeq L$$

The coarse comparison constants of (‡) are independent of the choices of L and γ, so, given N we can choose L large enough that (‡) implies $\neg(\dagger)$, which guarantees that γ' contains a subsegment γ'' consisting of at least N–many consecutive edges dual to walls of $\mathcal{M}'$. Thus, γ contains arbitrarily long subsegments $\alpha := \gamma''$ such that there exists a wide parabolic subcomplex Υ such that $\alpha \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$ is an edge path, every edge of which is dual to a wall that cuts Υ.

For the second step, suppose a and b are the first and last vertices of α, respectively. Replace Υ, α, a, and b by their $\mathfrak{p}_\Upsilon(a)^{-1}$–translates, so that $\Upsilon^{(0)} = W_T$ for some wide $T \subset S$ and $\mathfrak{p}_\Upsilon(a) = \mathbb{1}$. Suppose the vertices of α are $a = x_0, x_1, \ldots, x_n = b$ and e_i is the edge from x_{i-1} to x_i. Suppose the label of e_i is $s_i \in S$. Let $U := \{s_1, \ldots, s_n\}$, so that the vertices of α are contained in the parabolic subcomplex $a\Sigma_U$.

Since the wall dual to e_i cuts Υ, its reflection $r_i := x_i s_i x_i^{-1}$ is in W_T.

Now, $x_1 = r_1 x_0 \in W_T x_0$. By induction, $x_j = r_j x_{j-1} = r_j \cdots r_1 x_0 \in W_T x_0$. Plugging this expression for x_j into the definition of r_j gives $r_j = r_j \cdots r_1 x_0 s_j x_0^{-1} r_1 \cdots r_j$. Rearranging gives $s_j = x_0^{-1} r_1 \cdots r_{j-1} r_j r_{j-1} \cdots r_1 x_0 \in x_0^{-1} W_T x_0 = a^{-1} W_T a$.

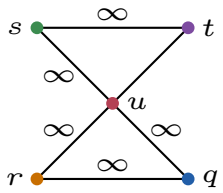
Since $\mathfrak{p}_\Upsilon(a) = \mathbb{1}$, the minimal length element of $W_T a$ is a, so for all $u \in W_T$ we have $|ua| = |u| + |a|$ [52, Lemma 4.3.3]. Since $W_T \ni w_j := a s_j a^{-1}$, we have $w_j a = a s_j$. The left-hand side is an element of $W_T a$, so the minimal length words in S representing this element have length $|w_j| + |a|$. The right-hand side is the same group element, expressed as a word of length $|a| + 1$. Thus $|w_j| = 1$, which means $w_j \in T$. We have shown $U \subset S \cap a^{-1} T a$.

Once $|\alpha| > \Delta$ then U cannot be spherical. Our goal is to show that U is contained in a wide subset of S, establishing $\neg(3)$. This is true, in particular, if U itself is wide, so suppose not. Thus, the canonical product decomposition of W_U is $W_{U_0} \times W_{U_1}$, where U_0 is irreducible nonspherical and not higher rank affine and U_1 is spherical. This implies $U_1 \subset U_0^\perp$, so $W_U \leq W_{U_0} \times W_{U_0^\perp}$. Thus, it suffices to show $U_0^\perp$ is nonspherical.

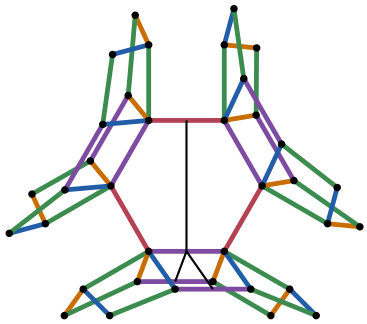
By Proposition 2.13 (1), since $U_0 \subset S$ is nonspherical irreducible, $a U_0 a^{-1} \subset T \subset S$ implies $U_0 = a U_0 a^{-1}$. Let T_0 be the irreducible component of T containing U_0. If T_0 is higher rank affine then $T_0 = U_0$, since irreducible affine parabolics have no proper nonspherical subparabolics. This is impossible, since we already said that U_0 is not higher rank affine. Since T is wide, the alternative is that $T_1 := T \smallsetminus T_0$ is nonspherical. But then $U_0 \subset T_0$ implies $T_1 \subset T_0^\perp \subset U_0^\perp$, so $U_0^\perp$ is nonspherical. □

The next example illustrates that knowing that an edge is dual to a wall that cuts a wide parabolic subcomplex does not imply that that edge belongs to a wide parabolic subcomplex. This explains the case split U spherical/nonspherical at the end of the proof of Theorem 3.31. This phenomenon cannot happen in the right-angled case, since in RACGs distinct generators are never conjugate.

Example 3.32. Consider the Coxeter diagram shown in Figure 6a. Let $T := \{q, r, s, t\}$ and $U := \{u\}$. The subset T is wide; $W_T \cong \mathcal{D}_\infty \times \mathcal{D}_\infty$. The generator u is not contained in any wide subset, but it is conjugate to a generator t that is: $tut = utu \implies ututu = t \in T$. The Davis complex, a portion of which is shown in Figure 6b, is a "tree of flats" where three flats "meet" by containing nonconsecutive edges of a hexagon, a translate of $\Sigma_{\{t,u\}}$. As shown in Figure 6b, a wall dual to a u–edge cuts the hexagon containing it, as well as the opposite t–edge, then branches into the flat containing that t–edge, so it cuts a wide parabolic subcomplex, but it also cuts an edge not contained in any wide parabolic subcomplex.



(A) The Coxeter diagram.



(B) A portion of the Davis complex with part of one wall shown.

FIGURE 6. The Coxeter diagram of Example 3.32 and its Davis complex.

4. Hyperbolic models

In Section 4.1 we use fine chains to define a metric d_f on $X^{(0)}_{\text{NR}}$, which we show is hyperbolic. Since it is defined in terms of walls, it also gives a hyperbolic metric on $\Sigma^{(0)}$.

A hyperbolic metric on $\Sigma^{(0)}$ must somehow eliminate the known sources of non-hyperbolicity, the wide parabolics. In fact, in Lemma 4.8 we observe that it collapses wide parabolics to diameter 1 subsets. This has the side effect, noted in Proposition 4.9, that it collapses the Niblo-Reeves cube complex to within bounded distance of the canonical copy of $\Sigma^{(0)}$ in $X^{(0)}_{\text{NR}}$.

Using Theorem 3.30 to relate Morse geodesics with fine chains along them, we deduce that $(\Sigma^{(0)}, d_f)$ is a *Morse recognizing space*, in Section 4.2.

A common strategy for producing hyperbolic spaces is to cone off known non-hyperbolic regions to make them have bounded diameter. In Section 4.3 we construct such a *coned-off space* $\widehat{\Sigma}$ by coning off the wide parabolic subcomplexes of Σ and show that it is W–equivariantly quasiisometric to $(\Sigma^{(0)}, d_f)$. This gives a more concrete construction of a hyperbolic Morse recognizing space for W.

4.1. **The fine chain metric.** Recall, from Definition 3.29, that a chain $\mathcal{V} := \{V_i\}_{i\in I}$ is *fine* if for all $i \neq j$, or, equivalently, for all consecutive i, j, every chain transverse to $\{V_i, V_j\}$ has length less than 2ℓ.

Definition 4.1. Define $d_f\colon \Sigma^{(0)} \times \Sigma^{(0)} \to \mathbb{R}$ by $d_f(x, x) := 0$ and for $x \neq y$:

$$d_f(x, y) := \max\{|\mathcal{V}| \mid \mathcal{V} \text{ is a fine chain separating } x \text{ and } y\}$$

Make the same definition for $d_f\colon X^{(0)}_{\text{NR}} \times X^{(0)}_{\text{NR}} \to \mathbb{R}$.

In this section we will show that d_f defines a hyperbolic metric on $X^{(0)}_{\text{NR}}$. This is a variation of a construction of Genevois. We further show that combinatorial geodesics are unparameterized quasirulers that can be reparameterized to be rough geodesic quasirulers, and that d_f restricts to a hyperbolic metric on $\Sigma^{(0)}$.

Lemma 4.2. *The fine chain metric d_f is a metric on $\Sigma^{(0)}$ and on $X^{(0)}_{\text{NR}}$. The canonical inclusion $\Sigma^{(0)} \hookrightarrow X^{(0)}_{\text{NR}}$ induces an isometric embedding $(\Sigma^{(0)}, d_f) \hookrightarrow (X^{(0)}_{\text{NR}}, d_f)$ that takes the vertex sequence of a combinatorial geodesic in Σ to the vertex sequence of a combinatorial geodesic in X_{NR}.*

Proof. A single wall is a fine chain, so $x \neq y \in X^{(0)}_{\text{NR}}$ implies $d_f(x, y) \geq 1$.

Vertices of X_{NR} do not lie on walls, so for all x, y, z in $X^{(0)}_{\text{NR}}$, if $\mathcal{V}$ is a chain realizing $d_f(x, z)$ then $\mathcal{V} = \mathcal{V}' \sqcup \mathcal{V}''$ where $\mathcal{V}'$ is an initial subchain separating x from y and $\mathcal{V}''$ is a terminal subchain separating y from z. Both $\mathcal{V}'$ and $\mathcal{V}''$ are fine, so $d_f(x, z) = |\mathcal{V}| = |\mathcal{V}'| + |\mathcal{V}''| \leq d_f(x, y) + d_f(y, z)$.

This shows that d_f is a metric on $X^{(0)}_{\text{NR}}$. The proof for $\Sigma^{(0)}$ is identical.

Recall that a vertex of $\Sigma^{(0)}$ defines a principal ultrafilter on the halfspace system that for each wall picks the complementary halfspace containing that vertex. This gives a canonical inclusion $\iota\colon \Sigma^{(0)} \hookrightarrow X^{(0)}_{\text{NR}}$ with the additional property that a wall separates two vertices of $\Sigma^{(0)}$ if and only if the corresponding wall in X_{NR} separates the $X^{(0)}_{\text{NR}}$ images of those vertices. Thus, for $x, y \in \Sigma^{(0)}$ the walls separating them correspond to the walls separating their images in $X^{(0)}_{\text{NR}}$, and fine chains are preserved by this identification, so $d_f(x, y) = d_f(\iota(x), \iota(y))$.

The vertex sequence $v_0, \ldots, v_n$ of a combinatorial geodesic, in either Σ or X_{NR}, is characterized by the properties that there is a unique wall separating v_i and v_{i+1} for each $0 \leq i < n$ and for every wall M, if for some j we have $v_j \in M^-$ and $v_{j+1} \in M^+$ then for all $i \leq j < k$ we have $v_i \in M^-$ and $v_k \in M^+$. □

Lemma 4.3 (cf [62, Lemma 6.54])**.** *Combinatorial geodesics in X_{NR} are unparameterized τ–quasirulers in $(X^{(0)}_{\text{NR}}, d_f)$, for constant τ depending only on ℓ.*

Proof. Let x, y, and z be vertices on a combinatorial geodesic of X_{NR} such that y occurs between x and z. Let $\mathcal{V} := \{V_1, \ldots, V_m\}$ be a longest fine chain separating x and y and let $\mathcal{H} := \{H_1, \ldots, H_n\}$ be a longest fine chain separating y and z. Assume that indices and halfspaces are chosen so that for all i and j, $y \in V_i^- \subset V_{i+1}^-$ and $y \in H_j^- \subset H_{j+1}^-$.

Suppose for some i_0 and j_0 that $V_{i_0} \cap H_{j_0} = \emptyset$. Since $y \in V_{i_0}^- \cap H_{j_0}^-$, we have $V_{i_0}^+ \cap H_{j_0}^+ = \emptyset$. Thus, for all $i \geq i_0$ and $j \geq j_0$ we have V_i and H_j are disjoint, since $V_i \subset V_{i_0}^+$ and $H_j \subset H_{j_0}^+$. Conversely, if $V_{i_0} \pitchfork H_{j_0}$ then $V_i \pitchfork H_j$ for all $i \leq i_0$ and all $j \leq j_0$.

If $m \geq 3$ and $n \geq 2\ell$ then set $\mathcal{K} := \{V_m, \ldots, V_3, H_{2\ell}, \ldots, H_n\}$. We claim $\mathcal{K}$ is a fine chain. This follows because if $V_2 \pitchfork H_{2\ell}$ then $\{V_1, V_2\} \pitchfork \{H_1, \ldots, H_{2\ell}\}$, contradicting fineness of $\mathcal{V}$, so V_2 separates V_3 from $H_{2\ell}$. Moreover, the only consecutive pair of walls at which $\mathcal{K}$ could fail to be fine is $\{V_3, H_{2\ell}\}$, but walls transverse to $\{V_3, H_{2\ell}\}$ are also transverse to $\{V_2, V_3\}$, so fineness of $\mathcal{V}$ forces fineness of $\mathcal{K}$.

The hypothesis that y is between x and z in X_{NR} implies that any wall separating y from one of x or z in X_{NR} also separates x and z, so $\mathcal{K}$ is a fine chain separating x and z, which gives a bound:

$$d_f(x, z) \geq |\mathcal{K}| = |\mathcal{V}| - 2 + |\mathcal{H}| - (2\ell - 1) = d_f(x, y) + d_f(y, z) - (2\ell + 1)$$

If $\min\{m,n\} < 2\ell$ then take $\mathcal{K}$ to be the longer of $\mathcal{V}$ and $\mathcal{H}$. Then $d_f(x,z) \geq |\mathcal{K}| \geq d_f(x,y)+d_f(y,z)-(2\ell-1)$. Thus, it suffices to take $\tau := 2\ell+1$.

The bounded jump condition is trivial, since combinatorial geodesics are continuous. □

The next result says the vertex sequence of a combinatorial geodesic can be parameterized as a rough geodesic in $(X^{(0)}_{\text{NR}}, d_f)$. The point is that along a combinatorial geodesic the d_f–distance from the initial vertex is nondecreasing, but it can be constant over long subintervals. However, since it is discrete valued, we can reparameterize so that those constant valued intervals are each traversed in time 1.

Lemma 4.4. *Let $\alpha\colon [0,n] \to X_{\text{NR}}$ be a combinatorial geodesic, and let $T := d_f(\alpha(0),\alpha(n))$. There exist numbers $0 = b_0 < b_1 < \cdots < b_n = T$ such that the map $\beta\colon [0,T] \to X^{(0)}_{\text{NR}}$ defined by sending T to $\alpha(n)$ and $t < T$ to $\alpha(i)$ when $b_i \leq t < b_{i+1}$ is a $(\tau+2)$–rough geodesic in $(X^{(0)}_{\text{NR}}, d_f)$ that traverses the vertices of the image of α in the same order as α, where τ is the quasiruler constant of Lemma 4.3.*

Proof. For each $i \in [0,n] \cap \mathbb{Z}$, define:

$$\text{FineProgress}(i) = \text{FP}(i) := d_f(\alpha(0),\alpha(i))$$

Since α is a combinatorial geodesic, for all $i \leq j$ in $[0,n]\cap\mathbb{Z}$, every wall separating $\alpha(0)$ and $\alpha(i)$ also separates $\alpha(0)$ and $\alpha(j)$, and the only wall separating $\alpha(0)$ and $\alpha(i+1)$ that does not also separate $\alpha(0)$ and $\alpha(i)$ is the wall transverse to the edge between $\alpha(i)$ and $\alpha(i+1)$. Thus, $\text{FP}(i+1) - \text{FP}(i) \in \{0,1\}$.

Lemma 4.3 says α is an unparameterized τ–quasiruler. Rearranging the almost degeneracy condition gives that for integers $i \leq j$:

$$\text{FP}(j) - \text{FP}(i) \leq d_f(\alpha(i),\alpha(j)) \leq \text{FP}(j) - \text{FP}(i) + \tau \tag{4}$$

Define $b_0 := 0$. For each integer D in $[1,T]$ consider the indices $\{i \mid \text{FP}(i) = D\}$, which is an integer interval $[j, j+k]$ for some j and k, by monotonicity of FP. For $i \in [j, j+k]$ define $b_i := D + \frac{i-j-k}{k+1}$, so that $b_j = D - 1 + \frac{1}{k+1}$ and $b_{j+k} = D$, with the other values b_i linearly interpolating between these two values. By construction, $i \mapsto b_i$ is strictly increasing and for all i we have:

$$0 \leq \text{FP}(i) - b_i < 1 \tag{5}$$

For integers $i \leq j$ in $[0,n]$, relations (4) and (5) give:

$$b_j - b_i - 1 \leq d_f(\beta(b_i),\beta(b_j)) \leq b_j - b_i + \tau + 1$$

Since $0 < b_{i+1} - b_i \leq 1$, if $x \leq y$ with $b_i \leq x < b_{i+1}$ and $b_j \leq y < b_{j+1}$ then $d_f(\beta(x),\beta(y)) = d_f(\beta(b_i),\beta(b_j))$, by definition of β, and $y - x \in b_j - b_i \pm 1$, so:

$$y - x - 2 \leq b_j - b_i - 1 \leq d_f(\beta(x),\beta(y)) \leq b_j - b_i + \tau + 1 \leq y - x + \tau + 2$$ □

Corollary 4.5. *For every combinatorial geodesic segment α in X_{NR} there is a parameterized $(1, \tau+2, \tau)$–quasiruler in $(X^{(0)}_{\text{NR}}, d_f)$ that has the same vertex sequence as α. The same is true for combinatorial geodesic segments in Σ.*

Proof. In X_{NR}, this is a combination of Lemma 4.3 and Lemma 4.4, since the almost degeneracy of the triangle inequality does not depend on the parameterization.

The same is true in Σ because Lemma 4.2 says the inclusion of $\Sigma^{(0)}$ into $X^{(0)}_{\text{NR}}$ is a d_f–isometric embedding that takes combinatorial geodesics to combinatorial geodesics. □

The following is essentially Genevois's proof that his δ_L metric on a CAT(0) cube complex defines a non-geodesic hyperbolic metric, [62, Proposition 6.53]. The difference is that his metric is defined in terms of two disjoint walls being L–well-separated if every *set of walls* transverse to both of them and containing no facing triple has cardinality at most L, whereas we are saying two disjoint walls make a fine chain if every *chain* transverse to both of them contains at most $2\ell-1$ walls. A pair of $(2\ell-1)$–well-separated walls makes a fine chain, and a two-wall fine chain gives a $(2\ell-1)\cdot\dim X_{\text{NR}}$–well-separated pair, so these two notions are related, but it turns out to be fairly easy to just rerun the hyperbolicity argument using d_f, as opposed to trying to compare d_f to $\delta_{2\ell-1}$ and $\delta_{(2\ell-1)\cdot\dim X_{\text{NR}}}$.

Proposition 4.6 (cf [62, Lemma 2.9 and Proposition 6.53])**.** *$(X^{(0)}_{\text{NR}}, d_f)$ is $(10\ell+3)$–hyperbolic.*

Proof. Let x_0, x_1, x_2, and x_3 be vertices of X_{NR}. Since X_{NR} is a CAT(0) cube complex, its 1–skeleton is a median graph. For each i, let y_i be the median of the triple $(x_{i+1}, x_{i+2}, x_{i+3})$, and let z_i be the median of the triple $(y_{i+1}, y_{i+2}, y_{i+3})$, with subscripts mod 4. Figure 7 shows the idealized version when all of these points are distinct. A key fact from median algebras is that sides of the median cube in Figure 7 have a parallelism

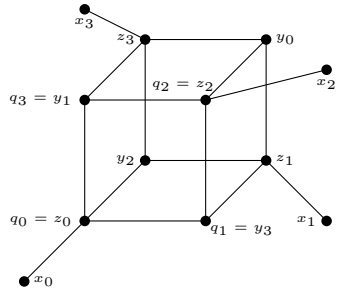


FIGURE 7. Setup for Proposition 4.6 pictured as the free median algebra on $\{x_0, x_1, x_2, x_3\}$.

relation such that a wall that separates the endpoints of one edge in the cube also separates the endpoints of the 3 edges parallel to it.

The intuition for the hyperbolicity proof is to show that in any configuration as in Figure 7 at least two of the dimensions of the cube are small, so the figure is uniformly tree-like.

Formally, there is a four-point δ–hyperbolicity condition [31, Definition III.H.1.20]:

$$(Q(\delta)) \qquad \begin{aligned} d_f(x_0, x_2) + d_f(x_1, x_3) \leq \\ \max\{d_f(x_0, x_1) + d_f(x_2, x_3), d_f(x_0, x_3) + d_f(x_1, x_2)\} + 2\delta \end{aligned}$$

We will establish condition $Q(10\ell + 3)$.

It will be enough in the argument to focus on one face of the cube. We arbitrarily choose the front face in the figure, and define its corners to be $q_0 := z_0$, $q_1 := y_3$, $q_2 := z_2$, and $q_3 := y_1$. The parallelism relation implies $a := d_f(q_0, q_1) = d_f(q_2, q_3)$ and $b := d_f(q_0, q_3) = d_f(q_1, q_2)$. Furthermore, every wall separating q_0 and q_1 crosses every wall separating q_0 and q_3. In particular, if $\mathcal{V}$ is a fine chain realizing $a = d_f(q_0, q_1)$ and $\mathcal{H}$ is a fine chain realizing $b = d_f(q_0, q_3)$ then $\mathcal{V} \pitchfork \mathcal{H}$, so fineness demands:

$$\min\{a, b\} = \min\{|\mathcal{V}|, |\mathcal{H}|\} \leq 1 \quad \text{or} \quad \max\{a, b\} = \max\{|\mathcal{V}|, |\mathcal{H}|\} \leq 2\ell - 1 \tag{6}$$

Let $[p_0, p_1]$ denote the *interval* between p_0 and p_1, which is all vertices that lie on combinatorial geodesics between p_0 and p_1. From the median point definitions we have, for each i:

$$[x_i, q_i] \cup [q_i, q_{i+1}] \cup [q_{i+1}, x_{i+1}] \subset [x_i, x_{i+1}]$$

Applying Lemma 4.3 twice:

$$\begin{aligned} d_f(x_3, x_0) &\geq d_f(x_3, q_3) + d_f(q_3, x_0) - \tau \\ &\geq d_f(x_3, q_3) + d_f(q_3, q_0) + d_f(q_0, x_0) - 2\tau \\ &= d_f(x_0, q_0) + d_f(x_3, q_3) + b - 2\tau \end{aligned}$$

A similar estimate holds for $d_f(x_1, x_2)$. Combining them gives:

$$d_f(x_3, x_0) + d_f(x_1, x_2) \geq 2b - 4\tau + \sum_{i=0}^{3} d_f(x_i, q_i) \tag{7}$$

A similar argument gives:

$$d_f(x_0, x_1) + d_f(x_2, x_3) \geq 2a - 4\tau + \sum_{i=0}^{3} d_f(x_i, q_i) \tag{8}$$

On the other hand, the triangle inequality gives:

$$d_f(x_0, x_1) + d_f(x_2, x_3) \leq 2a + \sum_{i=0}^{3} d_f(x_i, q_i) \tag{9}$$

Suppose, without loss of generality, that:

$$d_f(x_0, x_3) + d_f(x_1, x_2) \leq d_f(x_0, x_1) + d_f(x_2, x_3) \tag{10}$$

Then (7), (9), and (10) combine to give:

$$b - a \leq 2\tau \tag{11}$$

Now, from the triangle inequality:

$$d_f(x_0, x_2) + d_f(x_1, x_3) \leq 2a + 2b + \sum_{i=0}^{3} d_f(x_i, q_i)$$

Combined with (7):

$$d_f(x_0, x_2) + d_f(x_1, x_3) \leq d_f(x_0, x_3) + d_f(x_1, x_2) + 4\tau + 2a \tag{12}$$

Similarly, using (8):

$$d_f(x_0, x_2) + d_f(x_1, x_3) \leq d_f(x_0, x_1) + d_f(x_2, x_3) + 4\tau + 2b \tag{13}$$

Recall from (6) that if $b \geq 2$ then $a \leq 2\ell - 1$, so (11) says $b \leq 2\tau + 2\ell - 1$, whereas if $b < 2$ then $b \leq 2\tau + 2\ell - 1$ is still true, so (13) gives:

$$d_f(x_0, x_2) + d_f(x_1, x_3) \leq d_f(x_0, x_1) + d_f(x_2, x_3) + 8\tau + 4\ell - 2 \tag{14}$$

The assumption (10) says that (14) realizes the maximum appearing in $Q(\delta)$, so (14) establishes $Q(\delta)$ with $\delta := 4\tau + 2\ell - 1$. Using the value $\tau := 2\ell + 1$ in the proof of Lemma 4.3, this gives $\delta = 10\ell + 3$. □

Theorem 4.7. *$(\Sigma^{(0)}, d_f)$ is a quasiruled hyperbolic space.*

Proof. By Lemma 4.2, the canonical inclusion $\Sigma^{(0)} \hookrightarrow X^{(0)}_{\text{NR}}$ is a d_f–isometric embedding. Thus, the 4–point hyperbolicity condition in the proof of Proposition 4.6 also applies to $(\Sigma^{(0)}, d_f)$. Quasiruledness is established by Corollary 4.5. □

Lemma 4.8. *If Υ is a wide parabolic subcomplex of Σ then $\Upsilon^{(0)}$ has diameter 1 in $(\Sigma^{(0)}, d_f)$.*

Proof. Recall from the analysis in the proof of Lemma 3.24 that fine chains in a wide Coxeter group are single walls, since every pair of disjoint walls is mutually transverse to a biinfinte chain. It follows that fine chains that cut a wide parabolic subcomplex of Σ consist of single walls. □

Recall that the W–action on the Niblo-Reeves cube complex X_{NR} is not cocompact if and only if W contains irreducible higher rank affine parabolic subgroups, [40]. It turns out that since the d_f metric collapses such parabolics, it collapses the difference between $\Sigma^{(0)}$ and $X^{(0)}_{\text{NR}}$:

Proposition 4.9. *The vertices of $\Sigma^{(0)}$ are coarsely dense in $(X^{(0)}_{\text{NR}}, d_f)$.*

Proof. We identify $\Sigma^{(0)}$ with its canonical image in $X^{(0)}_{\text{NR}}$. For $x \in X^{(0)}_{\text{NR}}$, choose $v_0 \in \Sigma^{(0)}$ minimizing the combinatorial distance in $X^{(1)}_{\text{NR}}$ from x to $\Sigma^{(0)}$.

We first recall that x lies in the cubical chamber containing v_0, as in [34, Section 7.5]: let $\Psi(v_0)$ be the set of halfspaces of X_{NR} that contain v_0 but not one of its neighbors in $\Sigma^{(0)}$. If $x \notin \psi \in \Psi(v_0)$, then there exists a combinatorial geodesic in X_{NR} from x to v_0 that crosses the wall $\partial\psi$ in its last edge. Let v_1 be the neighbor of v_0 in $\Sigma^{(0)}$ not contained in ψ. Then v_1 sits in the interior of a combinatorial geodesic from x to v_0, so it is a vertex of $\Sigma^{(0)}$ that is closer to v than v_0 is, contradicting the choice of v_0. Thus, x belongs to the cubical chamber of v_0.

Let $\mathcal{F}$ be a longest fine chain separating v_0 from x. The walls of $\mathcal{F}$ are pairwise disjoint and contained in the set of walls $\mathcal{M}(v_0, x)$ separating v_0 from x.

By [34, Proposition 24] there is a constant $K = K(W, S)$ such that for any set of pairwise disjoint walls $\mathcal{M}$ contained in $\mathcal{M}(v_0, x)$ with $|\mathcal{M}| > K$, we have $W(\mathcal{M}) \cong \mathcal{D}_\infty$ and $\text{Pc}(\mathcal{M})$ is irreducible higher rank affine.

Thus, $|\mathcal{F}| > K$ implies $\text{Pc}(\mathcal{F})$ is irreducible higher rank affine, which implies $|\mathcal{F}| = 1$, which is a contradiction, since K is clearly at least 1.

Conclude that $d_f(x, v_0) = |\mathcal{F}| \leq K$. □

4.2. **Recognizing Morse quasigeodesics and parabolics.** In Section 4.2.1 we show that quasigeodesics in (Σ, d_2) are Morse if and only if they are (parameterized) quasigeodesics in $(\Sigma^{(0)}, d_f)$. In Section 4.2.2 we discuss related notions in the literature. In Section 4.2.3 we characterize Morse and stable parabolic subgroups.

4.2.1. *The fine chain metric is Morse recognizing.* For the next theorem, choose a map $\Pi\colon \Sigma \to \Sigma^{(0)}$ sending each point in Σ to a vertex contained in a common closed chamber. Every chamber is a W–translate of the fundamental chamber, which is compact, and every chamber contains one vertex, so Π is the identity map on $\Sigma^{(0)}$, and is coarsely equivalent to the identity map on Σ. For all $x \in \Sigma$, no wall separates x and $\Pi(x)$.

Theorem 4.10. *A quasigeodesic γ is Morse in (Σ, d_2) if and only if $\Pi \circ \gamma$ is quasigeodesic in $(\Sigma^{(0)}, d_f)$. More specifically, since Σ is CAT(0), Morse is equivalent to strong contraction, by Proposition 2.32, and we show:*

1. *For all λ, ϵ, χ there exist λ' and ϵ' such that for every χ–strongly contracting (λ, ϵ)–quasigeodesic $\gamma\colon I \to (\Sigma, d_2)$ the map $\Pi \circ \gamma\colon I \to (\Sigma^{(0)}, d_f)$ is a (λ', ϵ')–quasigeodesic.*
2. *For all λ, ϵ, λ', ϵ' there exists χ such that for every (λ, ϵ)–quasigeodesic $\gamma\colon I \to (\Sigma, d_2)$ such that $\Pi \circ \gamma\colon I \to (\Sigma^{(0)}, d_f)$ is (λ', ϵ')–quasigeodesic, γ is χ–strongly contracting in (Σ, d_2).*

Proof. It suffices to assume the interval $I \subset \mathbb{R}$ is compact, because if γ restricted to compact subintervals is uniformly contracting or uniformly quasigeodesic then so is γ on all of I.

First we prove Item (1). By Lemma 2.35 there are χ' and D depending on χ, λ, and ϵ such that the geodesic α between the endpoints of $\gamma(I)$ is χ'–strongly contracting and has Hausdorff distance at most D from $\gamma(I)$.

Suppose we know Item (1) for d_2–geodesics, so can show that $\Pi \circ \alpha$ is (λ'', ϵ'')–quasigeodesic in $(\Sigma^{(0)}, d_f)$, for some λ'' and ϵ'' depending on χ', hence on χ, λ, and ϵ. Let ϕ be a map from the domain of γ to the domain of α such that $d_2(\gamma(t), \alpha(\phi(t))) \leq D$ for all $t \in I$. Since there is no wall separating x from $\Pi(x)$, any fine chain separating $\Pi(x)$ from $\Pi(y)$ also separates x from y, so $d_f(\Pi(x), \Pi(y)) \leq d_w(x, y) \asymp d_2(x, y)$. Thus, $d_f(\Pi \circ \gamma(t), \Pi \circ \alpha(\phi(t))) \leq D'$, for some uniform D'.

Since γ is a d_2–quasigeodesic, for $s < t$ in I:

$$d_f(\Pi \circ \gamma(s), \Pi \circ \gamma(t)) \preceq d_2(\gamma(s), \gamma(t)) \preceq t - s$$

Conversely, we need the d_f–quasigeodesic lower bound for $\Pi \circ \gamma$. We have the d_2–quasigeodesic lower bound:

$$\frac{1}{\lambda}(t - s) - \epsilon \leq d_2(\gamma(s), \gamma(t)) \leq d_2(\alpha(\phi(s)), \alpha(\phi(t))) + 2D = |\phi(t) - \phi(s)| + 2D$$

Thus:

$$|\phi(t) - \phi(s)| \geq \frac{1}{\lambda}(t - s) - \epsilon - 2D \tag{15}$$

Since we supposed that $\Pi \circ \alpha$ is a d_f–quasigeodesic:

$$d_f(\Pi \circ \alpha(\phi(s)), \Pi \circ \alpha(\phi(t))) \geq \frac{1}{\lambda''}|\phi(t) - \phi(s)| - \epsilon'' \tag{16}$$

Combining (15) and (16) gives:

$$\begin{aligned} d_f(\Pi \circ \gamma(s), \Pi \circ \gamma(t)) &\geq d_f(\Pi \circ \alpha(\phi(s)), \Pi \circ \alpha(\phi(t))) - 2D' \\ &\geq \frac{1}{\lambda''}|\phi(t) - \phi(s)| - \epsilon'' - 2D' \\ &\geq \frac{1}{\lambda\lambda''}(t - s) - \frac{2D + \epsilon}{\lambda''} - (2D' + \epsilon'') \end{aligned}$$

This shows that $\Pi \circ \gamma$ is a d_f–quasigeodesic with constants depending only on λ, ϵ, and χ. Thus, it suffices to prove Item (1) in the case that γ is a d_2–geodesic. One direction is easy, and is independent of χ:

$$d_f(\Pi \circ \gamma(t_0), \Pi \circ \gamma(t_1)) \preceq d_2(\Pi \circ \gamma(t_0), \Pi \circ \gamma(t_1)) \preceq d_2(\gamma(t_0), \gamma(t_1)) = |t_1 - t_0|$$

In the other direction, since d_2–geodesics have connected wall preimages, Proposition 2.32 and the equivalence Theorem 3.30 (4) $\iff$ (2), imply that there exist L and $\rho > 0$ depending on χ such that for every subinterval $I' = [t_0, t_1] \subset I$ of length at least L there is a fine chain $\mathcal{V}$ separating the endpoints of $\gamma(I')$ with

$|\mathcal{V}|/|I'| \geq \rho$. Since no wall separates $\gamma(t_i)$ from $\Pi(\gamma(t_i))$, the fine chain $\mathcal{V}$ separates $\Pi(\gamma(t_0))$ and $\Pi(\gamma(t_1))$, so $d_f(\Pi\circ\gamma(t_0), \Pi\circ\gamma(t_1)) \geq |\mathcal{V}| \geq \rho|t_1 - t_0|$. If $|t_1 - t_0| < L$ then $\rho|t_1 - t_0| - \rho L < 0$, so for all $t_0, t_1 \in I$ we have:

$$d_f(\Pi\circ\gamma(t_0), \Pi\circ\gamma(t_1)) \geq \rho|t_1 - t_0| - \rho L$$

Now we prove Item (2). In view of Lemma 2.35 and connect-the-dots trick, it suffices to assume that γ is continuous.

Let $J := [a, b] \subset I$. Let $\mathcal{V} = \{V_i\}_{0\leq i\leq n+1}$ be a longest fine chain separating $\Pi\circ\gamma(a)$ and $\Pi\circ\gamma(b)$, oriented from $\Pi\circ\gamma(a)$ to $\Pi\circ\gamma(b)$. No wall separates $\gamma(a)$ from $\Pi(\gamma(a))$ or separates $\gamma(b)$ from $\Pi(\gamma(b))$, but conceivably $\gamma(a)$ could be a point on V_0 and/or $\gamma(b)$ could be a point on V_{n+1}. Even so, $\mathcal{V}' := \{V_i\}_{1\leq i\leq n}$ is a fine chain separating $\gamma(a)$ from $\gamma(b)$. Combined with the hypothesis that $\Pi\circ\gamma$ is d_f–quasigeodesic:

$$\frac{|\mathcal{V}'|}{|J|} = \frac{|\mathcal{V}| - 2}{|J|} = \frac{d_f(\Pi\circ\gamma(a), \Pi\circ\gamma(b)) - 2}{|J|} \geq \frac{|J|/\lambda' - \epsilon' - 2}{|J|} = \frac{1}{2\lambda'} + \left(\frac{1}{2\lambda'} - \frac{\epsilon' + 2}{|J|}\right)$$

Thus, there exists a fine chain separating $\gamma(a)$ and $\gamma(b)$ of density at least $\rho := \frac{1}{2\lambda'}$, provided:

$$|J| \geq 2\lambda'(\epsilon' + 2) \tag{17}$$

Proposition 3.11 says γ has at least quadratic divergence, with gauge depending on λ, ϵ, ℓ, the density $1/2\lambda'$, and the lower bound of (17), hence on λ, ϵ, λ', and ϵ'. Proposition 2.32 says this divergence gauge determines an upper bound on a strong contraction constant, so γ is χ–strongly contracting for some χ bounded above in terms of λ, ϵ, λ', and ϵ'. □

4.2.2. *Morse recognizing is equivalent to stability recognizing.* The purpose of this subsection is to give definitions of "Morse recognizing" and "stability recognizing", compare them to related notions that appear in the literature, and show that these two recognition properties are equivalent.

Definition 4.11. $\phi\colon X \to Y$ is *Morse recognizing* if:

- For all λ, ϵ, μ there exist λ' and ϵ' such that if $\gamma\colon I \to X$ is a μ–Morse (λ, ϵ)–quasigeodesic then $\phi\circ\gamma\colon I \to Y$ is a (λ', ϵ')–quasigeodesic.
- For all λ, ϵ, λ', ϵ' there exists μ such that if $\gamma\colon I \to X$ is a (λ, ϵ)–quasigeodesic and $\phi\circ\gamma\colon I \to Y$ is a (λ', ϵ')–quasigeodesic then $\gamma\colon I \to X$ is μ–Morse.

Definition 4.12. A subset $Z \subset X$ is (λ, ϵ)–*quasigeodesically connected* if for every pair of points z_0 and z_1 in Z there exists a (λ, ϵ)–quasigeodesic $\gamma\colon I \to X$ whose image is in Z and whose endpoints are z_0 and z_1.

Definition 4.13 (cf [58, 48, 2][4])**.** A subset $Z \subset X$ is (λ, ϵ, μ)–*stable* if for every pair of points z_0 and z_1 in Z there exists a μ–Morse (λ, ϵ)–quasigeodesic $\gamma\colon I \to X$ whose image is in Z and has endpoints z_0 and z_1.

Definition 4.14. $\phi\colon X \to Y$ is *stability recognizing* if:

- For all λ, ϵ, μ there exist λ' and ϵ' such that for every (λ, ϵ, μ)–stable $Z \subset X$ the map $\phi|_Z$ is a (λ', ϵ')–quasiisometric embedding of $(Z, d_X|_Z)$ into Y.
- For all λ, ϵ, λ', ϵ' there exists μ such that for every (λ, ϵ)–quasigeodesically connected set $Z \subset X$ if $\phi|_Z$ is a (λ', ϵ')–quasiisometric embedding of $(Z, d_X|_Z)$ into Y then Z is (λ, ϵ, μ)–stable in X.

Proposition 4.15. *$\phi\colon X \to Y$ is Morse recognizing if and only if it is stability recognizing.*

Usually one also requires that the space Y is hyperbolic.[5] In that case the Morse property, stability, and quasiconvexity are equivalent, so the recognition properties can be restated. When Y is hyperbolic, "$\phi\colon X \to Y$ is Morse recognizing" is equivalent to "a quasigeodesic γ in X is Morse if and only if $\phi\circ\gamma$ is a Morse quasigeodesic in Y, with quantitative control on the quasigeodesic constants and Morse gauges in both directions." Similarly, "$\phi\colon X \to Y$ is stability recognizing" is equivalent to "a quasigeodesically connected subspace Z is stable in X if and only if $\phi|_Z$ is a quasiisometric embedding with quasiconvex image, with quantitative control between the stability parameters and the quasiisometric embedding/quasiconvexity parameters, in both directions." We do not need Y to be hyperbolic for Proposition 4.15.

[4]We are treating stability as a property of a subspace Z of X with the subspace metric. It is not hard to show this is equivalent to the standard definition in which Z is considered as a separate metric space and stability is a property of a quasiisometric embedding of Z into X.

[5]A notable early example of a recognition-type theorem where the target space is not hyperbolic is the characterization of convex cocompact subgroups of the mapping class group in terms of having quasiconvex orbits in Teichmüller space [61].

Russell, Spriano, and Tran [96, Definition 4.17] defined a *Morse detectable space* X to be one such that there exists a Morse recognizing map $\phi\colon X \to Y$ as above, with the additional conditions that Y is hyperbolic and ϕ is coarse Lipschitz. This gave a name to the property characterizing contracting quasigeodesics in hierarchically hyperbolic spaces described by Abbott, Behrstock, and Durham [2]. This terminology has been picked up in subsequent work related to the Morse-local-global property [5, 90].

Durham and Taylor [58] defined a notion of *stable subgroup* abstracting the properties of what were called convex cocompact subgroups of mapping class groups of hyperbolic surfaces [61]. A subgroup H of a finitely generated group G is stable[6] if it is stable as a subspace of a Cayley graph of G. Kent and Leininger [73] (see also [67]) proved what we would now call a stable subgroup recognition theorem, showing that convex cocompact subgroups of the mapping class group are those for which the orbit map into the curve graph is a quasiisometric embedding of the subgroup. Analogous stable subgroup recognition theorems exist for groups hyperbolic relative to peripherals with linear divergence [12], right-angled Artin groups [76], hierarchically hyperbolic groups [2], etc [92, 93, 103, 17].

Definition 4.14 for stable *subspace* recognition is formally stronger than stable *subgroup* recognition, since if $\phi\colon G \to Y$ recognizes stable subgroups of G it is conceivable that there are stable subspaces of G that are not close to any finitely generated subgroup and for which ϕ does not give a quasiisometric embedding. We do not know any examples.

Proof of Proposition 4.15. Suppose $\phi\colon X \to Y$ is Morse recognizing.

Suppose $Z \subset X$ is (λ, ϵ, μ)–stable. For all $z_0, z_1 \in Z$ there exists a μ–Morse (λ, ϵ) quasigeodesic $\gamma\colon [t_0, t_1] \to X$ with image in Z and $\gamma(t_i) = z_i$ for $i \in \{0, 1\}$. Since $\phi\colon X \to Y$ is Morse recognizing, $\phi \circ \gamma\colon [t_0, t_1] \to Y$ is (λ', ϵ')–quasigeodesic, so:

$$d_Y(\phi(z_0), \phi(z_1)) = d_Y(\phi(\gamma(t_0)), \phi(\gamma(t_1))) \overset{\lambda', \epsilon'}{\asymp} |t_1 - t_0| \overset{\lambda, \epsilon}{\asymp} d_X(z_0, z_1)$$

Thus, $\phi|_Z$ is a quasiisometric embedding with constants determined by λ, ϵ, λ', ϵ', and λ' and ϵ' were determined by λ, ϵ, and μ.

Conversely, suppose $Z \subset X$ is (λ, ϵ)–quasigeodesically connected and $\phi|_Z$ is a (λ', ϵ')–quasiisometric embedding of Z into Y. For all $z_0, z_1 \in Z$ there exists a (λ, ϵ)–quasigeodesic $\gamma\colon [t_0, t_1] \to X$ with image in Z and $\gamma(t_i) = z_i$ for $i \in \{0, 1\}$. Since $\phi|_Z$ is a (λ', ϵ')–quasiisometric embedding, $\phi \circ \gamma$ is a quasigeodesic with constants determined by λ, ϵ, λ', and ϵ'. Since $\phi\colon X \to Y$ is Morse recognizing, there exists μ such that γ is μ–Morse, for μ depending on λ, ϵ and the quasigeodesic constants of $\phi \circ \gamma$, hence on λ, ϵ, λ', and ϵ'. Thus, the transitive family of uniform quasigeodesics verifying quasigeodesic connectivity of Z is also uniformly Morse, so Z is stable.

We have shown that $\phi\colon X \to Y$ is Morse recognizing implies that it is stability recognizing. Now suppose $\phi\colon X \to Y$ is stability recognizing.

Suppose $\gamma\colon I \to X$ is a μ–Morse (λ, ϵ)–quasigeodesic. Then $Z := \gamma(I)$ is a (λ, ϵ, μ)–stable subset of X. Since $\phi\colon X \to Y$ is stability recognizing, $\phi|_Z$ is a (λ', ϵ')–quasiisometric embedding. Thus, $\phi \circ \gamma$ is a composition of quasiisometric embeddings, so $\phi \circ \gamma\colon I \to Y$ is a quasigeodesic with constants determined by λ, ϵ, λ', and ϵ', hence by λ, ϵ, and μ.

Conversely, suppose $\gamma\colon I \to X$ is (λ, ϵ)–quasigeodesic and $\phi \circ \gamma\colon I \to Y$ is (λ', ϵ')–quasigeodesic. Then $Z := \gamma(I)$ is (λ, ϵ)–quasigeodesically connected in X. Furthermore, for any $z_0, z_1 \in Z$ there exist $t_0, t_1 \in I$ such that $\gamma(t_i) = z_i$, and:

$$d_X(z_0, z_1) = d_X(\gamma(t_0), \gamma(t_1)) \overset{\lambda, \epsilon}{\asymp} |t_1 - t_0| \overset{\lambda', \epsilon'}{\asymp} d_Y(\phi(\gamma(t_0)), \phi(\gamma(t_1))) = d_Y(\phi(z_0), \phi(z_1))$$

This says $\phi|_Z$ is a quasiisometric embedding of $(Z, d_X|_Z)$ into (Y, d_Y), with constants depending on λ, ϵ, λ', and ϵ'. Since $\phi\colon X \to Y$ is stability recognizing, there exists μ depending on λ, ϵ, λ', and ϵ' such that Z is (λ, ϵ, μ)–stable. Consider an interval $[t_0, t_1] \subset I$. Since Z is stable, there exists a μ–Morse (λ, ϵ)–quasigeodesic $\delta\colon J \to X$ with image in Z and endpoints at $\gamma(t_0)$ and $\gamma(t_1)$. So $\gamma([t_0, t_1])$ and δ are (λ, ϵ)–quasigeodesics with the same endpoints such that δ is μ–Morse. It follows that the Hausdorff distance between δ and $\gamma([t_0, t_1])$ can be bounded in terms of λ, ϵ, and μ. Then it follows that there is μ' depending on λ, ϵ, and μ such that $\gamma([t_0, t_1])$ is μ'–Morse. Since this is true for every subsegment of γ, conclude that γ is μ'–Morse. □

[6]Quasigeodesic connectivity of H already forces it to be finitely generated and undistorted in G.

4.2.3. *Recognizing Morse and stable parabolics.* The combination of Theorem 4.10 and Proposition 4.15 gives:

Proposition 4.16. *A subgroup $H \leq W$ is stable if and only if it is finitely generated and the orbit map is a quasiisometric embedding of H with respect to a word metric into $(\Sigma^{(0)}, d_f)$.*

It is not immediately clear from Theorem 4.10 how to recognize Morse subgroups of W. However, for parabolic subgroups, equivalently, for special subgroups, there are nice criteria for being stable and Morse. Given a finite rank Coxeter system (W, S) and $T \subset S$, let Γ_T denote the subdiagram of the Coxeter diagram of (W, S) spanned by the vertices T. Call the subdiagram "spherical" if it defines a spherical Coxeter group.

Theorem 4.17. *Let (W, S) be a finite rank Coxeter system, and let $T \subset S$.*

W_T is stable if and only if $\Gamma_{T\cap U}$ is spherical for every wide $U \subset S$.

W_T is Morse if and only if for every wide $U \subset S$, one of the following is true:

1. *The intersection of every connected component of Γ_U with Γ_T is spherical.*
2. *Every nonspherical connected component of Γ_U is contained in Γ_T.*

Proof. The stable case follows immediately from Theorem 3.31: if U is wide and $T \cap U$ is nonspherical then there is a biinfinite d_1–geodesic γ contained in $\Sigma_{T\cap U} = \Sigma_T \cap \Sigma_U$, so the entire γ has edges with labels in the wide subset U, so γ is not Morse. On the other hand, if $T \cap U$ is spherical for every wide U, then, since there are only finitely many spherical subsets of S, there is a uniform diameter bound on $\Sigma_{T\cap U}$, so for every d_1–geodesic γ in Σ_T there is a uniform bound on the lengths of its subsegments having label in some wide subset of S. Theorem 3.31 then says that geodesics in Σ_T are uniformly Morse.

Now consider Morseness of Σ_T. Condition (1) is equivalent to: $W_{T\cap U} = W_T \cap W_U$ is spherical. Since nonspherical irreducible Coxeter groups do not have finite index proper parabolic subgroups, condition (2) is equivalent to saying that the index of $W_{T\cap U} = W_T \cap W_U$ in W_U is finite. The two conditions together say that every wide special subgroup is either virtually contained in W_T or virtually has trivial intersection with W_T.

Suppose $U \subset S$ is wide and neither condition (1) nor (2) is true. Let $U = U_1 \sqcup \cdots \sqcup U_k \sqcup K$ be the decomposition of U into irreducible nonspherical components U_i and spherical K. Since U is wide, if $k = 1$ then U_1 is irreducible higher rank affine, so it has no proper nonspherical subparabolics. Thus $T \cap U_1$ is either spherical or all of U_1. This contradicts the hypothesis, so we must have $k \geq 2$, and there is some i such that $W_{T\cap U_i}$ has infinite index in W_{U_i}. If $T \cap U_i$ is spherical then there is some $j \neq i$ such that $T \cap U_j$ is nonspherical. If $T \cap U_i$ is nonspherical then consider any $j \neq i$. If $T \cap U_j$ is spherical then $W_{T\cap U_j}$ has infinite index in W_{U_j}; in this case swap i and j. Thus, we may assume we have indices $i \neq j$ such that $W_{T\cap U_i}$ has infinite index in W_{U_i} and $T \cap U_j$ is nonspherical. For $n \in \mathbb{N}$, choose $b_n \in W_{U_i}$ such that $d_1(b_n, \Sigma_{T\cap U_i}) \xrightarrow{n\to\infty} \infty$, which is possible since $W_{T\cap U_i}$ has infinite index in W_{U_i}. Then Lemma 2.46 implies that $d_1(b_n, \Sigma_T) \to \infty$. Pick $a_n \in W_{T\cap U_j}$ with $|a_n|_S \geq |b_n|_S$. Since W_{U_i} and W_{U_j} commute, the geodesics $[\mathbb{1}, a_n] \subset \Sigma_T$ have endpoints that are connected by $(3, 0)$–quasigeodesics $[\mathbb{1}, b_n] + [b_n, a_n b_n] + [a_n b_n, a_n]$ that contain points b_n arbitrarily far from Σ_T. Thus, Σ_T is not Morse.

Now suppose that for every wide $U \subset S$ either (1) or (2) is true. We derive a contradiction by supposing that Σ_T is not Morse. Let D be the diameter of a chamber of Σ. Since (Σ, d_2) is CAT(0), Proposition 2.32 and Proposition 2.33 combine to say that if Σ_T is not Morse then it does not have the Bounded Geodesic Image Property. Thus, for every $n \in \mathbb{N}$ there exists a d_2–geodesic $[a_n, b_n]$ that does not come n–close to Σ_T, but $\operatorname{diam}(\pi_{\Sigma_T}([a_n, b_n])) > n$. By passing to a subsegment, we may assume $d_2(\pi_{\Sigma_T}(a_n), \pi_{\Sigma_T}(b_n)) > n$. Let x_n be a vertex closest to a_n, and let y_n be a vertex closest to b_n. Then $d_2(a_n, x_n) \leq D$, $d_2(b_n, y_n) \leq D$, and the Hausdorff distance between $[a_n, b_n]$ and $[x_n, y_n]$ is at most D, so $d_2(\pi_{\Sigma_T}(x_n), \pi_{\Sigma_T}(y_n)) \geq d_2(\pi_{\Sigma_T}(a_n), \pi_{\Sigma_T}(b_n)) - 2D \geq n - 2D$ and $d_2(\Sigma_T, [x_n, y_n]) \geq n - D$. Recalling the definition of the gate map from Section 2.4.2, set $x'_n := \mathfrak{p}_{\Sigma_T}(x_n)$ and $y'_n := \mathfrak{p}_{\Sigma_T}(y_n)$. Lemma 2.49 implies $d_2(\pi_{\Sigma_T}(x_n), x'_n) \leq D$ and $d_2(\pi_{\Sigma_T}(y_n), y'_n) \leq D$, so $d_2(x'_n, y'_n) \geq n - 4D$.

Fix n large enough that d_2–distance at least $n - 4D$ implies d_c distance at least 4ℓ in Lemma 2.22 and Lemma 2.24. The former implies $d_c(x'_n, y'_n) \geq 4\ell$, so there is a chain $\mathcal{V}^- \sqcup \mathcal{V}^+$ oriented from x'_n to y'_n such that the walls of $\mathcal{V}^-$ come first, the walls of $\mathcal{V}^+$ come second, and each of the two subchains contains at least 2ℓ–many walls.

Every wall of $\mathcal{V}^- \sqcup \mathcal{V}^+$ cuts Σ_T, so $\operatorname{Pc}(\mathcal{V}^-)$ and $\operatorname{Pc}(\mathcal{V}^+)$ are both irreducible nonspherical subgroups of W_T. By Corollary 2.43, no wall of $\mathcal{V}^- \sqcup \mathcal{V}^+$ separates x_n from x'_n or separates y_n from y'_n, so they all cut $\gamma := [x_n, y_n]$. Let p be a point on γ that occurs between the last wall of $\mathcal{V}^-$ and the first wall of $\mathcal{V}^+$ and

is not contained in any wall. Let γ^- be the subsegment of γ from x_n to p, and let γ^+ be the subsegment of γ from p to y_n. Let p' be a vertex closest to p. Then $d_2(p,p') \leq D$, no wall separates p and p', and $d_2(p', \Sigma_T) \geq n - 2D$. Then the choice of n with respect to Lemma 2.24 gives a chain $\mathcal{H}_0$ of length at least 4ℓ separating p' and Σ_T. Since no walls separate p and p', the chain $\mathcal{H}_0$ also separates p and Σ_T. The walls of $\mathcal{H}_0$ do not cut Σ_T, but all of them must cut the paths between Σ_T and p given by $[x'_n, x_n] + \gamma^-$ and $\gamma^+ + [y_n, y'_n]$. Since any wall cuts γ at most once, each wall of $\mathcal{H}_0$ cuts at least one of $[x'_n, x_n]$ and $[y_n, y'_n]$. Let $\mathcal{H}^-$ be the subchain of $\mathcal{H}_0$ consisting of walls that cut $[x_n, x'_n]$, and let $\mathcal{H}^+ := \mathcal{H}_0 \smallsetminus \mathcal{H}^-$. Without loss of generality, assume $|\mathcal{H}^-| \geq 2\ell$, and set $\mathcal{V} := \mathcal{V}^-$ and $\mathcal{H} := \mathcal{H}^-$. This is possible because otherwise $|\mathcal{H}^+| \geq 2\ell$, so then we would take $\mathcal{V} := \mathcal{V}^+$ and $\mathcal{H} := \mathcal{H}^+$ and swap x_n and x'_n with y_n and y'_n in what follows.

Each wall $H \in \mathcal{H}$ cuts $[x_n, x'_n]$ and separates p and Σ_T. Since H separates p from Σ_T and separates x_n from $x'_n \in \Sigma_T$, we have that x_n and p are on the same side of H. Since the open halfspaces of H are convex, γ^- lies in the same open halfspace of H as x_n and p. For each $V \in \mathcal{V}$ there is a path from p to Σ_T that follows γ^- from p to $V \cap \gamma$ and then follows V to Σ_T. Since H cuts this path but does not cut γ^-, we must have $H \pitchfork V$.

Since this is true for all $H \in \mathcal{H}$ and $V \in \mathcal{V}$, we have a grid $(\mathcal{V}, \mathcal{H})$ with $|\mathcal{V}| \geq 2\ell$ and $|\mathcal{H}| \geq 2\ell$. Corollary 2.16 and Proposition 3.14 say that $N_W(\mathrm{Pc}(\mathcal{V})) = \mathrm{Pc}(\mathcal{V}) \times \mathrm{Pc}(\mathcal{V})^\perp$ is a wide parabolic, and either $\mathrm{Pc}(\mathring{\mathcal{H}}) \leq \mathrm{Pc}(\mathcal{V})^\perp$ or $\mathrm{Pc}(\mathcal{V}) = \mathrm{Pc}(\mathcal{H})$ is irreducible higher rank affine.

Since the walls of $\mathcal{H}$ do not cut Σ_T, their reflections are not in W_T.

The affine case is impossible because it yields $W_T \geq \mathrm{Pc}(\mathcal{V}) = \mathrm{Pc}(\mathcal{H}) \not\leq W_T$.

The commuting case is also impossible. Up to the W_T–action, we may assume $\mathrm{Pc}(\mathcal{V}) = W_U$ for some $U \subset T$, and it follows that $N_W(\mathrm{Pc}(\mathcal{V})) = W_U \times W_{U^\perp}$. Now, U is a nonspherical irreducible component of the wide set $U \sqcup U^\perp$ that is contained in T, so condition (1) is not satisfied for $U \sqcup U^\perp$. By hypothesis, condition (2) must be satisfied, meaning that T contains every nonspherical irreducible component of $U \sqcup U^\perp$, hence of $U^\perp$. However, this is not true, because $\mathrm{Pc}(\mathring{\mathcal{H}})$ is a nonspherical, irreducible reflection subgroup of $W_{U^\perp}$, so by Lemma 2.12 it is contained in $W_{\mathcal{C}}$ for some connected component of $\Gamma_{U^\perp}$. The component $\mathcal{C}$ is necessarily nonspherical, since $\mathrm{Pc}(\mathring{\mathcal{H}})$ is. We get the contradiction $W_T \not\geq \mathrm{Pc}(\mathring{\mathcal{H}}) \leq W_{\mathcal{C}} \leq W_T$.

Since both cases lead to contradictions, Σ_T is Morse. □

4.3. **The coned-off space.**

Definition 4.18. The *coned-off space* $(\widehat{\Sigma}, \hat{d})$ is the metric graph $\widehat{\Sigma}$ and geodesic metric $\hat{d}$ obtained from $\Sigma^{(1)}$ by adding a new vertex v_Υ for each wide parabolic subcomplex Υ of Σ, and connecting v_Υ to each vertex of Υ by an edge of length $1/2$.

In the case that (W, S) is a right-angled Coxeter group, $\widehat{\Sigma}$ is W–equivariantly quasiisometric to the universal acylindrical hyperbolic space for W constructed by Abbott, Behrstock, and Durham [2]. In the 2-dimensional, right-angled, strongly CFS case this also coincides, again up to W–equivariant quasiisometry, with the intersection complex of Oh [88], see [44, Section 6.2]. We also note that Genevois [63] has results producing hyperbolic spaces by coning off certain convex subcomplexes of CAT(0) cube complexes, but Σ is not a cube complex outside of the right-angled case.

The goal of this subsection is to prove:

Theorem 4.19. *The identity map on $\Sigma^{(0)}$ induces a quasiisometry $(\Sigma^{(0)}, d_f) \to (\widehat{\Sigma}, \hat{d})$.*

Corollary 4.20. *$(\widehat{\Sigma}, \hat{d})$ is a geodesic hyperbolic space and $(\Sigma^{(1)}, d_1) \xrightarrow{\mathrm{Id}_{\Sigma^{(1)}}} (\widehat{\Sigma}, \hat{d})$ is Morse recognizing.*

Proof. $(\widehat{\Sigma}, \hat{d})$ is a geodesic space by construction. Hyperbolicity is preserved by quasiisometries between quasiruled spaces [27, Section 2.1]. We have that $(\Sigma^{(0)}, d_f)$ is a quasiruled hyperbolic space, Theorem 4.7, and quasiisometric to the geodesic space $(\widehat{\Sigma}, \hat{d})$, Theorem 4.19, so $(\widehat{\Sigma}, \hat{d})$ is hyperbolic. The quasiisometry transfers the Morse recognizing property of $(\Sigma^{(0)}, d_f)$, Theorem 4.10, to $(\widehat{\Sigma}, \hat{d})$. □

Since $\Sigma^{(0)}$ is coarsely dense in $\widehat{\Sigma}$, the identity map on $\Sigma^{(0)}$ is a coarse surjection $\Sigma^{(0)} \to \widehat{\Sigma}$, so to prove Theorem 4.19 we show $d_f \asymp \hat{d}$ on $\Sigma^{(0)}$. One direction is easy: Lemma 4.21 says $d_f \leq \hat{d}$, because d_f gives wide parabolic subcomplexes diameter 1, so it collapses as much as $\hat{d}$. In the other direction we need to argue that d_f does not collapse too much more than $\hat{d}$. That is Proposition 4.25. To get the idea of the proof, recall that to show that a combinatorial geodesic γ in Σ is an unparameterized d_f quasiruler, we defined a FineProgress

function measuring the distance travelled by γ in terms of d_f against its parameterization. We compressed parameter intervals on which FineProgress was constant, so that it had roughly constant growth rate relative to the reparameterization. Now we look at γ in $\widehat{\Sigma}$ and replace subsegments of γ by cone path shortcuts, when possible, to get a new path $\overline{\gamma}^*$, essentially making a geometric compression of γ, instead of just changing its speed. This compression is uniformly bounded in terms of d_f, by Lemma 4.24, whose proof produces wide parabolic subcomplexes from grids to show that if d_f collapses the length of a combinatorial geodesic segment enough then the geodesic has a long subsegment in a uniform neighborhood of a wide parabolic subcomplex, so that subsegment has bounded $\hat{d}$–diameter. The proof of Proposition 4.25 cuts γ along walls of a longest fine chain separating its endpoints and uses the quasiruler property to bound the d_f distance between endpoints of the resulting subsegments, and then applies Lemma 4.24 to control the $\hat{d}$–length of the subsegments. It will turn out after the fact, Corollary 4.26, that the paths $\overline{\gamma}^*$ are uniform quasigeodesics in $\widehat{\Sigma}$.

There are two kinds of edges in $\widehat{\Sigma}$, those coming from $\Sigma^{(1)}$ and those that connect a vertex $a \in \Upsilon^{(0)}$, for some wide parabolic subcomplex Υ, to the cone vertex v_Υ. Furthermore, v_Υ is only connected by an edge to vertices from $\Upsilon^{(0)}$, so a $\hat{d}$–geodesic γ between x and y in $\Sigma^{(0)}$ can be decomposed as a concatenation $\beta_0 + c_1 + \beta_1 + \cdots + c_m + \beta_m$ where the β_i are disjoint (possibly singleton) d_1–geodesics and the c_i are cone paths of the form $a_i - v_{\Upsilon_i} - b_i$ for some $a_i \neq b_i \in \Upsilon_i^{(0)}$. Moreover, two vertices of any β_i are contained in a common wide parabolic subcomplex only if they are adjacent in β_i. Compare with the "electric paths" and "electric path length" in the relatively hyperbolic case [60].

Lemma 4.21. *On $\Sigma^{(0)}$, $d_f \leq \hat{d}$.*

Proof. Let $x, y \in \Sigma^{(0)}$, and let γ be a $\hat{d}$–geodesic connecting them, decomposed as above. Since γ is $\hat{d}$–geodesic, so are its subpaths. Thus, if β_i is a d_1–geodesic subsegment from β_i^- to β_i^+ then $|\beta_i| = \hat{d}(\beta_i^-, \beta_i^+)$. Since it is a combinatorial geodesic, we also have $|\beta_i| = d_w(\beta_i^-, \beta_i^+) \geq d_f(\beta_i^-, \beta_i^+)$. For cone paths $c_i = a_i - v_{\Upsilon_i} - b_i$ Lemma 4.8 says $d_f(a_i, b_i) = 1$. Thus:

$$d_f(x,y) \leq \sum_{i=0}^{m} d_f(\beta_i^-, \beta_i^+) + \sum_{i=1}^{m} d_f(a_i, b_i) \leq \sum_{i=0}^{m} \hat{d}(\beta_i^-, \beta_i^+) + \sum_{i=1}^{m} \hat{d}(a_i, b_i) = \hat{d}(x,y) \qquad \square$$

Definition 4.22 (Shortcut graph, greedy path, realization)**.** Let γ be a d_1–geodesic segment from x to y in $\Sigma^{(1)}$ with vertex sequence $x = z_0, z_1, \ldots, z_n = y$. Define its *shortcut graph* $\widehat{\gamma}$ to be the graph with vertices z_i, with an "ordinary edge" between z_i and z_j if they are adjacent in $\Sigma^{(1)}$, and a "shortcut edge" between z_i and z_j if they are not adjacent in $\Sigma^{(1)}$, but are contained in a common wide parabolic subcomplex.

Define the *greedy path* γ^* to be the path in $\widehat{\gamma}$ with vertex sequence $z_{\phi(i)}$ where ϕ is defined inductively by $\phi(0) := 0$ and:

$$\phi(i+1) := \max\{m \mid z_{\phi(i)} \text{ and } z_m \text{ are adjacent in } \widehat{\gamma}\}$$

Define a *realization* $\overline{\gamma}^*$ of γ^* to be a path in $\widehat{\Sigma}$ with vertices and ordinary edges corresponding to those of γ^*, and additional vertices and edges defined as follows. When γ^* has a shortcut edge from z_i to z_j there exist wide parabolic subcomplexes containing z_i and z_j. Pick one such parabolic subcomplex Υ and add a cone shortcut $z_i - v_\Upsilon - z_j$ to $\overline{\gamma}^*$.

It follows from the definitions that $\hat{d}(x,y) \leq d_{\widehat{\gamma}}(x,y) \leq |\gamma^*| = |\overline{\gamma}^*|$.

Observe that we have identified the vertices of γ with those of $\widehat{\gamma}$, and we have identified the vertices of γ^* with the $\Sigma^{(0)}$ vertices of $\overline{\gamma}^*$, which are a subset of the vertices of γ. The distinction is that γ^* is a canonical path in $\widehat{\gamma}$, whereas $\overline{\gamma}^*$ is a path in $\widehat{\Sigma}$ whose $\Sigma^{(0)}$ vertices are canonical, but whose cone vertices may not be.

Lemma 4.23. *The greedy path is a shortest path from z_0 to z_n in $\widehat{\gamma}$.*

Proof. Let ϕ be the indexing function of the greedy path, and inductively define ϕ' by $\phi'(0) := 0$ and:

$$\phi'(i+1) := \max\{m \mid \exists k \leq \phi'(i),\ z_k \text{ and } z_m \text{ are adjacent in } \widehat{\gamma}\}$$

We claim that $\phi(i) = \phi'(i)$ for all i. By definition, $\phi(0) = \phi'(0) = 0$. Suppose inductively that $\phi = \phi'$ on $[0, i]$, and consider $i+1$. By definition, the maximum defining $\phi'(i+1)$ is taken over a superset of the maximum defining $\phi(i+1)$, so $\phi'(i+1) \geq \phi(i+1)$. To derive the opposite bound, suppose $j = \phi'(i+1)$. Then there is some $k \leq \phi'(i)$ such that z_k and z_j are adjacent in $\widehat{\gamma}$. If they are adjacent via a $\Sigma^{(1)}$ edge then

$j = k+1$, so $\phi'(i+1) - 1 = k \leq \phi'(i) = \phi(i) < \phi(i+1)$ implies $\phi'(i+1) \leq \phi(i+1)$. If they are adjacent via a shortcut through a wide parabolic subcomplex Υ then since parabolic subcomplexes are d_1–convex and $k \leq \phi'(i) = \phi(i) < \phi(i+1) \leq \phi'(i+1) = j$, the vertex $z_{\phi(i)}$ is also in Υ. Since there is an edge from $z_{\phi(i)}$ to z_j in $\widehat{\gamma}$, it follows that $\phi(i+1) \geq j = \phi'(i+1)$. Thus $\phi \equiv \phi'$.

Now take any injective edge path from z_0 to z_n in $\widehat{\gamma}$, and let $q_0, q_1, \ldots, q_m$ be its vertex sequence, and define ψ by $q_i = z_{\psi(i)}$. It follows immediately that $\psi(i) \leq \phi'(i)$ for all i, so by our previous argument $\psi(i) \leq \phi(i)$. Thus, at every step the greedy path is at least as far along γ as the q path. In particular, since $\psi(m) = n$, the greedy path takes at most m steps, so it is at least as short as every edge path from z_0 to z_n in $\widehat{\gamma}$. □

Lemma 4.24. *There is a function* Collapse$\colon \mathbb{N} \to \mathbb{N}$ *defined by:*

$$\text{Collapse}(N) := \sup\{d_{\widehat{\gamma}}(x,y) \mid d_f(x,y) \leq N \text{ and } \gamma \text{ is a } d_1\text{–geodesic from } x \text{ to } y\}$$

Proof. We must show Collapse$(N) < \infty$. For $n \in \mathbb{N}$ with $n \leq$ Collapse(N) there exists a d_1–geodesic γ_n between some vertices x_n and y_n such that $d_f(x_n, y_n) \leq N$ and $d_{\widehat{\gamma}_n}(x_n, y_n) \geq n$. For such n, let $p_0, p_1, \ldots, p_q$ be the vertex sequence of the greedy path γ_n^* in $\widehat{\gamma}_n$. Call these the "greedy vertices" of γ_n. By Lemma 4.23, γ_n^* is a $\widehat{\gamma}_n$-geodesic, so $q = d_{\widehat{\gamma}_n}(x_n, y_n) \geq n$.

Since γ_n is a d_1–geodesic, the walls dual to the edges of γ_n are distinct and separate x_n and y_n. For each $0 \leq i < q$, call the wall dual to the first edge of γ_n after p_i a "greedy wall". Dilworth's Theorem implies there is a subset of these walls forming a chain $\mathcal{V} = \{V_i\}_{0 \leq i \leq I}$ such that $I + 1 = |\mathcal{V}| \geq n/\dim(X_{\text{NR}})$, where the indexing and orientation are as usual: $x_n \in V_i^- \subset V_{i+1}^-$ for all $i < I$.

Suppose $n \geq (N+1)\dim(X_{\text{NR}})$ so that $E := \lfloor \frac{|\mathcal{V}|-1}{N} \rfloor \geq 1$. Consider the subchain of $\mathcal{V}$ given by $\mathcal{V}' := \{V_0, V_E, V_{2E}, \ldots, V_{NE}\}$. Then $d_f(x_n, y_n) \leq N < N+1 = |\mathcal{V}'|$, so $\mathcal{V}'$ is not fine, so there exists $0 \leq j < N$ such that $\{V_{jE}, V_{(j+1)E}\}$ is transverse to a chain $\mathcal{H}$ of length 2ℓ. Let $\mathcal{V}'' := \{V_i\}_{jE \leq i \leq (j+1)E}$. Then $(\mathcal{V}'', \mathcal{H})$ is a grid with chains of lengths $E + 1 \geq |\mathcal{V}|/(N+1)$ and 2ℓ.

By Proposition 3.27 there exists a constant C such that $|\mathcal{V}''| \geq C(2\ell - 1)$ implies there exists a wide parabolic subcomplex Υ and a convex subchain $\mathcal{V}'''$ of $\mathcal{V}''$ such that $|\mathcal{V}'''| \geq |\mathcal{V}''|/C$ and the subsegment β of γ_n between the extreme walls of $\mathcal{V}'''$ is contained in $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$. We have $|\mathcal{V}''| \geq n/(N+1)\dim(X_{\text{NR}})$, so when $n \geq (2\ell-1)(N+1)C\dim(X_{\text{NR}})$ this result applies and gives $|\mathcal{V}'''| \geq n/(N+1)C\dim(X_{\text{NR}})$.

Now proceed as in the proof of Theorem 3.31: Since $\beta \subset \bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon)$ and Υ is convex, the d_w–version of the 4-gon trick (Lemma 2.23) implies that, for β^- and β^+ the endpoints of β, if $\mathcal{M}$ is the set of all walls transverse to β and $\mathcal{M}'$ is the subset of those that cut Υ, we have:

$$D := |\mathcal{M}| - |\mathcal{M}'| \overset{+}{\preceq} d_w(\beta^-, \Upsilon) + d_w(\beta^+, \Upsilon) \asymp d_2(\beta^-, \Upsilon) + d_2(\beta^+, \Upsilon) \leq 2\mathfrak{R}$$

Thus, there exists D_0, depending only on (W, S), such that $D \leq D_0$ for all n.

The count $D = |\mathcal{M}| - |\mathcal{M}'|$ gives us a decomposition of β as D–many single edges whose dual walls do not cut Υ separating $(D+1)$–many (possibly trivial) subpaths consisting of edges all of whose dual walls cut Υ. Consider one such subsegment β'. By the same argument as in the proof of Theorem 3.31, its edge labels belong to a set $U = S \cap a^{-1}Ta$ for some a, and if U is nonspherical then it is contained in a wide subset of S, which then implies that β' is contained in a wide parabolic subcomplex, so its diameter in $\widehat{\gamma}_n$ is 1. It is also possible that U is spherical and not contained in a wide subset, recall Example 3.32, but in this case $|\beta'| \leq \Delta$, the maximal d_1–diameter of a spherical special subgroup, so its diameter in $\widehat{\gamma}_n$ is also bounded above by $\Delta \geq 1$. Thus, $\text{diam}_{\widehat{\gamma}_n}(\beta) \leq D + (D+1)\Delta \leq D_0 + (D_0+1)\Delta$. Let $D_1 := D_0 + (D_0+1)\Delta$.

For all but the first wall of $\mathcal{V}'''$, the edge path β contains the vertex of γ_n immediately prior to that wall. Since the walls of $\mathcal{V}'''$ are greedy walls, this means that β contains $(|\mathcal{V}'''| - 1)$–many greedy vertices. Since the greedy path is a $\widehat{\gamma}_n$ geodesic, this implies that:

$$\frac{n}{(N+1)C\dim(X_{\text{NR}})} - 2 \leq |\mathcal{V}'''| - 2 \leq \text{diam}_{\widehat{\gamma}_n}(\beta) \leq D_1$$

Thus, $n \leq (D_1+2)(N+1)C\dim(X_{\text{NR}})$ is uniformly bounded. We conclude:

$$\text{Collapse}(N) \leq (N+1)C\dim(X_{\text{NR}}) \cdot \max\{2\ell - 1, D_1 + 2\}$$ □

Proposition 4.25. *There exists C such that for every d_1–geodesic γ in Σ with endpoints $x, y \in \Sigma^{(0)}$:*

$$\hat{d}(x,y) \leq d_{\widehat{\gamma}}(x,y) \leq C d_f(x,y) + C$$

Proof. Let τ be the quasiruler constant of Lemma 4.3. By Lemma 4.24, $C := \mathrm{Collapse}(2\tau) + 1 < \infty$.

Let $\gamma\colon [0,T] \to \Sigma^{(1)}$ be the arc length parameterization with $x := \gamma(0)$ and $y := \gamma(T)$. Let $\mathcal{V} := \{V_i\}_{1\le i\le n}$ be a longest fine chain separating x and y, with $x \in V_i^- \subset V_{i+1}^-$ for all i. Let t_i be the half-integral time such that $\gamma(t_i) \in V_i$, and set $t_0 := -1/2$ and $t_{n+1} := T + 1/2$. For each $0 \le i \le n$, let γ_i be the maximal edge-subpath of $\gamma \smallsetminus \mathcal{V}$ with endpoints $a_i := \gamma(t_i + 1/2)$ and $b_i := \gamma(t_{i+1} - 1/2)$.

For each i, $\{V_j\}_{1\le j\le i}$ is a fine chain separating x and a_i, so $d_f(x, a_i) \ge i$, and $\{V_j\}_{i+1\le j\le n}$ is a fine chain separating b_i and y, so $d_f(b_i, y) \ge n - i$. Applying Lemma 4.3 and Corollary 4.5 twice:

$$n = d_f(x,y) \ge d_f(x,a_i) + d_f(a_i,y) - \tau \ge d_f(x,a_i) + d_f(a_i,b_i) + d_f(b_i,y) - 2\tau \ge i + d_f(a_i,b_i) + n - i - 2\tau$$

Thus, $d_f(a_i, b_i) \le 2\tau$, which implies $d_{\widehat\gamma_i}(a_i, b_i) \le \mathrm{Collapse}(2\tau)$.

Since $\widehat\gamma_i$ is a subgraph of $\widehat\gamma$, this implies $d_{\widehat\gamma}(a_i, b_i) \le \mathrm{Collapse}(2\tau)$ for every i.

Also, for each $1 \le i \le n$ the vertices b_{i-1} and a_i are the endpoints of the edge of γ crossing V_i, so they are adjacent in $\Sigma^{(1)}$, so they are adjacent in $\widehat\gamma$.

Concatenating these edges with shortest paths in each $\widehat\gamma_i$, we obtain a path in $\widehat\gamma$ from x to y of length at most $(n+1)\,\mathrm{Collapse}(2\tau) + n$. Thus:

$$\hat d(x,y) \le d_{\widehat\gamma}(x,y) \le (n+1)\,\mathrm{Collapse}(2\tau) + n \le Cn + C = Cd_f(x,y) + C \qquad \square$$

Corollary 4.26 (Realizations of greedy paths are uniform $\hat d$–quasigeodesics)**.** *There exists C such that for every d_1–geodesic segment γ in $\Sigma^{(1)}$, any realization $\overline\gamma^*$ in $\widehat\Sigma$ of the greedy path γ^* in $\widehat\gamma$ is a (C,C)–quasigeodesic.*

Proof. Let C_0 be the constant of Proposition 4.25, and let $p_0, \dots, p_m$ be the vertex sequence of γ^*, which is the same as the $\Sigma^{(0)}$–vertex sequence of $\overline\gamma^*$. For $i < j$ let γ_{ij} be the subsegment of γ from p_i to p_j. By Lemma 4.23, the subpath of γ^* from p_i to p_j is a geodesic in $\widehat\gamma$. Moreover, the vertices of the greedy path are vertices of γ occurring in the same order as they do in γ, so they all belong to γ_{ij}. Since $\widehat\gamma_{ij}$ is naturally identified with a subgraph of $\widehat\gamma$, it follows that the subpath of γ^* from p_i to p_j is identified with γ^*_{ij}. Thus, $j - i = d_{\widehat\gamma_{ij}}(p_i, p_j)$. By Proposition 4.25, $d_{\widehat\gamma_{ij}}(p_i,p_j) \le C_0 d_f(p_i,p_j) + C_0$. Since $d_f \le \hat d$, by Lemma 4.21, conclude $j - i \le C_0\hat d(p_i,p_j) + C_0$. On the other hand, the subpath of $\overline\gamma^*$ from p_i to p_j has length $j - i$, so $j - i \ge \hat d(p_i, p_j)$.

We have shown that $i \mapsto p_i$ is a discrete (C_0, C_0)–quasigeodesic in $\widehat\Sigma$. Since every point on $\overline\gamma^*$ is within distance $1/2$ of some p_i, we can increase C_0 a uniformly bounded amount such that $\overline\gamma^*$ is a continuous (C,C)–quasigeodesic. $\square$

5. Extending the MCG analogy

We do not know if the Davis complex of a finite rank Coxeter system is a hierarchically hyperbolic space. However, there are similarities. In Section 5.1 we develop Coxeter group counterparts to some of the machinery from the mapping class groups/hierarchically hyperbolic space (MCG/HHS) settings. In the following two subsections we use these tools to prove Coxeter versions of two MCG/HHS theorems:

- $W \curvearrowright (\widehat\Sigma, \hat d)$ is an acylindrical action, Section 5.2.
- The stable subgroups of W can be characterized as those that are undistorted and purely loxodromic with respect to $W \curvearrowright (\widehat\Sigma, \hat d)$, Section 5.3.

The MCG/HHS tools (see [20, 22]) for which we obtain analogues include: one-sided distance formula, large link condition, bounded geodesic image, descent by induction on a notion of complexity, which for us is rank of a parabolic subgroup. We do not have analogues for hierarchy paths and the two-sided distance formula. For the two target applications we do not need the full power of those tools; where they would normally occur in an MCG/HHS proof we can replace them with arguments specific to walls and parabolic subgroups in Coxeter groups.

5.1. Parabolic projections and hierarchical features.

5.1.1. *Projections.* Recall from Section 2.4.2 that given a parabolic subcomplex Υ there is a gate projection $\mathfrak{p}_\Upsilon\colon \Sigma^{(0)} \to \Upsilon^{(0)}$, and it agrees with d_1–closest point projection.

Definition 5.1. Given a parabolic subcomplex Υ, define its projection pseudometric on $\Sigma^{(0)}$:

$$d_\Upsilon(x,y) := d_1(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$$

Now we want to make similar definitions of gate map and projection pseudometric directly for a nonspherical irreducible parabolic subgroup P. Recall from Proposition 2.13 and Corollary 2.16 that there exists a parabolic subgroup $P^\perp$ such that $Z_W(P) = P^\perp$ and $N_W(P) = P \times P^\perp$.

Definition 5.2. For a parabolic subgroup P, define:

$$\mathrm{Stab}^{-1}(P) := \{\text{parabolic subcomplexes } \Upsilon \mid \mathrm{Stab}(\Upsilon) = P\}$$

When P is irreducible and nonspherical, let $\Upsilon_{N(P)}$ denote the unique element of $\mathrm{Stab}^{-1}(N_W(P))$, recall Corollary 2.17.

Lemma 5.3. *Let P be a nonspherical irreducible parabolic subgroup of (W, S).*

1. *$P^\perp$ acts freely and transitively by left multiplication on $\mathrm{Stab}^{-1}(P)$, and the action induces isomorphisms between the corresponding subcomplexes.*
2. *Distinct elements of $\mathrm{Stab}^{-1}(P)$ are disjoint, and $\coprod_{\Upsilon \in \mathrm{Stab}^{-1}(P)} \Upsilon^{(0)} = \Upsilon^{(0)}_{N(P)}$.*
3. *For every $\Upsilon \in \mathrm{Stab}^{-1}(P)$, the composition of the inclusion $\Upsilon^{(0)} \hookrightarrow \Upsilon^{(0)}_{N(P)}$ with the quotient $\Upsilon^{(0)}_{N(P)} \twoheadrightarrow P^\perp \backslash \Upsilon^{(0)}_{N(P)}$ is a bijection.*

Proof. By Corollary 2.15, there is a unique $T \subset S$ such that P is conjugate to W_T. Fix $w_0 \in W$ such that $P = w_0 W_T w_0^{-1}$. Then $w_0 \Sigma_T \in \mathrm{Stab}^{-1}(P)$. By Corollary 2.16, $N_W(P) = P \times P^\perp$, with $P^\perp = w_0 W_{T^\perp} w_0^{-1}$, so $\Upsilon_{N(P)} = w_0 \Sigma_{T \sqcup T^\perp}$.

For (1), the action is well-defined, because if $z \in P^\perp$ and $\Upsilon \in \mathrm{Stab}^{-1}(P)$ then $\mathrm{Stab}(z\Upsilon) = z\,\mathrm{Stab}(\Upsilon) z^{-1} = zPz^{-1} = P$, since $z \in P^\perp = Z_W(P)$. Transitivity and the fact that the multiplication induces an isomorphism of subcomplexes is Lemma 2.48. Freeness follows because $P \cap P^\perp$ is trivial.

By (1), $\mathrm{Stab}^{-1}(P) = \{z w_0 \Sigma_T \mid z \in P^\perp\}$, so:

$$\bigcup_{\Upsilon \in \mathrm{Stab}^{-1}(P)} \Upsilon^{(0)} = P^\perp w_0 W_T = w_0 W_{T^\perp} W_T = w_0 W_{T \sqcup T^\perp} = \Upsilon^{(0)}_{N(P)}$$

Suppose $z, z' \in P^\perp$ and $x, y \in W_T$ satisfy $z w_0 x = z' w_0 y$. Then $P^\perp \ni (z')^{-1} z = w_0 y x^{-1} w_0^{-1} \in w_0 W_T w_0^{-1} = P$, so $z = z'$ and $x = y$. The former gives that two members of $\mathrm{Stab}^{-1}(P)$ with a common vertex agree, so the union written above is a disjoint union. Moreover, since members of $\mathrm{Stab}^{-1}(P)$ are full subcomplexes, this implies that distinct members of $\mathrm{Stab}^{-1}(P)$ are disjoint. The latter gives (3), since the $P^\perp$–orbit of a vertex $z w_0 x$ of $\Upsilon_{N(P)}$ is $\{z' w_0 x \mid z' \in P^\perp\}$, which meets $z' w_0 \Sigma_T$ of $\mathrm{Stab}^{-1}(P)$ in the vertex $z' w_0 x$. □

We define $\mathfrak{p}_P$ by fixing any $\Upsilon \in \mathrm{Stab}^{-1}(P)$ and setting:

$$\mathfrak{p}_P \colon \Sigma^{(0)} \to P^\perp \backslash \Upsilon^{(0)}_{N(P)} : x \mapsto P^\perp \mathfrak{p}_\Upsilon(x)$$

By Corollary 2.44 and Lemma 2.45, if $g \in P^\perp$ and $\Upsilon \in \mathrm{Stab}^{-1}(P)$ then $\mathfrak{p}_{g\Upsilon} = g\mathfrak{p}_\Upsilon$. Thus, for all $x \in \Sigma^{(0)}$ and $\Xi \in \mathrm{Stab}^{-1}(P)$, we have $\mathfrak{p}_P(x) = P^\perp \mathfrak{p}_\Upsilon(x) = P^\perp \mathfrak{p}_\Xi(x)$, so $\mathfrak{p}_P(x)$ is a well defined $P^\perp$–orbit of vertices in $\Upsilon^{(0)}_{N(P)}$, independent of the choice of $\Upsilon \in \mathrm{Stab}^{-1}(P)$.

Definition 5.4. Given a nonspherical irreducible parabolic subgroup P, define its projection pseudometric on $\Sigma^{(0)}$:

$$d_P(x, y) := d_1(\mathfrak{p}_P(x), \mathfrak{p}_P(y))$$

The definition of d_P is in terms of the minimal d_1–distance between the $P^\perp$–orbits $\mathfrak{p}_P(x)$ and $\mathfrak{p}_P(y)$. Since $\Upsilon_{N(P)}$ is a product and the $P^\perp$–action induces isomorphisms between P–fibers, an equivalent formulation is to choose $\Upsilon \in \mathrm{Stab}^{-1}(P)$, set $d_P = d_\Upsilon$, and observe that the result is independent of the choice of Υ. This shows that d_P actually is a pseudometric.

Definition 5.5. For any parabolic subcomplex Υ and $x, y \in \Sigma$:

$$\mathcal{M}_\Upsilon(x, y) := \{M \in \mathcal{M}(x, y) \mid M \text{ cuts } \Upsilon\} = \{\text{walls } M \mid M \text{ separates } x \text{ and } y \text{ and cuts } \Upsilon\}$$

Lemma 5.6. *For any parabolic subcomplex Υ and $x, y \in \Sigma^{(0)}$:*

$$\mathcal{M}_\Upsilon(x, y) = \mathcal{M}(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$$

In particular, this implies $d_\Upsilon(x, y) = |\mathcal{M}_\Upsilon(x, y)|$.

Proof. Corollary 2.43 says that a wall M separates x from $\mathfrak{p}_\Upsilon(x)$ if and only if it separates x from Υ, so for any wall M that cuts Υ, vertices x and $\mathfrak{p}_\Upsilon(x)$ are on the same side of M.

For $M \in \mathcal{M}_\Upsilon(x,y)$, suppose $x \in M^-$. Then $y \in M^+$ and M cuts Υ, so $\mathfrak{p}_\Upsilon(x) \in M^-$ and $\mathfrak{p}_\Upsilon(y) \in M^+$ so, $M \in \mathcal{M}(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$. Conversely, for $M \in \mathcal{M}(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$, suppose $\mathfrak{p}_\Upsilon(x) \in M^-$. Then $\mathfrak{p}_\Upsilon(y) \in M^+$, so M cuts Υ, so $x \in M^-$ and $y \in M^+$, so $M \in \mathcal{M}_\Upsilon(x,y)$. □

5.1.2. *Coned-off graphs of parabolic subgroups.* The following definition is very close to that of the *Artin complex* of [51]. It is also going to play a role similar to that of the *factored contact graph* [20, 2].

Definition 5.7 (Parabolic coned-off graph)**.** Given a parabolic subcomplex Υ, let $C(\Upsilon)$ be the graph obtained from $\Upsilon^{(1)}$ by coning off every proper parabolic subcomplex $\Xi \subsetneq \Upsilon$ by adding a vertex v_Ξ and an edge of length $1/2$ from v_Ξ to every vertex of $\Xi^{(0)}$.

Given a nonspherical irreducible parabolic subgroup P, define:

$$\widetilde{C(P)} := \coprod_{\Upsilon \in \mathrm{Stab}^{-1}(P)} C(\Upsilon)$$

Then $P^\perp$ acts on $\widetilde{C(P)}$ by left-multiplication, with $z \in P^\perp$ taking $v \in C(\Upsilon)$ to $zv \in zC(\Upsilon) = C(z\Upsilon)$. Define $C(P)$ to be the quotient of $\widetilde{C(P)}$ by the left $P^\perp$–action.

Remark. In contrast to the curve complex/HHS setting, the graphs $C(\Upsilon)$ and $C(P)$ **need not be hyperbolic**. Indeed, suppose that $\Upsilon = \Sigma_T$ for $T \subset S$ irreducible higher rank affine. Then W_T has no proper nonspherical parabolics. Since special spherical subgroups have uniformly bounded diameter, $\Upsilon^{(1)} \hookrightarrow C(\Upsilon)$ is a quasiisometry. Thus, $C(\Upsilon)$ is quasiisometric to $\mathbb{Z}^{|T|-1}$.

Lemma 5.8. *If P is a nonspherical irreducible parabolic subgroup and $\Upsilon \in \mathrm{Stab}^{-1}(P)$, then the composition $C(\Upsilon) \hookrightarrow \widetilde{C(P)} \to C(P)$ is a graph isomorphism. Moreover, for any two such parabolic subcomplexes Υ and Υ', the resulting isomorphism $C(\Upsilon) \to C(P) \to C(\Upsilon')$ is left-multiplication by the unique $z \in P^\perp$ such that $z\Upsilon = \Upsilon'$.*

There is a canonical assignment of a proper parabolic subgroup of P to each cone vertex of $C(P)$ by taking the common stabilizer of the corresponding $P^\perp$–orbit of cone vertices of the $C(\Upsilon)$.

Proof. This is essentially the same argument as in the construction of $\mathfrak{p}_P$ and d_P in the previous subsection. The "ordinary" vertices of $C(P)$ can be identified with $P^\perp$–orbits of pairs (Υ, v), where $\Upsilon \in \mathrm{Stab}^{-1}(P)$ and $v \in \Upsilon^{(0)}$. The "cone" vertices of $C(P)$, which are distinguished by the fact that all of their incident edges have length $1/2$, can be identified with $P^\perp$–orbits of pairs (Υ, Ξ), where $\Upsilon \in \mathrm{Stab}^{-1}(P)$ and Ξ is a parabolic subcomplex contained in Υ.

Note that if $z \in P^\perp$ and (Υ, Ξ) is a parabolic subcomplex pair with $\Upsilon \in \mathrm{Stab}^{-1}(P)$ then $\mathrm{Stab}(z\Xi) = z\,\mathrm{Stab}(\Xi)z^{-1} = \mathrm{Stab}(\Xi)$, so we can associate the subgroup $\mathrm{Stab}(\Xi)$ to the cone vertex of $C(P)$ corresponding to the $P^\perp$–orbit of (Υ, Ξ), independent of the choice of representative pair. □

Just as we defined projection pseudometrics d_Υ and d_P in Definition 5.1 and Definition 5.4, we now define their $C(\Upsilon)$ and $C(P)$ analogues:

Definition 5.9. If Υ is a parabolic subcomplex, let $d_{C(\Upsilon)}$ be the distance in the graph $C(\Upsilon)$. Define a projection pseudometric on $\Sigma^{(0)}$:

$$\delta_\Upsilon(x,y) := d_{C(\Upsilon)}(\mathfrak{p}_\Upsilon(x), \mathfrak{p}_\Upsilon(y))$$

Here we implicitly compose the gate map $\mathfrak{p}_\Upsilon \colon \Sigma^{(0)} \to \Upsilon^{(0)}$ with the inclusion $\Upsilon^{(0)} \to C(\Upsilon)^{(0)}$.

Similarly, if P is a nonspherical irreducible parabolic subgroup, define:

$$\delta_P(x,y) := d_{C(P)}(\mathfrak{p}_P(x), \mathfrak{p}_P(y))$$

An equivalent formulation of δ_P is to choose $\Upsilon \in \mathrm{Stab}^{-1}(P)$, take $\delta_P := \delta_\Upsilon$, and observe that the resulting pseudometric is independent of the choice of Υ.

Lemma 5.10. *If P is a nonspherical irreducible parabolic subgroup and Υ is a parabolic subcomplex:*

$$\mathrm{diam}_{\delta_P}(\Upsilon) > 1 \implies P \leq \mathrm{Stab}(\Upsilon)$$

Proof. Choose any parabolic subcomplex $\Xi \in \mathrm{Stab}^{-1}(P)$. By Lemma 2.46, $\mathfrak{p}_\Xi(\Upsilon) \subset \Xi^{(0)}$ spans a parabolic subcomplex of Ξ, so if $\mathfrak{p}_\Xi(\Upsilon) \subsetneq \Xi^{(0)}$ then it spans a proper parabolic subcomplex of Ξ, which is coned off in $C(\Xi)$, so $\mathfrak{p}_\Xi(\Upsilon)$ has δ_P–diameter at most 1. Thus, $\mathfrak{p}_\Xi(\Upsilon) \subsetneq \Xi^{(0)}$ implies $\mathrm{diam}_{\delta_P}(\Upsilon) \leq 1$, or, contrapositively, $\mathrm{diam}_{\delta_P}(\Upsilon) > 1$ implies $\mathfrak{p}_\Xi(\Upsilon) = \Xi^{(0)}$. This gives, by Corollary 2.47 and Lemma 2.6, that:

$$P = \mathrm{Stab}(\Xi) = \mathrm{Stab}(\mathfrak{p}_\Xi(\Upsilon)) = \mathrm{Stab}(\mathfrak{p}_\Upsilon(\Xi)) \leq \mathrm{Stab}(\Upsilon)$$ □

Definition 5.11. A nonspherical irreducible parabolic subgroup is *pre-wide* if its normalizer is wide.

Lemma 5.12. *Every nonspherical irreducible parabolic subgroup of the stabilizer of a wide parabolic subcomplex is pre-wide.*

Proof. Let Υ be a wide parabolic subcomplex, and let $Q_1 \times \cdots \times Q_n \times F$ be the canonical decomposition of $\mathrm{Stab}(\Upsilon)$ into irreducible nonspherical factors Q_i and a spherical factor F. Any nonspherical irreducible parabolic subgroup P of $\mathrm{Stab}(\Upsilon)$ is a reflection subgroup, so Lemma 2.12 says it splits as a direct product of its intersections with the factors of $\mathrm{Stab}(\Upsilon)$. Since it is nonspherical and irreducible, this implies there exists i such that $P \leq Q_i$. Assume, by renumbering, that $i = 1$. Since $\mathrm{Stab}(\Upsilon)$ is wide, either $n > 1$ or $n = 1$ and Q_1 is higher rank affine. In the former case, $Q_2 \leq Q_1^\perp \leq P^\perp$, so $P^\perp$ is nonspherical, so $N_W(P) = P \times P^\perp$ is wide. In the latter case, since proper parabolic subgroups of an irreducible affine Coxeter group are spherical, $P = Q_1$, so P is irreducible higher rank affine, which also implies $N_W(P)$ is wide. □

5.1.3. *Large links and bounded geodesic image for wide parabolics.* Recall $\Delta := \max_{\text{spherical } T \subset S} \mathrm{diam}_{d_1} W_T$.

Lemma 5.13. *For any wide special subgroups P and Q and element $w \in W$, if $|P \cap wQw^{-1}| = \infty$, then $\hat{d}(\mathbb{1}, w) \leq \Delta + 2$.*

Proof. Suppose $P \cap wQw^{-1}$ is infinite. Intersections of parabolic subgroups are parabolic, and an infinite parabolic has at least one nonspherical factor in its canonical decomposition as a product of irreducible factors. Thus there are nonspherical irreducible subsets $T, U \subset S$ and elements $p \in P$ and $q \in Q$ such that $W_T \leq P$, $W_U \leq Q$, and $pW_Tp^{-1} = wqW_Uq^{-1}w^{-1}$. By Corollary 2.15, $T = U$, so $h := p^{-1}wq \in N_W(W_T)$. By Proposition 2.13 (2), $N_W(W_T) = W_T \times W_{T^\perp}$, so $h = ab$ for some $a \in W_T$ and $b \in W_{T^\perp}$, and then $w = pabq^{-1}$.

We now bound the $\hat{d}$–length of these factors uniformly. Since $W_T \subseteq P$ and P is wide, every vertex of Σ_T has $\hat{d}$–distance at most 1 from $\mathbb{1}$, so $\hat{d}(\mathbb{1}, pa) \leq 1$. Similarly, $\hat{d}(\mathbb{1}, q) \leq 1$. If $W_{T^\perp}$ is finite then $\hat{d}(\mathbb{1}, b) \leq d_1(\mathbb{1}, b) \leq \Delta$, so $\hat{d}(\mathbb{1}, pabq^{-1}) \leq \Delta + 2$. If $W_{T^\perp}$ is not finite then $W_T \times W_{T^\perp}$ is wide, so $\hat{d}(\mathbb{1}, ab) \leq 1$ and $\hat{d}(\mathbb{1}, pabq^{-1}) \leq 3 \leq \Delta + 2$. □

Corollary 5.14. *If Υ and Ξ are wide parabolic subcomplexes of Σ with corresponding cone vertices v_Υ and v_Ξ in $\widehat{\Sigma}$, and $|\mathrm{Stab}(\Upsilon) \cap \mathrm{Stab}(\Xi)| = \infty$, then $\hat{d}(v_\Upsilon, v_\Xi) \leq \Delta + 3$.*

Proof. Write $\Upsilon = a\Sigma_T$ and $\Xi = b\Sigma_U$, where T and U are wide subsets of S. We have $\mathrm{Stab}(\Upsilon) = aW_Ta^{-1}$ and $\mathrm{Stab}(\Xi) = bW_Ub^{-1}$. Thus:

$$|\mathrm{Stab}(\Upsilon) \cap \mathrm{Stab}(\Xi)| = \infty \Longrightarrow |W_T \cap a^{-1}bW_Ub^{-1}a| = \infty$$

By Lemma 5.13 $\hat{d}(a, b) = \hat{d}(\mathbb{1}, a^{-1}b) \leq \Delta + 2$. Then, by the triangle inequality:

$$\hat{d}(v_\Upsilon, v_\Xi) \leq \hat{d}(v_\Upsilon, a) + \hat{d}(a, b) + \hat{d}(b, v_\Xi) \leq \Delta + 3$$ □

Lemma 5.15. *Let Υ be a wide parabolic subcomplex, and let $\mathcal{V}$ be a chain of walls cutting Υ with $|\mathcal{V}| \geq 2$. There exists a biinfinite chain $\mathcal{H}$ cutting Υ such that $(\mathcal{V}, \mathcal{H})$ is a grid.*

Proof. See the proof of Lemma 3.24. □

Lemma 5.16. *Let $x, y \in \Sigma^{(0)}$, and let γ be a geodesic in $\widehat{\Sigma}$ from x to y. Denote by $\mathcal{P}(\gamma)$ the collection of wide parabolic subcomplexes whose cone vertices occur on γ. For any parabolic subcomplex Υ satisfying $d_\Upsilon(x, y) > \dim(X_{\mathbb{NR}}) \cdot \hat{d}(x, y)$, there exists $\Xi \in \mathcal{P}(\gamma)$ such that there is a pair of disjoint walls M and M' that cut both Υ and Ξ. This implies, in particular, that $|\mathrm{Stab}(\Upsilon) \cap \mathrm{Stab}(\Xi)| = \infty$.*

Proof. Let $x = z_0, z_1, \dots, z_m = y$ be the sequence of vertices of $\Sigma^{(0)}$ occurring in γ, so that at each step $z_{i-1} - z_i$ either z_{i-1} and z_i are adjacent in γ and in $\Sigma^{(1)}$, or γ contains a cone shortcut $z_{i-1} - v_\Xi - z_i$, where Ξ is a wide parabolic subcomplex containing z_{i-1} and z_i.

Since ordinary edges have length 1 and cone edges have length 1/2, we have:

$$|\mathcal{P}(\gamma)| \leq \hat{d}(x,y) = m$$

By Lemma 5.6, $|\mathcal{M}_\Upsilon(x,y)| = d_\Upsilon(x,y)$. Dilworth's theorem implies $\mathcal{M}_\Upsilon(x,y)$ contains a chain $\mathcal{V}$ of length at least $d_\Upsilon(x,y)/\dim(X_{\text{NR}})$. Thus, supposing $d_\Upsilon(x,y) > \dim(X_{\text{NR}}) \cdot \hat{d}(x,y)$, we have $|\mathcal{V}| > m$.

Every wall $M \in \mathcal{V}$ separates x from y, so some step of γ has endpoints in opposite halfspaces of M. Assign M to one such step. An ordinary Davis edge crosses exactly one wall, so the "excess" $(|\mathcal{V}| - m)$–many walls come from cone shortcut steps. In particular, there is some cone shortcut step $z_{i-1} - v_\Xi - z_i$ such that z_{i-1} and z_i are separated by distinct walls $M, M' \in \mathcal{V}$. Because $z_{i-1}, z_i \in \Xi$, both walls cut Ξ. By construction, both walls also cut Υ. Furthermore, they are disjoint, since they came from the chain $\mathcal{V}$. □

Proposition 5.17 (Intersection large links for wide parabolics)**.** *There are constants λ, D_0, and K_0 such that for every pair of vertices $x, y \in \Sigma^{(0)}$ there exists a collection of wide parabolic subcomplexes $\mathcal{P}(x,y)$ satisfying the following conditions:*

- $|\mathcal{P}(x,y)| \leq \lambda \hat{d}(x,y) + \lambda^2$.
- *For all $\Xi \in \mathcal{P}(x,y)$ and any geodesic α in $\widehat{\Sigma}$ from x to y, $\hat{d}(v_\Xi, \alpha) \leq D_0$.*
- *If Υ is a wide parabolic subcomplex and $d_\Upsilon(x,y) > K_0$, then there exists $\Xi \in \mathcal{P}(x,y)$ such that a pair of disjoint walls cut both Υ and Ξ. In particular, $|\operatorname{Stab}(\Upsilon) \cap \operatorname{Stab}(\Xi)| = \infty$.*

The name of Proposition 5.17 is meant to invoke [20, Proposition 9.4 "large link lemma"], where the analogous conclusion would be that Υ is parallel into Ξ. Our proposition is intentionally coarser: wide parabolic subcomplexes are not the analogues of HHS domains. Indeed, even in RACGs there are examples, see Example 5.18, containing maximal wide parabolic subcomplexes whose intersection is nonspherical but not wide, so their unbounded overlap is not captured by a lower "domain".

Example 5.18. Take the Coxeter diagram to be a hexagon with vertices 1,2,3,4,5,6. Add one more vertex 0 adjacent to 1. Label all edges ∞. Consider the complement of $\{2,5\}$ and the complement of $\{3,6\}$. These are maximal wide, and their intersection $\{0,1,4\}$ is nonspherical and nonwide, since $W_{\{0,1,4\}} \cong \mathcal{D}_\infty \times \mathbb{Z}/2\mathbb{Z}$.

Proof of Proposition 5.17. Let γ be a d_1–geodesic in $\Sigma^{(1)}$ from x to y, and let γ^* be the greedy path in the shortcut graph $\widehat{\gamma}$ of γ, and let $\overline{\gamma}^*$ be a choice of realization of γ^* in $\widehat{\Sigma}$. Recall that this means that for each shortcut edge $\gamma^*(i) - \gamma^*(i+1)$, we choose some wide parabolic subcomplex Ξ containing those two vertices, and $\overline{\gamma}^*$ contains a cone shortcut $\gamma^*(i) - v_\Xi - \gamma^*(i+1)$. Let $\mathcal{P}(x,y) = \mathcal{P}(\overline{\gamma}^*)$ be the collection of wide parabolic subcomplexes Ξ so chosen.

It follows from the construction in the proof of Lemma 4.23 that the order of vertices in γ^* is the same as their order in γ, hence the order of $\Sigma^{(0)}$ vertices of $\overline{\gamma}^*$ is the same as it is in γ.

By Corollary 4.26, there is a λ such that $\overline{\gamma}^*$ is a uniform (λ,λ)–quasigeodesic from x to y in $\widehat{\Sigma}$, so:

$$|\mathcal{P}(x,y)| \leq |\overline{\gamma}^*| \leq \lambda \hat{d}(x,y) + \lambda^2$$

Since $(\widehat{\Sigma}, \hat{d})$ is δ–hyperbolic, the quasigeodesic $\overline{\gamma}^*$ is contained in a fixed radius D_0 neighborhood of any $\hat{d}$–geodesic α from x to y in $\widehat{\Sigma}$, where the radius D_0 depends only on λ and δ.

Let C and $\mathfrak{R}$ be the constants of Proposition 3.27. Let $K = K(\mathfrak{R})$ be such that whenever $d_2(u,v) \leq \mathfrak{R} + 1/2 + \operatorname{diam}(\text{chamber})$, then $d_1(u,v) \leq K$. Since $\hat{d} \leq d_1$, this also gives $\hat{d}(u,v) \leq K$. We set $D := \lambda(2K+3+\lambda)$. By enlarging λ, if necessary, we may assume $D \geq (2\ell - 1)/\dim(X_{\text{NR}})$. Finally, let:

$$K_0 := CD \dim(X_{\text{NR}})^2$$

Now let Υ be a wide parabolic subcomplex such that $d_\Upsilon(x,y) > K_0$. By Lemma 5.6, $d_\Upsilon(x,y) = |\mathcal{M}_\Upsilon(x,y)|$. By Dilworth's Theorem, $\mathcal{M}_\Upsilon(x,y)$ contains a chain $\mathcal{V}$ with:

$$|\mathcal{V}| \geq \frac{d_\Upsilon(x,y)}{\dim(X_{\text{NR}})} > \frac{K_0}{\dim(X_{\text{NR}})} = CD \dim(X_{\text{NR}}) \geq C(2\ell - 1)$$

By Lemma 5.15 there is a chain $\mathcal{H}$ of 2ℓ–many walls cutting Υ such that $(\mathcal{V}, \mathcal{H})$ is a grid. Proposition 3.27 applied to that grid produces a wide parabolic subcomplex Υ' and a convex subchain $\mathcal{V}' \subset \mathcal{V}$ with $|\mathcal{V}'| \geq \frac{|\mathcal{V}|}{C}$,

and a subpath γ' of γ between the extremal walls of $\mathcal{V}'$ lying within $\bar{\mathcal{N}}_{\mathfrak{R}}(\Upsilon')$. Let a be the last vertex of γ before γ', and let b be the first vertex of γ after γ', so that $d_2(a,\Upsilon') \le \mathfrak{R}+1/2$, $d_2(b,\Upsilon') \le \mathfrak{R}+1/2$, and every wall of $\mathcal{V}'$ separates a and b. Let a' and b' be vertices of Υ' contained in a common chamber with $\pi_{\Upsilon'}(a)$ and $\pi_{\Upsilon'}(b)$, respectively. Then:

$$\hat{d}(a,b) \le \hat{d}(a,a') + \hat{d}(a',v_{\Upsilon'}) + \hat{d}(v_{\Upsilon'},b') + \hat{d}(b',b) \le 2K+1$$

Let $\gamma^*(l)$ be the vertex of γ^* closest to a in $\hat{\gamma}$ lying on the x–side of a in γ. Similarly, let $\gamma^*(r)$ be the vertex of γ^* closest to b in $\hat{\gamma}$ lying on the y–side of b along γ. We have $\hat{d}(a,\overline{\gamma}^*(l)) \le 1$ and $\hat{d}(b,\overline{\gamma}^*(r)) \le 1$, so $\hat{d}(\overline{\gamma}^*(l),\overline{\gamma}^*(r)) \le 2K+3$. In particular, the quasigeodesic condition gives:

$$r - l \le \lambda(2K+3) + \lambda^2 = D$$

Thus, the segment of $\overline{\gamma}^*$ from $\overline{\gamma}^*(l)$ to $\overline{\gamma}^*(r)$ has $\hat{d}$–length at most D.

Since every wall of $\mathcal{V}'$ separates a and b, our choices of $\overline{\gamma}^*(l)$ and $\overline{\gamma}^*(r)$ imply that every wall of $\mathcal{V}'$ separates $\overline{\gamma}^*(l)$ and $\overline{\gamma}^*(r)$ in Σ. Since $|\mathcal{V}'| > D\dim(X_{\mathrm{NR}})$ and $d_{\hat{\gamma}}(\gamma^*(l),\gamma^*(r)) = r-l \le D$, there is some $s \in \{l,\dots r-1\}$ such that there are more than $\dim(X_{\mathrm{NR}})$–many walls of $\mathcal{V}'$ separating $\overline{\gamma}^*(s)$ and $\overline{\gamma}^*(s+1)$. Since $\mathcal{V}' \subset \mathcal{V}$, those walls cut Υ, so Lemma 5.6 yields $d_\Upsilon(\overline{\gamma}^*(s),\overline{\gamma}^*(s+1)) > \dim(X_{\mathrm{NR}})$.

Note that since $\overline{\gamma}^*(s)$ and $\overline{\gamma}^*(s+1)$ are separated by multiple walls, they are not adjacent in $\Sigma^{(1)}$, so their adjacency in $\hat{\gamma}$ is due to a shortcut edge corresponding to one of our chosen wide parabolic subcomplexes $\Xi \in \mathcal{P}(x,y)$. Thus:

$$d_\Upsilon(\overline{\gamma}^*(s),\overline{\gamma}^*(s+1)) > \dim(X_{\mathrm{NR}}) = \dim(X_{\mathrm{NR}})\cdot 1 = \dim(X_{\mathrm{NR}})\cdot \hat{d}(\overline{\gamma}^*(s),\overline{\gamma}^*(s+1))$$

Since v_Ξ is the only cone vertex of the $\hat{d}$–geodesic segment $\overline{\gamma}^*(s) - v_\Xi - \overline{\gamma}^*(s+1)$, Lemma 5.16 implies there is a pair of disjoint walls cutting both Υ and Ξ, so $\mathrm{Stab}(\Upsilon)\cap\mathrm{Stab}(\Xi)$ is infinite. □

Theorem 5.19 (Bounded Geodesic Image)**.** *Let D_0 and K_0 be the constants of Proposition 5.17. Define $B_0 := D_0 + \Delta + 3$. Let γ be a $\hat{d}$–geodesic between $x,y \in \Sigma^{(0)}$, and let Υ be a wide parabolic subcomplex.*

$$\hat{d}(v_\Upsilon,\gamma) > B_0 \implies d_\Upsilon(x,y) \le K_0$$

Proof. By Proposition 5.17, if $d_\Upsilon(x,y) > K_0$, then there exist $\Xi \in \mathcal{P}(x,y)$ with $|\mathrm{Stab}(\Upsilon)\cap\mathrm{Stab}(\Xi)| = \infty$. Corollary 5.14 together with the same proposition yields:

$$\hat{d}(v_\Upsilon,\gamma) \le \hat{d}(v_\Upsilon,v_\Xi) + \hat{d}(v_\Xi,\gamma) \le \Delta + 3 + D_0 = B_0$$ □

Lemma 5.20 (One-sided distance formula)**.** *There is a constant C such that for all $x,y \in \Sigma^{(0)}$:*

$$d_1(x,y) \le C(1+\hat{d}(x,y))\cdot\max\{1,\sup_\Upsilon d_\Upsilon(x,y)\}$$

where the supremum is taken over wide parabolic subcomplexes Υ.

Proof. It suffices to take C to be the constant of Proposition 4.25.

Let γ be a d_1–geodesic in $\Sigma^{(1)}$ from x to y, and let $x = z_0,\dots,z_{|\gamma^*|} = y$ be the vertex sequence of the greedy path γ^*. By Proposition 4.25 and Lemma 4.21, $|\overline{\gamma}^*| \le C\hat{d}(x,y) + C$.

If the step from z_i to z_{i+1} in $\overline{\gamma}^*$ is an edge of $\Sigma^{(1)}$ then $d_1(z_i,z_{i+1}) = 1$. Otherwise it is a cone path shortcut, so z_i and z_{i+1} are contained in a common wide parabolic subcomplex Υ. By d_1–convexity, the subsegment γ_i of γ between z_i and z_{i+1} is a d_1–geodesic in Υ. By definition, $d_\Upsilon(x,y)$ is the d_1–distance between the gate projections $\mathfrak{p}_\Upsilon(x)$ and $\mathfrak{p}_\Upsilon(y)$ of x and y to Υ. Now:

$$\begin{aligned} d_1(x,z_i)+|\gamma_i| + d_1(z_{i+1},y) &= |\gamma| = d_1(x,y) \\ &\le d_1(x,\mathfrak{p}_\Upsilon(x)) + d_1(\mathfrak{p}_\Upsilon(x),\mathfrak{p}_\Upsilon(y)) + d_1(\mathfrak{p}_\Upsilon(y),y)\end{aligned}$$

Since the gate map agrees with d_1–closest point projection, this gives:

$$0 \le d_1(x,z_i) - d_1(x,\mathfrak{p}_\Upsilon(x)) + d_1(z_{i+1},y) - d_1(\mathfrak{p}_\Upsilon(y),y) \le d_1(\mathfrak{p}_\Upsilon(x),\mathfrak{p}_\Upsilon(y)) - |\gamma_i|$$

Thus, $|\gamma_i| \le d_1(\mathfrak{p}_\Upsilon(x),\mathfrak{p}_\Upsilon(y)) = d_\Upsilon(x,y)$. The claim follows. □

5.1.4. *Pre-wide witnesses for large projections.*

Lemma 5.21. *Let $P = W_T$ be nonspherical and irreducible, and take $\Upsilon \in \mathrm{Stab}^{-1}(P)$. Given $u, v \in \Upsilon^{(0)}$, let $m := \delta_P(u, v)$. The element $w = u^{-1}v$ admits a factorization $w = h_1 \cdots h_m$, where $h_i \in W_{J_i}$ for $J_i \subsetneq T$. Moreover, this factorization can be chosen so that each h_i is a reduced word in the corresponding W_{J_i} and $h_1 \cdots h_m$ is a reduced word in W.*

Proof. Let $u = z_0, z_1, \ldots, z_m = v$ be the $\Upsilon^{(0)}$–vertices along a geodesic in $C(\Upsilon)$. Each step z_{i-1} to z_i joins vertices lying in a common proper parabolic subcomplex of Υ, say $z_{i-1}, z_i \in a_i W_{J_i}$ for some $J_i \subsetneq T$. Thus $h_i = z_{i-1}^{-1} z_i \in W_{J_i}$. This is true even if z_{i-1} and z_i are adjacent in $\Sigma^{(1)}$, since then their difference is in S, so J_i can be taken to be a singleton, which is a proper subset of T, since P nonspherical implies $|T| \geq 2$.

Among $C(\Upsilon)$–geodesics from u to v, choose one that minimizes $\sum_{i=1}^m |h_i|$. Choose a reduced word for each h_i. If their concatenation was not reduced, then the Coxeter deletion condition gives two letters whose deletion leaves a word representing the same element w. Recall that a reduced word representing an element of W_J only uses letters in J; therefore, the remaining letters in the i-th block still belong to J_i, and hence represent an element of W_{J_i}. Thus, after further reducing each remaining W_{J_i}–word, either some block becomes trivial, contradicting the definition of m, or all m factors remain nontrivial, but the total word length decreases, contradicting minimality of the sums of the word lengths. □

Lemma 5.22. *If P is a nonspherical irreducible parabolic subgroup and $x, y \in \Sigma^{(0)}$ with $d_P(x, y) > \Delta \delta_P(x, y)$ then there is a nonspherical irreducible parabolic subgroup $Q \lneq P$ such that:*

$$d_Q(x, y) \geq \frac{\frac{d_P(x,y)}{\delta_P(x,y)} - \Delta}{\lfloor (\mathrm{rank}(P) - 1)/2 \rfloor}$$

Note that if $\mathrm{rank}(P) = 2$ then $d_P = \delta_P$, so the hypothesis is impossible to satisfy, so there is no division-by-zero problem.

Proof. Up to the W–action, we may assume $P = W_T$ and $\Upsilon = \Sigma_T$ for some $T \subset S$ with $|T| = r := \mathrm{rank}(P)$. Let $m := \delta_P(x, y)$. Let $u := \mathfrak{p}_\Upsilon(x)$ and $v := \mathfrak{p}_\Upsilon(y)$. By Lemma 5.21, the element $w = u^{-1}v$ admits a factorization $w = h_1 \cdots h_m$ satisfying $h_i \in W_{J_i}$ for $J_i \subsetneq T$ and:

$$d_P(x, y) = d_P(u, v) = \sum_{i=1}^m |h_i|$$

For at least one i the word h_i has at least average length, $|h_i| \geq d_P(x, y)/m > \Delta$. Take the canonical decomposition of W_{J_i} into a product of nonspherical irreducible factors $W_{J_{i,j}}$ and spherical factor F_i:

$$W_{J_i} = W_{J_{i,1}} \times W_{J_{i,2}} \times \cdots \times W_{J_{i,k}} \times F_i$$

There must be at least one nonspherical factor, since $|h_i| > \Delta$ implies $h_i \notin F_i$. Since $2 \leq |J_i| \lneq |T| = r$, and since the irreducible nonspherical components of J_i are disjoint sets containing at least two vertices each, the number of non-spherical components is $0 < k \leq \lfloor \frac{r-1}{2} \rfloor$.

Since different factors commute, we can further factorize:

$$h_i = h_{i,1} h_{i,2} \cdots h_{i,k} \cdot f_i, \quad \text{where } h_{i,j} \in W_{J_{i,j}},\ f_i \in F_i, \text{ and } |h_i| = \sum_{j=1}^k |h_{i,j}| + |f_i|$$

Thus, there is at least one j such that:

$$|h_{i,j}| \geq \frac{|h_i| - |f_i|}{k} \geq \frac{d_P(x, y)/\delta_P(x, y) - \Delta}{\lfloor (r-1)/2 \rfloor}$$

Because the full concatenated expression for w is reduced, we may commute the component words within h_i and obtain a reduced word for w in which a reduced expression for $h_{i,j}$ occurs as a contiguous subword. The corresponding subpath of the combinatorial geodesic lies in a parabolic subcomplex Ξ whose stabilizer Q is conjugate to $W_{J_{i,j}}$. Moreover, the prefix in w preceding this subword is a word in T, so $\Xi \subsetneq \Upsilon$ and $Q \lneq P$. Every wall crossed by this subpath separates the gates of x and y in Υ, hence separates x and y and cuts Ξ. Thus:

$$d_Q(x, y) \geq |h_{i,j}| \geq \frac{d_P(x, y)/\delta_P(x, y) - \Delta}{\lfloor (r-1)/2 \rfloor}$$

□

Proposition 5.23 (Pre-wide witness)**.** *There is a function $M\colon \mathbb{N}^{|S|-1} \to \mathbb{N}$ such that for all $x, y \in \Sigma^{(0)}$ and all wide parabolic subcomplexes Υ, if $d_\Upsilon(x,y) > M(c_2, \dots, c_{|S|})$, then there exists a pre-wide $P \leq \mathrm{Stab}(\Upsilon)$ such that $\delta_P(x,y) > c_{\mathrm{rank}(P)}$.*

Proof. We will choose $M'_2 < M'_3 < \ldots < M'_{|S|}$ recursively. Set $M'_1 := 0$. Then, for $2 \leq r \leq |S|$, choose:

$$L_r := \Delta + \lfloor \frac{r-1}{2} \rfloor M'_{r-1} \qquad M'_r := c_r L_r \tag{18}$$

We first prove a statement for irreducible parabolics by induction on rank.

Claim 5.23.1. Suppose that P is a nonspherical irreducible parabolic subgroup with $d_P(x,y) > M'_{\mathrm{rank}(P)}$. Then either $\delta_P(x,y) > c_{\mathrm{rank}(P)}$ or there exists a nonspherical irreducible parabolic $Q \lneq P$ such that $d_Q(x,y) > M'_{\mathrm{rank}(Q)}$.

Proof of Claim: Let $r := \mathrm{rank}(P)$, and suppose $\delta_P(x,y) \leq c_r$. Then $d_P(x,y) > c_r L_r \geq \delta_P(x,y)\Delta$, so Lemma 5.22 gives a nonspherical irreducible parabolic $Q \lneq P$ with:

$$d_Q(x,y) \geq \frac{\frac{d_P(x,y)}{\delta_P(x,y)} - \Delta}{\lfloor (r-1)/2 \rfloor} > \frac{\frac{M'_r}{c_r} - \Delta}{\lfloor (r-1)/2 \rfloor} = M'_{r-1} \geq M'_{\mathrm{rank}(Q)}$$

This proves the claim. ◊

Take the canonical decomposition $\mathrm{Stab}(\Upsilon) = P_1 \times \cdots \times P_k \times F$ as a product of nonspherical irreducible factors and a spherical factor. Since gate maps to nested parabolic subcomplexes compose, Corollary 2.42, gate distance splits as the sum of the component distances, while the spherical contribution is at most Δ. Therefore, setting $M(c_2, \dots, c_{|S|}) := \Delta + \lfloor \frac{|S|}{2} \rfloor M'_{|S|}$ guarantees that whenever $d_\Upsilon(x,y) > M(c_2, \dots, c_{|S|})$, then at least one of the $k \leq \lfloor \frac{|S|}{2} \rfloor$ many P_i will have $d_{P_i}(x,y) > M'_{\mathrm{rank}(P_i)}$. Applying the claim by inducting on $\mathrm{rank}(P_i)$, we see that there is an irreducible nonspherical $P \leq P_i$ such that $\delta_P(x,y) > c_{\mathrm{rank}(P)}$. Finally, P is pre-wide by Lemma 5.12. □

5.1.5. *Assigning cone vertices to pre-wide parabolics.*

Lemma 5.24. *There is a map ρ from pre-wide parabolics to cone vertices of $\widehat{\Sigma}$ defined by sending a pre-wide parabolic P to the cone vertex corresponding to the unique parabolic subcomplex $\Upsilon_{N(P)} \in \mathrm{Stab}^{-1}(N_W(P))$. The map has the following properties:*

1. *$g\rho(P) = \rho(gPg^{-1})$*
2. *If $P \leq Q$ are pre-wide parabolic subgroups, then $\hat{d}(\rho(P), \rho(Q)) \leq 1$.*
3. *If $P \leq Q$ are pre-wide parabolic subgroups, then for all $x, y \in \Sigma^{(0)}$ and $\Upsilon_Q \in \mathrm{Stab}^{-1}(Q)$, we have $\delta_P(x,y) = \delta_P(\mathfrak{p}_{\Upsilon_Q}(x), \mathfrak{p}_{\Upsilon_Q}(y))$.*
4. *For all $x, y \in \Sigma^{(0)}$, $\delta_P(x,y) \leq d_{\Upsilon_{N(P)}}(x,y)$.*

Proof. Item (1) is clear.

For Item (2), for any $\Upsilon_Q \in \mathrm{Stab}^{-1}(Q)$, choose $\Upsilon_P \in \mathrm{Stab}^{-1}(P)$ with $\Upsilon_P \subset \Upsilon_Q$. Then $\Upsilon_P \subset \Upsilon_{N(P)} \cap \Upsilon_{N(Q)}$. In particular, every vertex of Υ_P is a $\Sigma^{(0)}$ vertex at distance $1/2$ from both the cone vertex for $\Upsilon_{N(P)}$, which is $\rho(P)$, and the cone vertex for $\Upsilon_{N(Q)}$, which is $\rho(Q)$. Thus, $\hat{d}(\rho(P), \rho(Q)) \leq 1$.

Continuing with the same choices, for Item (3), apply Corollary 2.42:

$$\begin{aligned}
\delta_P(x,y) &= d_{C(\Upsilon_P)}(\mathfrak{p}_{\Upsilon_P}(x), \mathfrak{p}_{\Upsilon_P}(y)) \\
&= d_{C(\Upsilon_P)}(\mathfrak{p}_{\Upsilon_P} \circ \mathfrak{p}_{\Upsilon_Q}(x), \mathfrak{p}_{\Upsilon_P} \circ \mathfrak{p}_{\Upsilon_Q}(y)) \\
&= \delta_P(\mathfrak{p}_{\Upsilon_Q}(x), \mathfrak{p}_{\Upsilon_Q}(y))
\end{aligned}$$

Finally, for any $\Upsilon_P \in \mathrm{Stab}^{-1}(P)$, we have, from Corollary 2.42 and the Lipschitz property of gate maps:

$$\begin{aligned}
\delta_P(x,y) &= d_{C(\Upsilon_P)}(\mathfrak{p}_{\Upsilon_P}(x), \mathfrak{p}_{\Upsilon_P}(y)) \\
&\leq d_1(\mathfrak{p}_{\Upsilon_P}(x), \mathfrak{p}_{\Upsilon_P}(y)) \\
&= d_1(\mathfrak{p}_{\Upsilon_P} \circ \mathfrak{p}_{\Upsilon_{N(P)}}(x), \mathfrak{p}_{\Upsilon_P} \circ \mathfrak{p}_{\Upsilon_{N(P)}}(y)) \\
&\leq d_1(\mathfrak{p}_{\Upsilon_{N(P)}}(x), \mathfrak{p}_{\Upsilon_{N(P)}}(y)) \\
&= d_{\Upsilon_{N(P)}}(x,y)
\end{aligned}$$

□

Corollary 5.25. *If P is a pre-wide parabolic and γ is a $\hat{d}$–geodesic segment in $\widehat{\Sigma}$ between $x, y \in \Sigma^{(0)}$ then $\hat{d}(\gamma, \rho(P)) > B_0$ implies $\delta_P(x, y) \leq K_0$, where B_0 and K_0 are as in Theorem 5.19.*

Proof. Since $\rho(P)$ is the cone vertex corresponding to the unique parabolic subcomplex $\Upsilon_{N(P)}$ with stabilizer $N_W(P)$, we have $\delta_P(x, y) \leq d_{\Upsilon_{N(P)}}(x, y) \leq K_0$, where the first inequality is Lemma 5.24 (4) and the second is from Theorem 5.19. □

5.2. **Acylindricity.** Bowditch [30] defines a group action $G \curvearrowright X$ on a metric space to be *acylindrical* if for all $\epsilon \geq 0$ there exist R_ϵ and N_ϵ such that for all points x and y with $d(x, y) \geq R_\epsilon$, the set $\{g \in G \mid d(x, gx) \leq \epsilon \text{ and } d(y, gy) \leq \epsilon\}$ has cardinality at most N_ϵ.

Theorem 5.26 (Acylindricity)**.** *The action $W \curvearrowright (\widehat{\Sigma}, \hat{d})$ is acylindrical. That is, for all $\epsilon \geq 0$ there are constants R_ϵ and N_ϵ such that $\hat{d}(x, y) \geq R_\epsilon$ implies $|A_\epsilon(x, y)| \leq N_\epsilon$, where:*

$$A_\epsilon(x, y) := \{w \in W \mid \hat{d}(x, wx) \leq \epsilon \text{ and } \hat{d}(y, wy) \leq \epsilon\}$$

Most of this section is dedicated to proving Theorem 5.26, but before diving in we draw some further conclusions, assuming that Theorem 5.26 holds.

Osin [89] formalized the notion of an *acylindrically hyperbolic group* as a group G that admits an acylindrical, non-elementary action on a hyperbolic space. In that work, he also defined a *generalized loxodromic* (genlox) as an element $g \in G$ such that there exists some acylindrical action of G on a hyperbolic space X_g for which the element g acts by a loxodromic isometry. A *universal acylindrical action* $G \curvearrowright X$ is an acylindrical action of G on a hyperbolic space X such that every generalized loxodromic element $g \in G$ acts loxodromically on X. See also [6].

Generalized loxodromic elements $g \in G$ are Morse [98]; that is, $\mathbb{Z} \to G : z \mapsto g^z$ is a Morse quasigeodesic with respect to any word metric on G. The converse is not true in general; Morse elements need not be generalized loxodromic [6]. However, in Coxeter groups we do get equivalence:

Proposition 5.27. *The following are equivalent, for each $w \in W$:*

- *w is a Morse element of W.*
- *w is loxodromic in the action $W \curvearrowright (\widehat{\Sigma}, \hat{d})$.*
- *w is a generalized loxodromic element of W.*

Proof. An element $w \in W$ is Morse if and only if $n \mapsto w^n$ is a Morse quasigeodesic in $(\Sigma^{(1)}, d_1)$. This excludes finite order elements, which also cannot act loxodromically, so it suffices to consider w of infinite order. Since (Σ, d_2) is CAT(0), cyclic subgroups of W are undistorted, so $n \mapsto w^n$ is quasigeodesic in (Σ, d_2) for all infinite order w. Since quasiisometries preserve quasigeodesics, the same is true in $(\Sigma^{(1)}, d_1)$, so it remains only to decide when $n \mapsto w^n$ is Morse. By Corollary 4.20, this is true if and only if $n \mapsto w^n$ is a quasigeodesic in $(\widehat{\Sigma}, \hat{d})$. This is equivalent to w being loxodromic for $W \curvearrowright (\widehat{\Sigma}, \hat{d})$. Since this particular action is an acylindrical action on a hyperbolic space, every element that acts loxodromically here is a generalized loxodromic element of W. All generalized loxodromic elements of W are Morse elements of W [98]. □

There is a straightforward characterization of Coxeter groups for which $W \curvearrowright (\widehat{\Sigma}, \hat{d})$ is non-elementary:

Proposition 5.28. *The following are equivalent:*

- *$W \curvearrowright (\widehat{\Sigma}, \hat{d})$ is non-elementary.*
- *(W, S) is nonspherical, not wide, and not virtually infinite cyclic.*
- *The canonical decomposition $P_1 \times \cdots \times P_n \times F$ of (W, S) into a product of irreducible nonspherical factors P_i and a spherical factor F has $n = 1$ and P_1 is not affine.*

Proof. The action is acylindrical, by Theorem 5.26, so by [89, Theorem 1.1], there are three possibilities:

- The action is non-elementary.
- The action has bounded orbits.
- W is virtually infinite cyclic and contains a loxodromic element.

If (W, S) is spherical or wide then $\widehat{\Sigma}$ is bounded. Otherwise, W contains a Morse element [36], and such an element acts loxodromically on $\widehat{\Sigma}$, by Proposition 5.27. In particular, $\widehat{\Sigma}$ is not bounded. Therefore, outside the spherical and wide cases, the action contains a loxodromic element and the action does not have bounded orbits, so the only possibilities are W is virtually infinite cyclic or the action is non-elementary.

To see the equivalence with the product decomposition description, observe that (W, S) is spherical if and only if $n = 0$. It is wide if and only if either $n > 1$ or $n = 1$ and P_1 is higher rank affine. The remaining case to rule out is that $n = 1$ and P_1 is infinite dihedral, which is precisely the case that W is virtually infinite cyclic. □

Corollary 5.29. *The action $W \curvearrowright (\widehat{\Sigma}, \hat{d})$ is a universal acylindrical action. Moreover, if (W, S) is nonspherical, not wide, and not virtually infinite cyclic, then the action is non-elementary.*

Proof. It is an acylindrical action, by Theorem 5.26. The action is universal, by Proposition 5.27. With the additional conditions, the action is non-elementary, by Proposition 5.28. □

Abbott, Balasubramanya, and Osin [1] define a poset of hyperbolic structures on a group: if X and Y are (possibly infinite) generating sets of a group G then Y *dominates* X if the identity map on G induces a Lipschitz map from the Cayley graph $\Gamma(G, Y)$ of G with respect to Y to $\Gamma(G, X)$. Two generating sets are equivalent if they dominate each other. A *hyperbolic structure* is an equivalence class of generating sets of G for which the corresponding Cayley graphs are hyperbolic. An *acylindrically hyperbolic structure* is a hyperbolic structure with the additional property that the actions of G on the corresponding Cayley graphs are acylindrical. The *largest* acylindrically hyperbolic structure, if it exists, is the one that dominates all the others.

Theorem 5.30. *$\widehat{\Sigma}$ defines the largest acylindrically hyperbolic structure for W.*

It is noted in [1] that if a largest acylindrically hyperbolic structure exists then it is universal, so Theorem 5.30 also implies Corollary 5.29.

Proof. The 1–skeleton of the Davis complex is the Cayley graph $\Gamma(W, S)$ of W with respect to S. The coned-off space $\widehat{\Sigma}$ contains the graph $\Sigma^{(1)}$ and adds additional length $1/2$ edges and cone vertices whose effect is to make any two distinct vertices contained in a wide parabolic subcomplex have distance exactly 1. Restricted to $\Sigma^{(0)}$, which is coarsely dense in $\widehat{\Sigma}$, this is exactly the same metric as the word metric defined by the generating set $S' := S \cup \bigcup_{\text{wide } T \subset S} W_T$. Therefore, $\Gamma(W, S')$ is equivariantly quasiisometric to $\widehat{\Sigma}$, so it is hyperbolic and the W–action is acylindrical, by Corollary 4.20 and Theorem 5.26.

The remaining claim is that S' dominates every other acylindrically hyperbolic structure on W. This part of the argument is the same as in [1, Theorem 7.9]. Suppose that X is some other acylindrically hyperbolic structure on W. The restriction of $W \curvearrowright \Gamma(W, X)$ to W_T is also acylindrical, for any $T \subset S$.

We claim that when T is wide every infinite order element $w \in W_T$ has a power with non-virtually cyclic centralizer. If W_T is virtually irreducible higher rank affine then it has a finite index translation subgroup isomorphic to $\mathbb{Z}^n$ for $n \geq 2$, so some power of w is contained in that subgroup, hence has centralizer containing $\mathbb{Z}^2$. If $W_T = A \times B$ with A and B infinite then write $w = (a, b)$. If both a and b have infinite order then $\langle (a, 1), (1, b) \rangle \cong \mathbb{Z}^2 \leq Z_{W_T}(w)$. If one of them, say b, has finite order n then $w^n = (a^n, 1)$ has $\langle w^n \rangle \times B \leq Z_{W_T}(w^n)$.

Powers of loxodromic elements are loxodromic, and centralizers of loxodromic elements in acylindrical actions are virtually cyclic [89, Corollary 6.9], so this shows that $W_T \curvearrowright \Gamma(W, X)$ contains no loxodromic elements, and then Osin's trichotomy implies it has bounded orbits. Since there are only finitely many wide subsets of S, we can take such a bound uniformly. Since S is also finite, $\sup_{s \in S'} |s|_X = C < \infty$, which implies that the identity map on W from $\Gamma(W, S')$ to $\Gamma(W, X)$ is C–Lipschitz. Hence, S' dominates X. □

The remainder of this subsection builds towards the proof of acylindricity. The proof is strongly inspired by that of acylindricity for the action of a group on a hierarchically hyperbolic space [20, Theorem 14.3]. In particular, they define strong and weak sets $L_1(m)$ and $L_2(m)$ associated to the midpoint m of a geodesic. We make analogous definitions in Section 5.2.3. However, we do not know how to complete the direct analogue of their proof for only a midpoint m. Instead we take two points m^- and m^+ trisecting the geodesic, define strong and weak sets for both of them, and use the separation between m^- and m^+ and a Coxeter specific argument to conclude.

Note that acylindricity is trivial if $\widehat{\Sigma}$ is bounded—just take all R_ϵ bigger than the diameter of $\widehat{\Sigma}$—so we may assume throughout that (W, S) is not wide and nonspherical.

The setup for Theorem 5.26 is written assuming x and y are vertices of the Davis complex. Since every point of $\widehat{\Sigma}$ is distance at most $1/2$ from $\Sigma^{(0)}$, the extension to arbitrary $x, y \in \widehat{\Sigma}$ is an easy step at the very end of the proof.

5.2.1. *Some parallel geodesic estimates and constant roundup.*

Lemma 5.31. *Let X be a δ–hyperbolic metric space. Given ϵ there exists H_ϵ such that for any geodesic segments $\alpha\colon [0,L] \to X$ and $\beta\colon [0,L] \to X$, if $d(\alpha(0),\beta(0)) \le \epsilon$ and $d(\alpha(L),\beta(L)) \le \epsilon$, then $d(\alpha(t),\beta(t)) \le H_\epsilon$ for every $t \in [0,L]$.*

Proof. This is an elementary δ–hyperbolicity calculation. We claim it suffices to take $H_\epsilon := 4\delta + 3\epsilon$. □

Lemma 5.32. *There exists J such that if α is a $\hat{d}$–geodesic between $x, y \in \Sigma^{(0)} \subset \widehat{\Sigma}$ and $m \in \Sigma^{(0)}$ is a vertex of α then for every d_1–geodesic γ from x to y there exists a vertex $p \in \gamma$ such that $\hat{d}(m,p) \le J$.*

Proof. Let γ^* be the greedy path in the shortcut graph of γ, and let $\overline{\gamma}^*$ be a realization in $\widehat{\Sigma}$. By Corollary 4.26, realizations of greedy paths are uniformly quasigeodesic in $\widehat{\Sigma}$.

There is a δ such that $\widehat{\Sigma}$ is δ–hyperbolic, by Corollary 4.20.

Since $\widehat{\Sigma}$ is hyperbolic, quasigeodesics are Morse, with Morse gauge depending only on the quasigeodesic constants and the hyperbolicity constant, so $\overline{\gamma}^*$ is μ–Morse for a Morse gauge μ independent of γ. Since α is a geodesic with the same endpoints as $\overline{\gamma}^*$, every point of α is within $\hat{d}$–distance $\mu(1,0)$ of some point of $\overline{\gamma}^*$. Since every point of $\overline{\gamma}^*$ is within $\hat{d}$–distance $1/2$ of γ, it suffices to take $J := \mu(1,0) + 1/2$. □

We now collect the constants that have appeared so far and define some more.

- $\Delta := \max_{\text{spherical } T \subset S} \operatorname{diam}_{d_1} W_T$.
- δ is the hyperbolicity constant of $\widehat{\Sigma}$.
- ϵ is the acylindricity displacement parameter.
- H_ϵ is the fellow-traveler constant from Lemma 5.31.
- J is the $\hat{d}$–quasiconvexity constant of Lemma 5.32.
- D_0 and K_0 are the constants of Proposition 5.17.
- $B_0 = D_0 + \Delta + 3$, as in Theorem 5.19.

Note that Δ is an integer and D_0 can be taken to be an integer, which also makes B_0 an integer. For our later convenience we replace H_ϵ, J, and K_0 by their integer ceilings. Then we make the following proleptic choices of additional constants, which are also integers.

- $C_0 := H_\epsilon + B_0 + 3J$
- $C_1 := C_0 + 2$
- $C_2 := C_1 + H_\epsilon$
- $C_3 := C_2 + B_0 + 1$

Also for each $2 \le r \le |S|$, we choose positive integers $a_r < c_r < b_r$ such that:

- $a_r > 2K_0 + 2C_3$
- $c_r > a_r + 2K_0$
- $b_r > c_r + 2K_0$
- $a_s > b_r + 4K_0$ when $s < r$

Such choices exist, for example by choosing $a_{|S|} > 2K_0 + 2C_3$ first, then choosing $c_{|S|}$, $b_{|S|}$, and $a_{|S|-1}$ satisfying the required conditions, then continuing inductively.

- $M_\epsilon := M(c_2, \dots, c_{|S|})$, where M is the witness threshold function of Proposition 5.23, and $c_2, \dots, c_{|S|}$ are the constants c_r chosen above.

5.2.2. *Parabolic carriers.* Let $u, v \in \Sigma^{(0)}$. For $i = 1, \dots, |S| - 1$, we inductively define three collections $\mathcal{P}_i$, $\mathcal{D}_i$, and $\mathcal{F}_i$ of parabolic subgroups. Elements of $\mathcal{P}_i$ are called the *parent carriers* at level i; those of $\mathcal{D}_i$ are the *descendant carriers* at level i; and those of $\mathcal{F}_i$ are the *factor carriers* at level i.

Set $\mathcal{P}_1 := \{W\}$.

Choose a $\hat{d}$–geodesic γ from u to v in $\widehat{\Sigma}$.

Let $\mathcal{D}_1$ be the set of stabilizers of wide parabolic subcomplexes associated to the cone vertices that occur on γ.

Define $\mathcal{F}_1$ to be the set of irreducible nonspherical factors occurring in the canonical product decomposition of some $Q \in \mathcal{D}_1$.

Now assume for $i \geq 1$, the sets $\mathcal{P}_i$, $\mathcal{D}_i$, and $\mathcal{F}_i$ have been defined. Let:

$$\mathcal{P}_{i+1} := \{P \in \mathcal{F}_i \mid \delta_P(u,v) \leq b_{\mathrm{rank}(P)} + 2K_0\}$$

For each $P \in \mathcal{P}_{i+1}$, choose a geodesic $\hat{\gamma}_P$ in $C(P)$ from $\mathfrak{p}_P(u)$ to $\mathfrak{p}_P(v)$, where $C(P)$ and $\mathfrak{p}_P$ are as defined in Section 5.1.1 and Definition 5.7.

Let $\mathcal{D}_{i+1}$ be the set of parabolic subgroups associated to the cone vertices on $\hat{\gamma}_P$ for each $P \in \mathcal{P}_{i+1}$.

Let $\mathcal{F}_{i+1}$ be the set of irreducible nonspherical factors occurring in the canonical product decomposition of some $Q \in \mathcal{D}_{i+1}$.

We set $\mathrm{rank}(\mathcal{P}_i) := \sup\{\mathrm{rank}(P) \mid P \in \mathcal{P}_i\}$, which is a maximum unless $\mathcal{P}_i = \emptyset$, and define $\mathrm{rank}(\mathcal{D}_i)$ and $\mathrm{rank}(\mathcal{F}_i)$ similarly. By construction, if $\mathcal{P}_i \neq \emptyset$ then:

$$\mathrm{rank}(\mathcal{P}_i) > \mathrm{rank}(\mathcal{D}_i) \geq \mathrm{rank}(\mathcal{F}_i) \geq \mathrm{rank}(\mathcal{P}_{i+1})$$

In particular, $\mathrm{rank}(\mathcal{P}_i) \leq |S| + 1 - i$, so $\mathrm{rank}(\mathcal{P}_{|S|-1}) \leq 2$, which implies $\mathcal{P}_{|S|} \subset \mathcal{F}_{|S|-1} = \emptyset$.

Lemma 5.33. *For all i, each $F \in \mathcal{F}_i$ is pre-wide.*

Proof. Each $F \in \mathcal{F}_1$ is a nonspherical irreducible parabolic subgroup of a wide parabolic subgroup, by definition of $\mathcal{F}_1$ and $\mathcal{D}_1$, so F is pre-wide, by Lemma 5.12.

Suppose the lemma is true for all $F \in \mathcal{F}_i$ for all $i \leq i_0$. Consider $F \in \mathcal{F}_{i_0+1}$. By construction, F is a nonspherical irreducible factor in the canonical decomposition of some $Q \in \mathcal{D}_{i_0+1}$. In turn, Q is the parabolic subgroup associated to some cone vertex of $C(P)$ for some $P \in \mathcal{P}_{i_0+1} \subset \mathcal{F}_{i_0}$. By the induction hypothesis, P is pre-wide, so $N_W(P)$ is a wide parabolic subgroup. Thus, $F \leq Q \leq P \leq N_W(P)$. Since F is a nonspherical irreducible parabolic contained in the wide parabolic $N_W(P)$, it is pre-wide, again by Lemma 5.12. □

Lemma 5.34. *There is a function* $\mathrm{FactorSum}\colon \mathbb{N} \to \mathbb{N}$ *defined by:*

$$\mathrm{FactorSum}(L) := \sup_{\hat{d}(u,v)=2L} \sup \sum_{i=1}^{|S|-2} |\mathcal{F}_i|$$

The first supremum is taken over pairs $u, v \in \Sigma^{(0)}$ such that $\hat{d}(u,v) = 2L$. The second supremum is taken over all possible choices of carrier systems for the given u and v.

Proof. Take any $u, v \in \Sigma^{(0)}$ such that $\hat{d}(u,v) = 2L$. Construct parent, descendant, and factor carriers for each level, including the choice of $\hat{d}$–geodesic γ from u to v in $\widehat{\Sigma}$. We must show $\sum_{i=1}^{|S|-2} |\mathcal{F}_i|$ is bounded above, depending only on L.

By construction, elements of $\mathcal{F}_i$ are irreducible nonspherical factors of the canonical product decomposition of elements of $\mathcal{D}_i$. If $\mathcal{D}_i = \emptyset$ then $|\mathcal{F}_i| = 0$. Otherwise, since such a factor has rank at least 2, the number of such factors of $Q \in \mathcal{D}_i$ is at most $\left\lfloor \frac{\mathrm{rank}(Q)}{2} \right\rfloor$. Therefore:

$$|\mathcal{F}_i| \leq \left\lfloor \frac{\mathrm{rank}(\mathcal{D}_i)}{2} \right\rfloor \cdot |\mathcal{D}_i| < \frac{|S| \cdot |\mathcal{D}_i|}{2}$$

Since $\hat{d}(u,v) = 2L$, there can be at most $2L$ cone vertices on γ, so $|\mathcal{D}_1| \leq 2L$ and $|\mathcal{F}_1| < L|S|$. For $i \geq 1$, recall the collection of parent carriers:

$$\mathcal{P}_{i+1} := \{P \in \mathcal{F}_i \mid \delta_P(u,v) \leq b_{\mathrm{rank}(P)} + 2K_0\}$$

Each element $P \in \mathcal{P}_{i+1}$ has a corresponding path $\hat{\gamma}_P$ in $C(P)$ of length at most $b_{\mathrm{rank}(P)} + 2K_0$, therefore contributing at most that number of cone vertex parabolics to the descendant carriers $\mathcal{D}_{i+1}$. Since $b_2 > b_3 > \cdots$, we have for $i \geq 2$:

$$|\mathcal{D}_i| \leq (b_2 + 2K_0)\,|\mathcal{P}_i| \leq (b_2 + 2K_0)\,|\mathcal{F}_{i-1}|$$

Therefore, $|\mathcal{F}_i| \leq \frac{|S|}{2}(b_2 + 2K_0)\,|\mathcal{F}_{i-1}|$. Recursively, this yields:

$$|\mathcal{F}_i| \leq L|S| \left(\frac{|S|(b_2 + 2K_0)}{2} \right)^{i-1}$$

This gives a uniform upper bound on $\sum_{i=1}^{|S|-2} |\mathcal{F}_i|$, depending only on L and constants:

$$\sum_{i=1}^{|S|-2} |\mathcal{F}_i| \leq L|S| \sum_{j=0}^{|S|-3} \left(\frac{|S|(b_2 + 2K_0)}{2} \right)^j \qquad \square$$

5.2.3. *The strong and weak sets for trisection vertices.* Let $x, y \in \Sigma^{(0)}$, and let γ be a $\hat{d}$–geodesic from x to y in $\widehat{\Sigma}$. Choose two vertices m_- and m_+ in $\Sigma^{(0)}$ on γ that are nearest to the points $1/3$ and $2/3$ of the way along γ. For all $g \in A_\epsilon(x, y)$ and $m \in \{m_\pm\}$, we have $\hat{d}(m, gm) \leq H_\epsilon$, by definition of H_ϵ.

Definition 5.35 (The strong set L_1 and weak set L_2)**.** For fixed x, y, and γ as above, for each $m \in \{m_\pm\}$, let $L_1(m)$ be the collection of pre-wide parabolics P such that:

- $\delta_P(x, y) > c_{\mathrm{rank}(P)}$
- $\hat{d}(m, \rho(P)) \leq C_1$
- There is no pre-wide parabolic $Q \gneq P$ with $\delta_Q(x, y) > c_{\mathrm{rank}(Q)}$.

Similarly, let $L_2(m)$ be the collection of pre-wide parabolics P such that:

- $\delta_P(x, y) > a_{\mathrm{rank}(P)}$
- $\hat{d}(m, \rho(P)) \leq C_2$
- There is no pre-wide parabolic $Q \gneq P$ with $\delta_Q(x, y) > b_{\mathrm{rank}(Q)}$.

In this section we will show that $|L_2(m)|$ is uniformly finite (Proposition 5.36), that $A_\epsilon(x, y)$ conjugates $L_1(m)$ into $L_2(m)$ (Proposition 5.37), and that if $L_1(m) = \emptyset$ then there is a vertex of $\Sigma^{(0)}$ uniformly $\hat{d}$–close to m that has bounded d_1–displacement under the $A_\epsilon(x, y)$–action (Proposition 5.39).

Proposition 5.36 (Uniform finiteness of $L_2(m)$)**.** *Let C_3 be as in Section 5.2.1. Let $x, y \in \Sigma^{(0)}$ with $\hat{d}(x, y) \geq R_\epsilon > 3(C_3 + 1)$. Let γ be any $\hat{d}$–geodesic from x to y, and let $m_\pm$ be trisection vertices of γ. For $m \in \{m_\pm\}$ we have $|L_2(m)| \leq \mathrm{FactorSum}(C_3)$, where* FactorSum *is the function of Lemma 5.34.*

Proof. The lower bound on R_ϵ guarantees that the subsegment γ' of γ centered about m of radius C_3 exists and does not contain x or y. Let u and v be the end vertices of γ'. Let $\mathcal{P}_i$, $\mathcal{D}_i$, and $\mathcal{F}_i$ be parent, descendant, and factor carriers for u and v at levels $i = 1, \ldots, |S| - 1$, with the assumption that γ' is the geodesic used to define $\mathcal{D}_1$. By Lemma 5.33, all of the factor carriers are pre-wide, hence every parent carrier is pre-wide except $W \in \mathcal{P}_1$. Our goal is to show that $L_2(m) \subset \bigcup \mathcal{F}_i$. By Lemma 5.34, this would imply the desired bound $|L_2(m)| \leq \mathrm{FactorSum}(C_3)$.

Let $P \in L_2(m)$. Since $\hat{d}(m, \rho(P)) \leq C_2$ and $\hat{d}(m, u) = \hat{d}(m, v) = C_3 = C_2 + B_0 + 1$, both segments $[x, u]$ and $[v, y]$ are more than B_0 away from $\rho(P)$. Therefore, by Corollary 5.25, $\delta_P(x, u)$, $\delta_P(v, y) \leq K_0$. Hence:

$$\delta_P(u, v) \geq \delta_P(x, y) - 2K_0 > a_{\mathrm{rank}(P)} - 2K_0 \tag{19}$$

We now make the following claim.

Claim 5.36.1. Let $P \in L_2(m)$. If $P \lneq Q$ for some $Q \in \mathcal{P}_i$, then either $P \in \mathcal{F}_i$ or $P \lneq Q'$ for some $Q' \in \mathcal{P}_{i+1}$.

Proof of Claim: We deal with the case of $i = 1$ and $i > 1$ separately. They are similar, but the definition of $\mathcal{D}_1$ is slightly different than $\mathcal{D}_i$ for $i > 1$.

For $i = 1$, recall that $\mathcal{D}_1$ is the collection of wide parabolic subgroups $\mathrm{Stab}(\Upsilon)$, where v_Υ is a cone vertex on γ'. We show that $P \leq \mathrm{Stab}(\Upsilon)$ for some $\mathrm{Stab}(\Upsilon) \in \mathcal{D}_1$. If not, then by Lemma 5.10, the δ_P–diameter of Υ is at most 1, for every such Υ. Thus, every cone shortcut step in γ' has δ_P–diameter at most 1. Every $\Sigma^{(1)}$ step in γ' also has δ_P–diameter at most 1, so $\delta_P(u, v) \leq 2C_3$. Since $a_{\mathrm{rank}(P)} - 2K_0 > 2C_3$, this contradicts (19). Therefore, there exists a cone vertex v_Υ on γ' such that $P \leq \mathrm{Stab}(\Upsilon)$. Since P is irreducible and nonspherical, $P \leq Q$ for one of the canonical nonspherical irreducible factors Q of $\mathrm{Stab}(\Upsilon)$. By construction, such a Q belongs to $\mathcal{F}_1$, so if $P = Q$ we are done. Otherwise, $P \lneq Q$, so the definition of $L_2(m)$ requires that $\delta_Q(x, y) \leq b_{\mathrm{rank}(Q)}$. Since P and Q are pre-wide we can apply Lemma 5.24 (2) to get:

$$\hat{d}(m, \rho(Q)) \leq 1 + \hat{d}(m, \rho(P)) \leq 1 + C_2$$

Since $\hat{d}(m, \rho(P))$ is half-integral and C_2 is integral, this is actually a strict inequality, so the subsegments of γ from x to u and from v to y are both more than $\hat{d}$–distance B_0 away from $\rho(Q)$, so Corollary 5.25 gives

$\delta_Q(x,u), \delta_Q(y,v) \leq K_0$. Then the triangle inequality yields:

$$\delta_Q(u,v) \leq \delta_Q(x,y) + 2K_0 \leq b_{\mathrm{rank}(Q)} + 2K_0$$

This shows $Q \in \mathcal{P}_2$.

For $i > 1$, suppose $P \lneq Q \in \mathcal{P}_i$. By definition of $\mathcal{P}_i$, this implies $\delta_Q(u,v) \leq b_{\mathrm{rank}(Q)} + 2K_0$. Let $\hat{\gamma}_Q$ be the geodesic in $C(Q)$ connecting $\mathfrak{p}_Q(u)$ to $\mathfrak{p}_Q(v)$ used to define $\mathcal{D}_i$. It has length at most $b_{\mathrm{rank}(Q)} + 2K_0$. For any $\Upsilon_Q \in \mathrm{Stab}^{-1}(Q)$, Lemma 5.8, says $C(Q)$ is isomorphic to $C(\Upsilon_Q)$, and Lemma 5.24 (3) says $\delta_P(u,v) = \delta_P(\mathfrak{p}_{\Upsilon_Q}(u), \mathfrak{p}_{\Upsilon_Q}(v))$. The same argument as in the $i = 1$ case gives that if $P \not\leq \mathrm{Stab}(\Upsilon)$ for every cone vertex v_Υ of $\hat{\gamma}_Q$, then $\delta_P(\mathfrak{p}_{\Upsilon_Q}(u), \mathfrak{p}_{\Upsilon_Q}(v)) \leq b_{\mathrm{rank}(Q)} + 2K_0$. However, since $\mathrm{rank}(P) < \mathrm{rank}(Q)$ and $a_{\mathrm{rank}(P)} > b_{\mathrm{rank}(Q)} + 4K_0$, this contradicts (19). Thus, there is some cone vertex v_Υ on $\hat{\gamma}_Q$ with $P \leq \mathrm{Stab}(\Upsilon)$, and, by construction, $\mathrm{Stab}(\Upsilon) \in \mathcal{D}_i$. Let Q' be the irreducible nonspherical factor of $\mathrm{Stab}(\Upsilon)$ containing P. Then $Q' \in \mathcal{F}_i$. A similar argument to the $i = 1$ case gives that either $P = Q' \in \mathcal{F}_i$, or $Q' \in \mathcal{P}_{i+1}$. ◊

Now we will apply the claim to see that eventually $P \in \mathcal{F}_i$. At $i = 1$ we have $\mathcal{P}_1 = \{W\}$, so $P \leq Q_1 := W \in \mathcal{P}_1$, but P is certainly not all of W since that would mean W is wide and $\widehat{\Sigma}$ is bounded. Thus, $P \lneq Q_1 \in \mathcal{P}_1$, and the assumption of the claim holds at level 1, so the claim says that either $P \in \mathcal{F}_1$, in which case we are done, or $P \lneq Q_2$ for some $Q_2 \in \mathcal{P}_2$. Continue inductively. Since $\mathrm{rank}(\mathcal{P}_j) \leq |S| + 1 - j$, there is some first $j_0 \leq |S|$ such that $\mathcal{P}_{j_0} = \emptyset$. Thus, $P \in \mathcal{F}_i$ for some $i < j_0 \leq |S|$. □

Proposition 5.37. *Let $x, y \in \Sigma^{(0)}$ with $\hat{d}(x,y) \geq R_\epsilon > 3(C_3 + \epsilon + 1)$. For any $\hat{d}$–geodesic from x to y, let $m_\pm$ be its trisection vertices. For all $m \in \{m_\pm\}$, $g \in A_\epsilon(x,y)$, and $P \in L_1(m)$, we have $gPg^{-1} \in L_2(m)$.*

Proof. Let $P' := gPg^{-1}$. The definition of $L_1(m)$ gives $\hat{d}(m, \rho(P)) \leq C_1$. Since $g\rho(P) = \rho(P')$ and $\hat{d}(m, gm) \leq H_\epsilon$, we have $\hat{d}(m, \rho(P')) \leq H_\epsilon + C_1 = C_2$. This shows the second condition of $P' \in L_2(m)$.

By our assumption on R_ϵ, the $\hat{d}$–distance from $\rho(P')$ to any geodesic from x to gx is at least:

$$\hat{d}(m,x) - \hat{d}(m, \rho(P')) - \hat{d}(x, gx) \geq R_\epsilon/3 - 1 - C_2 - \epsilon > C_3 - C_2 = B_0 + 1 \tag{20}$$

Therefore, by Corollary 5.25, $\delta_{P'}(x, gx) \leq K_0$. Similarly, $\delta_{P'}(y, gy) \leq K_0$. Hence:

$$\begin{aligned}
\delta_{P'}(x,y) &\geq \delta_{P'}(gx, gy) - \delta_{P'}(gx, x) - \delta_{P'}(y, gy) \\
&\geq \delta_P(x,y) - 2K_0 \\
&> c_{\mathrm{rank}(P)} - 2K_0 \\
&> a_{\mathrm{rank}(P)} \\
&= a_{\mathrm{rank}(P')}
\end{aligned}$$

This shows the first condition of $P' \in L_2(m)$.

Lastly, suppose that $P' \lneq Q$, for some pre-wide Q. Suppose that $\delta_Q(x,y) > b_{\mathrm{rank}(Q)}$. Then, by Lemma 5.24 (2), $\hat{d}(\rho(P'), \rho(Q)) \leq 1$. The extra $+1$ slack in (20) accommodates an extra $\hat{d}(\rho(P'), \rho(Q))$ term in a similar computation, so again Corollary 5.25 implies $\delta_Q(x, gx), \delta_Q(y, gy) \leq K_0$ and then:

$$\begin{aligned}
\delta_{g^{-1}Qg}(x,y) &= \delta_Q(gx, gy) \\
&\geq \delta_Q(x,y) - \delta_Q(x, gx) - \delta_Q(y, gy) \\
&> b_{\mathrm{rank}(Q)} - 2K_0 > c_{\mathrm{rank}(Q)}
\end{aligned}$$

This contradicts $P \in L_1(m)$, so we must have had $\delta_Q(x,y) \leq b_{\mathrm{rank}(Q)}$, which gives the final condition for $P' \in L_2(m)$. □

Lemma 5.38. *Fix $x, y \in \Sigma^{(0)}$ and a $\hat{d}$–geodesic γ from x to y in $\widehat{\Sigma}$. Let $m_\pm$ be the trisection vertices of γ. If $L_1(m) = \emptyset$ for some $m \in \{m_\pm\}$ and Υ is a wide parabolic subcomplex with $\hat{d}(m, v_\Upsilon) \leq C_0$, then $d_\Upsilon(x,y) \leq M_\epsilon$.*

Proof. Suppose $d_\Upsilon(x,y) > M_\epsilon$. Then Proposition 5.23 says there exists some pre-wide witness $P \leq \mathrm{Stab}(\Upsilon)$ with $\delta_P(x,y) > c_{\mathrm{rank}(P)}$. If $\Upsilon = w\Sigma_T$ then $P \leq \mathrm{Stab}(\Upsilon)$ implies $P = wvW_U v^{-1} w^{-1}$ for some $v \in W_T$ and $U \subset T$. Then $\Xi := wv\Sigma_U$ has $\mathrm{Stab}(\Xi) = P$ and $\Xi \subset \Upsilon$. On the other hand, by Lemma 5.3, Ξ is a subcomplex of the unique parabolic subcomplex with stabilizer $N_W(P)$, which is the subcomplex corresponding to the cone vertex $\rho(P)$. Hence, the cone vertices $\rho(P)$ and v_Υ have every vertex of $\Xi^{(0)}$ as mutual neighbors, so $\hat{d}(v_\Upsilon, \rho(P)) \leq 1$.

Take a maximal Q among the set of pre-wide parabolics that contain P and satisfy $\delta_Q(x,y) > c_{\mathrm{rank}(Q)}$. The set is not empty, since it contains P, and chains are uniformly bounded in length by $|S|-2$, since $\mathrm{rank}(P) \geq 2$ and strict containment of parabolic subgroups forces strict increase in rank. Thus, a maximal element exists.

Since P and Q are nested, Lemma 5.24 (2) says $\hat{d}(\rho(P),\rho(Q)) \leq 1$. Therefore:

$$\hat{d}(m,\rho(Q)) \leq \hat{d}(m,v_\Upsilon) + \hat{d}(v_\Upsilon,\rho(P)) + \hat{d}(\rho(P),\rho(Q)) \leq C_0 + 2 = C_1$$

We have confirmed the three membership conditions for $Q \in L_1(m)$, but $L_1(m) = \emptyset$, so this is a contradiction and we must have had $d_\Upsilon(x,y) \leq M_\epsilon$. □

Proposition 5.39. *There exists D_ϵ such that if $\hat{d}(x,y) \geq R_\epsilon > 3(B_0+C_0+\epsilon+1)$, for any $\hat{d}$–geodesic γ from x to y with trisection vertices $m_\pm$, if $L_1(m) = \emptyset$ for some $m \in \{m_\pm\}$, then for any choice of a d_1–geodesic from x to y and a vertex p on that geodesic $\hat{d}$–closest to m, we have $d_1(p,gp) \leq D_\epsilon$ for all $g \in A_\epsilon(x,y)$.*

Proof. We show it suffices to take $D_\epsilon := C(1+H_\epsilon+2J)(2M_\epsilon+K_0)$, where C is the constant of Lemma 5.20, M_ϵ is as in Lemma 5.38, and H_ϵ, J, and K_0 are as in Section 5.2.1.

Suppose $L_1(m) = \emptyset$. For any choice of d_1–geodesic from x to y, take a vertex p that is $\hat{d}$–closest to m. Then $\hat{d}(m,p) \leq J$, by Lemma 5.32.

For all $g \in A_\epsilon(x,y)$, we have $\hat{d}(m,gm) \leq H_\epsilon$, by Lemma 5.31, so $\hat{d}(p,gp) \leq H_\epsilon + 2J$.

We claim $d_\Upsilon(p,gp) \leq 2M_\epsilon + K_0$, for all wide parabolic subcomplexes Υ. If $d_\Upsilon(p,gp) \leq K_0$ we are happy, so suppose not. Then Theorem 5.19 implies that the $\hat{d}$–distance from v_Υ to any $\hat{d}$–geodesic from p to gp is at most B_0. Since the length of any such geodesic is at most $H_\epsilon + 2J$, it follows that:

$$\hat{d}(v_\Upsilon, m) \leq \hat{d}(v_\Upsilon, p) + \hat{d}(p,m) \leq B_0 + H_\epsilon + 3J = C_0$$

The same estimate holds using gp and gm, so:

$$\hat{d}(m, v_{g^{-1}\Upsilon}) = \hat{d}(gm, v_\Upsilon) \leq \hat{d}(v_\Upsilon, gp) + \hat{d}(gp, gm) \leq C_0$$

Lemma 5.38 gives $d_\Upsilon(x,y) \leq M_\epsilon$ and $d_{g^{-1}\Upsilon}(x,y) \leq M_\epsilon$.

By Lemma 5.6, $d_\Upsilon(x,y) = |\mathcal{M}_\Upsilon(x,y)|$. Since p is a vertex on a d_1–geodesic between x and y:

$$d_\Upsilon(x,p) \leq d_\Upsilon(x,p) + d_\Upsilon(p,y) = d_\Upsilon(x,y) \leq M_\epsilon$$

Similarly:

$$d_\Upsilon(gx,gp) = d_{g^{-1}\Upsilon}(x,p) \leq d_{g^{-1}\Upsilon}(x,y) \leq M_\epsilon$$

Any $\hat{d}$–geodesic from x to gx has $\hat{d}$–distance from v_Υ at least:

$$\hat{d}(m,x) - \hat{d}(m,v_\Upsilon) - \hat{d}(x,gx) \geq R_\epsilon/3 - 1 - C_0 - \epsilon > B_0$$

Then Theorem 5.19 implies $d_\Upsilon(x,gx) \leq K_0$.

Putting the last three bounds together:

$$\begin{aligned} d_\Upsilon(p,gp) &\leq d_\Upsilon(p,x) + d_\Upsilon(x,gx) + d_\Upsilon(gx,gp) \\ &\leq M_\epsilon + K_0 + M_\epsilon \end{aligned}$$

This proves the claim.

Now apply Lemma 5.20:

$$\begin{aligned} d_1(p,gp) &\leq C(1+\hat{d}(p,gp)) \cdot \max\{1, \sup_{\text{wide } \Upsilon} d_\Upsilon(p,gp)\} \\ &\leq C(1+H_\epsilon+2J)(2M_\epsilon+K_0) \end{aligned}$$

□

5.2.4. *Proof of acylindricity.* Choose R_ϵ large enough so that:

$$R_\epsilon/3 - 1 > \max\{C_3+\epsilon,\ \Delta + 2C_1 + 4\} \tag{21}$$

The first condition guarantees R_ϵ is sufficiently large to apply Proposition 5.36, Proposition 5.37, and Proposition 5.39.

Proof of Theorem 5.26. Let D_ϵ be the constant of Proposition 5.39. Let R_ϵ satisfy (21). Let $x,y \in \Sigma^{(0)}$, with $\hat{d}(x,y) \geq R_\epsilon$, fix a $\hat{d}$–geodesic from x to y, and let $m_\pm$ be its trisection vertices. We have two cases:

Case A. $L_1(m) = \emptyset$ for some $m \in \{m_\pm\}$.

Then, by Proposition 5.39, for any choice of a d_1–geodesic from x to y and a vertex p on that geodesic $\hat{d}$–closest to m, we have $d_1(p, gp) \leq D_\epsilon$ for all $g \in A_\epsilon(x, y)$. Since W acts freely on $\Sigma^{(0)}$, which is locally finite, this implies:

$$|A_\epsilon(x, y)| \leq |\{w \in W \mid d_1(p, wp) \leq D_\epsilon\}| = |\bar{\mathcal{N}}_{D_\epsilon}(1)| < \infty$$

Case B. $L_1(m) \neq \emptyset$ for both $m \in \{m_\pm\}$.

Fix a pair $P_- \in L_1(m_-)$ and $P_+ \in L_1(m_+)$. By Proposition 5.37, for any $g \in A_\epsilon(x, y)$:

$$gP_-g^{-1} \in L_2(m_-) \quad \text{and} \quad gP_+g^{-1} \in L_2(m_+)$$

Define the map:

$$\Phi\colon\ A_\epsilon(x, y) \to L_2(m_-) \times L_2(m_+) : g \mapsto (gP_-g^{-1}, gP_+g^{-1})$$

By Proposition 5.36, $|L_1(m)| \leq |L_2(m)| \leq \mathrm{FactorSum}(C_3)$, so the cardinality of the image of Φ is bounded by $\mathrm{FactorSum}(C_3)^2$. We now bound the cardinality of each fiber. Suppose $g, h \in A_\epsilon(x, y)$ are in the same fiber, so:

$$gP_-g^{-1} = hP_-h^{-1} \quad \text{and} \quad gP_+g^{-1} = hP_+h^{-1}$$

Equivalently, $h^{-1}g \in N_W(P_-) \cap N_W(P_+)$. By Definition 5.35, $P_- \in L_1(m_-)$ implies $\hat{d}(m_-, \rho(P_-)) \leq C_1$ and, similarly, $\hat{d}(m_+, \rho(P_+)) \leq C_1$. Now:

$$\hat{d}(\rho(P_-), \rho(P_+)) \geq \hat{d}(m_-, m_+) - 2C_1 \geq \hat{d}(x, y)/3 - 2 - 2C_1 \geq R_\epsilon/3 - 2 - 2C_1 \overset{(21)}{>} \Delta + 3$$

Then the contrapositive of Corollary 5.14 implies $N_W(P_-) \cap N_W(P_+)$ is a spherical parabolic. It follows that $|A_\epsilon(x, y)|$ is bounded by $\mathrm{FactorSum}(C_3)^2$ times the maximum cardinality of a spherical parabolic subgroup of (W, S).

Acylindricity for $x, y \in \Sigma^{(0)}$ follows by taking:

$$N_\epsilon := \max\{|\bar{\mathcal{N}}_{D_\epsilon}(1)|, \mathrm{FactorSum}(C_3)^2 \cdot \max\{|W_T| \mid \text{spherical } T \subset S\}\}$$

For the general case, use the above R_ϵ and N_ϵ to define $R'_\epsilon := R_{\epsilon+1} + 1$ and $N'_\epsilon := N_{\epsilon+1}$. Suppose $x', y' \in \widehat{\Sigma}$ with $\hat{d}(x', y') \geq R'_\epsilon$. Choose closest $\Sigma^{(0)}$ vertices x to x' and y to y'. Then $\hat{d}(x, x'), \hat{d}(y, y') \leq 1/2$, so $\hat{d}(x, y) \geq R_{\epsilon+1}$ and $A_\epsilon(x', y') \subset A_{\epsilon+1}(x, y)$. Thus:

$$\hat{d}(x', y') \geq R'_\epsilon \implies |A_\epsilon(x', y')| \leq |A_{\epsilon+1}(x, y)| \leq N_{\epsilon+1} = N'_\epsilon \qquad \square$$

5.3. **Dynamical characterization of stability.** We already have a characterization of subgroup stability in terms of a geometric condition that is uniform over the whole subgroup:

Corollary 5.40. *A finitely generated subgroup $G \leq W$ is stable if and only if for some/every finite generating set S_G of G the orbit map of G into $\widehat{\Sigma}$, obtained by sending $g \in G$ to $g.\mathbb{1} \in \Sigma^{(0)}$ and then including into $\widehat{\Sigma}$, is a quasiisometric embedding $(G, d_{S_G}) \to (\widehat{\Sigma}, \hat{d})$, where d_{S_G} is the word metric on G with respect to the generating set S_G.*

This follows by combining Proposition 4.16 and Theorem 4.19.

In this section we give an element-wise characterization.

We first recall the mapping class group situation. Corollary 5.40 is analogous to the result that a subgroup of MCG is stable if and only if it is finitely generated and the orbit map is a quasiisometric embedding into the curve graph [73, 67, 58]. Bestvina, Bromberg, Kent, and Leininger [23] give a different characterization[7]: a finitely generated subgroup of the mapping class group is stable if and only if it is undistorted and purely pseudo-Anosov, where *purely pseudo-Anosov* means that every infinite order element of the subgroup is a pseudo-Anosov surface homeomorphism class. The pseudo-Anosovs are precisely the elements that are loxodromic for the action on the curve graph, so "purely pseudo-Anosov" could be equivalently described as being "purely curve-graph-loxodromic", or just "purely loxodromic", with the action on the curve graph implicit. Leaving the action implicit is natural here since the action on the curve graph gives the largest acylindrical hyperbolic structure on MCG. This is also the terminology used in the RAAG and graph product cases [76, 17]. We make the corresponding definition for Coxeter groups:

[7]Here we elide a difference between stability and convex cocompactness in exceptional low complexity cases.

Definition 5.41. Let $\Sigma = \Sigma(W, S)$ be the Davis complex of a finite rank Coxeter system. A subgroup G of W is *purely loxodromic* if every infinite order element of G acts loxodromically on the coned-off space $\widehat{\Sigma}$. Equivalently, for every infinite order element $g \in G$, the orbit map gives a quasiisometric embedding $\mathbb{Z} \to \widehat{\Sigma} : n \mapsto g^n.\mathbb{1}$.

Definition 5.42. A finitely generated subgroup of a finitely generated group is *undistorted* if the inclusion map is a quasiisometric embedding, with respect to word metrics for any choices of finite generating sets of the two groups.

The goal of this section is to prove the following analogue of [23, Main Theorem]:

Theorem 5.43. *A finitely generated subgroup of W is stable if and only if it is undistorted and purely loxodromic.*

Remark. It is not known if the analogue of [23, Main Theorem] is true for every HHG with a universal action on the maximal hyperbolic space of its HHG structure, [2, Question C]. Theorem 5.43 does not address that question in that generality, but, interestingly, provides some examples where the corresponding conclusion is true even though the Coxeter group is known *not* to be an HHG, recall footnote 1. These are not the first examples of this phenomenon; stability in many graph products can be characterized in terms of being purely loxodromic [17]. It is easy to make non-HHG graph products by taking one of the vertex groups to have a distorted finitely generated Abelian subgroup (Plummer, unpublished, cf [64, Proposition 2.17]). Our examples are somewhat more subtle; since Coxeter groups are CAT(0) groups, this kind of easy obstruction to being an HHG is not available.

Remark. In the MCG setting, it is an open question, asked by Farb and Mosher [61] in the free case, whether the characterization of stable subgroups requires "undistorted" as a separate hypothesis. Stable subgroups are undistorted, so this is really a question about the diversity of purely pseudo-Anosov subgroups. For arbitrary hyperbolic groups "undistorted" really is necessary as a hypothesis: consider a hyperbolic group constructed as the mapping torus of pseudo-Anosov homeomorphism of a closed hyperbolic surface. The surface group fiber is a hyperbolic subgroup of a hyperbolic group, so it is purely loxodromic with respect to its action on the ambient hyperbolic group, but it is distorted, hence not stable. Every Coxeter group acting as a cocompact geometric reflection group on $\mathbb{H}^3$ is virtually of this form [7, Theorem 5.2], so undistorted is necessary as a separate hypothesis in our theorem. Notice further that if W is hyperbolic then it has no wide parabolic subgroups to cone off, so $\widehat{\Sigma}$ is just $\Sigma^{(1)}$, which is the Cayley graph of W with respect to S, so loxodromic for the action on the ambient group and loxodromic for the action on $\widehat{\Sigma}$ coincide in the hyperbolic case.

Remark. There are several classes of finitely generated, purely pseudo-Anosov subgroups of MCG that are known to be stable without an additional undistortedness hypothesis, see, for example [79] and the references in its Section 1.1. It would be interesting to know if there are natural classes of examples of finitely generated, purely loxodromic subgroups of Coxeter groups in which undistortedness is automatic.

This subsection has the same architecture as [23]. Proposition 5.45 corresponds to [23, Proposition 3.1], Lemma 5.46 and Proposition 5.47 correspond to [23, Proposition 4.1], and the proof of Theorem 5.43 imitates the proof of [23, Main Theorem]. The main difference is that [23, Proposition 4.1] uses extensive MCG/subsurface machinery to decompose a word g into subwords with linearly summing projections. We replace that machinery with greedy paths and wall combinatorics, resulting in an expression of g as a combination of possibly overlapping subpaths, but with enough control on the overlaps to reach a similar conclusion.

Throughout this section we have a finitely generated subgroup $G \leq W$, for a finite rank Coxeter system (W, S). We will choose a finite generating set S_G of G, and let $|\cdot|_G$ denote word length in G with respect to S_G, and define $\lambda_G := \max_{s \in S_G} |s|_S$.

Lemma 5.44. *Let (W, S) be a finite rank Coxeter system. For all $g \in W$, we have $|\operatorname{BigProj}(g)| \leq 2^{|g|_S}$, where:*

$$\operatorname{BigProj}(g) := \{\textit{nonspherical irreducible parabolic subgroup } Q \mid \delta_Q(1, g) > 1\}$$

Proof. Suppose Υ is a parabolic subcomplex with stabilizer Q such that $\delta_Q(1,g) > 1$. Let $u := \mathfrak{p}_\Upsilon(1)$ and $v := \mathfrak{p}_\Upsilon(g)$. Then $d_{C(\Upsilon)}(u,v) = \delta_\Upsilon(1,g) = \delta_Q(1,g) > 1$, so there is no proper parabolic subcomplex of Υ containing both u and v. Since u and v are vertices of Υ, every wall separating them cuts Υ, hence has reflection in $Q = \mathrm{Stab}(\Upsilon)$. Thus, $\mathrm{Pc}(\mathcal{M}(u,v))$ is a parabolic subgroup of Q.

Choose a combinatorial geodesic from 1 to $u^{-1}v$, whose length is $n := d_1(u,v)$. Let $s_i \in S$ be the label of the ith edge, and let r_i be reflection in that edge. Then $r_i = s_1 \cdots s_{i-1} s_i s_{i-1} \cdots s_1$. Conversely, $s_i = r_1 \cdots r_{i-1} r_i r_{i-1} \cdots r_1$, so if $T := \{s_1, \ldots, s_n\} \subset S$ then $W_T = W(r_1, \ldots, r_n)$. Translating by u, the successive reflections along a d_1–geodesic from u to v are ur_iu^{-1}, so $v = (ur_nu^{-1}) \cdots (ur_1u^{-1})u$, so $u^{-1}v = r_n \cdots r_1 = s_1 \cdots s_n$. Thus, $v \in uW_T = uW(r_1, \ldots, r_n) = W(ur_1u^{-1}, \ldots, ur_nu^{-1})u \subset \mathrm{Pc}(\mathcal{M}(u,v))u \subset Qu$. Now, $uW_T \subset Qu \subset \Upsilon^{(0)}$, so $u\Sigma_T$ is a parabolic subcomplex of Υ. It contains both u and v, so it is not a proper subcomplex of Υ, so $\mathrm{Pc}(\mathcal{M}(u,v)) = uW_Tu^{-1} = \mathrm{Stab}(u\Sigma_T) = \mathrm{Stab}(\Upsilon) = Q$.

By Lemma 5.6, $\mathcal{M}(u,v) \subset \mathcal{M}(1,g)$. Thus, a parabolic subgroup $Q \in \mathrm{BigProj}(g)$ can be specified as the parabolic closure of a set of reflections corresponding to some subset of the walls separating 1 and g. There are only $2^{|g|_S} = 2^{|\mathcal{M}(1,g)|}$–many distinct subsets of $\mathcal{M}(1,g)$, so $|\,\mathrm{BigProj}(g)| \leq 2^{|g|_S}$. □

Proposition 5.45 (Criterion for reducibility/normalizing)**.** *Let (W,S) be a finite rank Coxeter system, and let $G \leq W$ be a subgroup generated by a finite set S_G. For all $c > 0$ there exists $R > 0$ such that if there exists $g \in G$ and a pre-wide parabolic subgroup P with $|g|_G \geq R$ and $d_P(1,g) \geq c|g|_G$, then G contains a nontrivial element normalizing some pre-wide parabolic subgroup.*

Proof. Suppose not, so there is some c for which the conditions are satisfied for arbitrarily large R, but G contains no nontrivial element normalizing a pre-wide parabolic subgroup. Let r be the smallest rank for which there exists $a > 0$ such that there are $g \in G$ of unbounded length for which there exists a pre-wide parabolic P of rank r with $d_P(1,g) \geq a|g|_G$. Since the ranks of pre-wide parabolics are between 2 and $|S|$, to establish that such a minimal r exists it is enough to argue that we are not minimizing over the empty set. To see this, take a to be the offending c. Then there are arbitrarily long $g \in G$ such that there exists a pre-wide P satisfying $d_P(1,g) \geq c|g|_G$. Since these elements g have unbounded lengths, there are infinitely many of them, but only finitely many possible ranks of the corresponding P, so some rank reoccurs infinitely often.

Notice that if $0 < a' \leq a$ then $d_P(1,g) \geq a|g|_G \geq a'|g|_G$, so we may assume $0 < a < 1$.

For every $q < r$ and every $b > 0$, define:

$$R(q,b) := \sup\{R \mid \exists g \in G,\ \text{pre-wide } P, \mathrm{rank}(P) = q \text{ and } d_P(1,g)/b \geq |g|_G > R\}$$

Minimality of r implies $R(q,b) < \infty$ for all $q < r$ and $b > 0$. Recalling that $\Delta := \max_{\text{spherical } T \subset S} \mathrm{diam}_{d_1} W_T$ and $\sup \emptyset = -\infty$, choose an integer:

$$L > \max\left\{\frac{4\Delta}{a}, \sup_{2 \leq q < r} R(q, \frac{a}{4\lfloor (r-1)/2 \rfloor})\right\}$$

By construction, there are arbitrarily long elements $g \in G$ such that there exists a pre-wide parabolic P of rank r with $d_P(1,g) \geq a|g|_G$. Take an S_G–geodesic word representing such a g, and cut it into $n := \lfloor |g|_G / L \rfloor$–many consecutive subwords of length L, plus a possibly empty remainder word:

$$g = g_1 \cdots g_n g_{n+1}$$

Let $p_0 := 1$, and for $i \leq n$, let $p_i := p_{i-1}g_i$ be the ith prefix of g, which has length Li. For $1 \leq i \leq n+1$ let $P_i := p_{i-1}^{-1} P p_{i-1}$.

Suppose for some $1 \leq i \leq n$ we have $d_{P_i}(1,g_i) \geq La/2$ and $\delta_{P_i}(1,g_i) = 1$. Then $d_{P_i}(1,g_i) = d_{P_i}(1,g_i)/\delta_{P_i}(1,g_i) > \Delta$, so Lemma 5.22 produces a nonspherical irreducible parabolic $Q \lneq P_i$ with:

$$d_Q(1,g_i) \geq \frac{d_{P_i}(1,g_i) - \Delta}{\lfloor (r-1)/2 \rfloor} > \frac{La/4}{\lfloor (r-1)/2 \rfloor}$$

Note in this case that $2 \leq \mathrm{rank}(Q) < \mathrm{rank}(P) = r$, so the denominators are positive. Rearranging gives:

$$\frac{d_Q(1,g_i)}{\frac{a}{4\lfloor (r-1)/2 \rfloor}} > L \tag{22}$$

Since P_i is pre-wide, Lemma 5.12 says Q is pre-wide, so (22) shows that $R(\mathrm{rank}(Q), \frac{a}{4\lfloor (r-1)/2 \rfloor}) \geq L$, which contradicts the definition of L. Thus:

$$\forall 1 \leq i \leq n,\ d_{P_i}(1,g_i) \geq La/2 \implies \delta_{P_i}(1,g_i) > 1 \tag{23}$$

Now consider:

$$d_P(1,g) \leq \sum_{i=1}^{n} d_P(p_{i-1},p_i) + d_P(p_n,g) = \sum_{i=1}^{n+1} d_{P_i}(1,g_i)$$

For all i we have $d_{P_i}(1,g_i) \leq |g_i|_S \leq \lambda_G |g_i|_G \leq \lambda_G L$.

Suppose there are $(n-k)$–many indices between 1 and n with $d_{P_i}(1,g_i) < La/2$. Then:

$$anL \leq a|g|_G \leq d_P(1,g) \leq (n-k)La/2 + (k+1)\lambda_G L$$

Since $\lambda_G \geq 1$ and $0 < a < 1$, we have $\lambda_G - a/2 > 0$, so this expression can be rearranged to bound k:

$$k \geq \frac{an/2 - \lambda_G}{\lambda_G - a/2} = \frac{(a/2)\lfloor \frac{|g|_G}{L} \rfloor - \lambda_G}{\lambda_G - a/2}$$

Conclude that the number of indices i for which $d_{P_i}(1,g_i) \geq La/2$ grows linearly with $|g|_G$. Applying (23), the number of indices i for which $\delta_{P_i}(1,g_i) > 1$ grows linearly with $|g|_G$.

Take the union of the sets $\mathrm{BigProj}(h)$ of Lemma 5.44 as h ranges over the L–sphere in G with respect to $|\cdot|_G$. The sphere contains less than $(2|S_G|)^L$–many elements h, and Lemma 5.44 implies $|\mathrm{BigProj}(h)| \leq 2^{\lambda_G L}$, so this is a finite collection of nonspherical irreducible parabolic subgroups whose cardinality depends on L, not on $|g|_G$. Thus, whenever $|g|_G$ is long enough and P is its corresponding parabolic, the decomposition $g = g_1 \cdots g_{n+1}$ has more indices $1 \leq i \leq n$ for which $\delta_{P_i}(1,g_i) > 1$ than there are potential distinct parabolics P_i, so there are indices $i < j$ such that $P_i = P_j$. Equivalently, $p_{i-1}^{-1} P p_{i-1} = p_{j-1}^{-1} P p_{j-1}$, so:

$$p_{j-1} p_{i-1}^{-1} \in N_W(P) \cap G$$

This element is nontrivial, since p_{i-1} and p_{j-1} are distinct prefixes of a geodesic word. This contradicts the supposition that G has no nontrivial element normalizing a pre-wide parabolic. □

For any $g \in W$, let α_g be a combinatorial geodesic from 1 to g in Σ, and let α_g^* be the greedy path in the shortcut graph of α_g. Choose a realization $\overline{\alpha}_g^*$ of α_g^* in $\widehat{\Sigma}$. Let x and y be a pair of adjacent vertices in α_g^*. Either x and y are adjacent in $\overline{\alpha}_g^*$ and in α_g, or there is a cone path shortcut $x - v_\Upsilon - y$ in $\overline{\alpha}_g^*$ for some wide parabolic subcomplex Υ containing x and y. Since parabolic subcomplexes are d_1–convex, the subsegment of α_g between x and y is contained in Υ. Call such a segment of α_g a *wide segment*.

The walls $\mathcal{M}(1,g)$ separating 1 and g form a finite poset under inclusion of the halfspaces containing 1. Consider a Dilworth partition of $\mathcal{M}(1,g)$ into $D \leq \dim(X_{\mathrm{NR}})$ many chains:

$$\mathcal{M}(1,g) = \mathcal{K}_1 \sqcup \cdots \sqcup \mathcal{K}_D$$

For each wide segment α of α_g, let $\mathcal{M}(\alpha)$ be the set of walls transverse to its edges. Pick an index j such that $\mathcal{M}(\alpha) \cap \mathcal{K}_j$ has maximal cardinality, and set $\mathcal{K}_\alpha := \mathcal{K}_j$. Maximality ensures $\mathcal{K}_\alpha$ contains at least the average number of $\mathcal{M}(\alpha)$ walls per chain:

$$\frac{|\mathcal{M}(\alpha)|}{D} \leq |\mathcal{M}(\alpha) \cap \mathcal{K}_\alpha| \tag{24}$$

Partition α_g into subsegments of the following types.

- E_0 consists of single ordinary $\Sigma^{(1)}$ edges corresponding to an edge in α_g^*.
- E_1 consists of wide segments α of α_g such that $|\mathcal{M}(\alpha) \cap \mathcal{K}_\alpha| = 1$.
- E_2 are the remaining wide subsegments α of α_g, which have $|\mathcal{M}(\alpha) \cap \mathcal{K}_\alpha| \geq 2$.

Order the segments in E_2 according to their appearance along α_g, say $E_2 = \{\alpha_1, \ldots, \alpha_m\}$. For each α_i, let $\mathcal{V}_i = \mathcal{M}(\alpha_i) \cap \mathcal{K}_{\alpha_i}$, and define:

$$P_i = \mathrm{Pc}(\mathcal{V}_i)$$

Because $\mathcal{V}_i$ is a chain with at least two walls, P_i is a nonspherical irreducible parabolic subgroup. Further, all walls of $\mathcal{V}_i$ cut any wide parabolic subcomplex Υ_i containing α_i, so their reflections lie in its stabilizer:

$$P_i \leq \mathrm{Stab}(\Upsilon_i)$$

By Lemma 5.12, each P_i is pre-wide.

Using the above notation, we establish the following statement.

Lemma 5.46. *Let (W, S) be a finite rank Coxeter system, let $G \leq W$ be a subgroup generated by a finite set S_G. Given $g \in G$, identify it with an S_G–shortest representative word, which is the label of a geodesic β_g from 1 to g in the Cayley graph of G with respect to S_G. Let α_g be a d_1–geodesic from 1 to g in $\Sigma^{(1)}$. With notation as above, for each $\alpha_i \in E_2$, there exists a subword g_i of g and a conjugate Q_i of P_i such that:*

(1) $\forall i$, $Dd_{Q_i}(1, g_i) \geq |\alpha_i|$
(2) $\sum_i |g_i|_G \leq (1 + \lambda_G/2)D|g|_G$

Proof. Let $1 = x_0, \ldots, x_n = g$ be the consecutive vertices of the geodesic β_g in the Cayley graph of G. For every wall $M \in \mathcal{M}(1, g)$, orient its halfspaces by choosing $1 \in M^-$ and $g \in M^+$. Define the first-entry index of M by:

$$s(M) := \min\{j \mid x_j \in M^+\}$$

Order any chain extracted from $\mathcal{M}(1, g)$ according to our choices for M^- and M^+, so that chain order is consistent with first-entry indices.

For each $\alpha_i \in E_2$, let M_i and N_i be the first and last walls of $\mathcal{V}_i$, and set:

$$a_i := s(M_i) - 1 \quad \text{and} \quad b_i = s(N_i)$$

Let $g_i := x_{a_i}^{-1} x_{b_i} \in G$ and $Q_i := x_{a_i}^{-1} P_i x_{a_i}$. Note that every wall of $\mathcal{V}_i$ separates x_{a_i} and x_{b_i} and has corresponding reflection in P_i, so Lemma 5.6 gives:

$$d_{Q_i}(1, g_i) = d_{P_i}(x_{a_i}, x_{b_i}) \geq |\mathcal{V}_i| \geq \frac{|\mathcal{M}(\alpha_i)|}{D} = \frac{|\alpha_i|}{D}$$

This proves Item (1).

Let β_i be the subsegment of β_g from x_{a_i} to x_{b_i}, which is nontrivial, since $a_i < b_i$. We can organize the segments β_i according to the D–many chains in $\mathcal{M}(1, g)$, as follows. Suppose β_i and β_j, with $i < j$, are defined using the same chain, i.e. $\mathcal{K}_{\alpha_i} = \mathcal{K}_{\alpha_j}$, with $\mathcal{V}_i = \mathcal{M}(\alpha_i) \cap \mathcal{K}_{\alpha_i}$ and $\mathcal{V}_j = \mathcal{M}(\alpha_j) \cap \mathcal{K}_{\alpha_j}$. Since α_g is a d_1–geodesic, the last wall N_i of $\mathcal{V}_i$ occurs in $\mathcal{K}_{\alpha_i}$ before the first wall M_j of $\mathcal{V}_j$, which implies $s(N_i) \leq s(M_j)$, so $b_i \leq a_j + 1$. A strict inequality $b_i < a_j + 1$ would mean that β_i and β_j have disjoint interiors, and equality would mean the last edge of β_i is the first edge of β_j. This could happen; a single S_G–edge has vertices whose d_1–distance in Σ is at most λ_G, so such vertices could be separated by as many as λ_G–many distinct walls—in particular by both N_i and M_j.

Pushing this reasoning further, if $i < j_1 < \cdots < j_m < k$ and β_i, β_{j_h}, and β_k are all defined using the same chain of $\mathcal{M}(1, g)$, and if β_i and β_k share an edge e, then a d_1–geodesic in Σ between the endpoints of e crosses the last wall of $\mathcal{V}_i$ and the first wall of $\mathcal{V}_k$, so also crosses every wall of every $\mathcal{V}_{j_h}$. Since $|\mathcal{V}_{j_h}| \geq 2$, such a d_1–geodesic, which has length at most λ_G, crosses at least $(2 + 2m)$–many walls, so $m \leq \lambda_G/2 - 1$. Thus, the edge e belongs to at most $(\lambda_G/2 + 1)$–many of the β–paths from a single chain. Summing the lengths of the β coming from a single chain, we have at most $|g|_G$–many edges each counted with multiplicity at most $\lambda_G/2 + 1$. There are D–many distinct chains to sum over, so we obtain the estimate of Item (2):

$$\sum_i |g_i|_G = \sum_i |\beta_i| \leq (1 + \lambda_G/2)D|g|_G \qquad \square$$

Proposition 5.47. *Let (W, S) be a finite rank Coxeter system, let $G \leq W$ be a subgroup generated by a finite set S_G. Assume G is undistorted. There exists a constant K such that the following statements hold. Given nontrivial $g \in G$, let $\mathcal{M}(1, g) = \mathcal{K}_1 \sqcup \cdots \sqcup \mathcal{K}_D$ be a partition of $\mathcal{M}(1, g)$ into $D \leq \dim(X_{\mathbb{NR}})$–many disjoint chains, and let E_0, E_1, and E_2 be the partition of a combinatorial geodesic α_g of g as discussed above. For each $\alpha_i \in E_2$, let P_i be the associated pre-wide parabolic subgroup, let $g_i \in G$ be the associated subword of a fixed S_G–minimal word representing g, and let Q_i be the conjugate of P_i, as given by Lemma 5.46.*

(1) $|E_0| + |E_1| + |E_2| \leq K\hat{d}(1, g)$
(2) $\frac{|g|_G}{KD} \leq |E_0| + |E_1| + \sum_{i=1}^{|E_2|} d_{Q_i}(1, g_i)$

Proof. Recall, from Corollary 4.26, that realizations of greedy paths are uniform quasigeodesics in $\widehat{\Sigma}$. Since G is undistorted, by hypothesis, we can choose a constant K such that (we can dispense the additive error by enlarging K):

$$\frac{1}{K}|g|_S \leq |g|_G \leq K|g|_S \quad \text{and} \quad |\overline{\alpha}_g^*| \leq K\hat{d}(1, g).$$

Since $|E_0| + |E_1| + |E_2| = |\overline{\alpha}_g^*|$, the latter gives Item (1).

For Item (2), we first observe:

$$|g|_S = |\alpha_g| = |E_0| + \sum_{\alpha\in E_1} |\alpha| + \sum_{\alpha\in E_2} |\alpha|$$

Recall that $\alpha \in E_1$ means α is a wide subsegment of α_g for which $|\mathcal{M}(\alpha) \cap \mathcal{K}_\alpha| = 1$, which, by (24), implies $|\alpha| = |\mathcal{M}(\alpha)| \le D$. For $\alpha_i \in E_2$, Lemma 5.46 (1) says $|\alpha_i| = |\mathcal{M}(\alpha_i)| \le D d_{Q_i}(1, g_i)$. This yields the second statement:

$$\frac{|g|_G}{K} \le |g|_S \le |E_0| + D|E_1| + D\sum_{i=1}^{|E_2|} d_{Q_i}(1, g_i) \le D\left(|E_0| + |E_1| + \sum_{i=1}^{|E_2|} d_{Q_i}(1, g_i)\right) \qquad \square$$

Proof of Theorem 5.43. Every stable subgroup $G \le W$ is necessarily undistorted and purely loxodromic. We prove the opposite direction: suppose G is undistorted and purely loxodromic. Since finitely generated Coxeter groups are virtually torsion-free [52, Corollary 6.12.12], every subgroup of W has a finite-index torsion-free subgroup. Since stability is a virtual property, up to replacing G by a finite-index subgroup, we may assume every nontrivial element of G has infinite order, hence is loxodromic. Supposing that G is not stable, we derive a contradiction.

Fix a finite generating set S_G of G. By Corollary 5.40, if G is not stable then the orbit map of G into $\widehat{\Sigma}$ is not a quasiisometric embedding. It is Lipschitz, so the failure is due to elements of G being too short in $\widehat{\Sigma}$; specifically, there exists a sequence $h_n \in G$ such that:

$$r_n := \frac{|h_n|_G}{\hat{d}(1, h_n)} \to \infty$$

The goal is to show that once r_n is large enough we can apply Proposition 5.45 to h_n.

Let $K \ge 1$ be the constant of Proposition 5.47. Choose $0 < c < \frac{1}{3(\lambda_G+2)K\dim(X_{\mathrm{NR}})^2}$, and let $R := R(c) \ge 1$ be the constant of Proposition 5.45.

For any n large enough that $r_n > 4K^3R\dim(X_{\mathrm{NR}})$, take $g := h_n$. We use the notation of Proposition 5.47 with respect to this g. Recall that $\mathcal{M}(1, g)$ was partitioned into D–many disjoint chains, for some D depending on g with $D \le \dim(X_{\mathrm{NR}})$. We claim:

$$\frac{|g|_G}{2KD} < \sum_{i=1}^{|E_2|} d_{Q_i}(1, g_i) \tag{25}$$

This follows from Proposition 5.47, because by Item (2), the alternative is that $|E_0| + |E_1|$ is at least half of $|g|_G/KD$, but then using Item (1) we could derive the following contradiction:

$$4K^3R\dim(X_{\mathrm{NR}}) < r_n = \frac{|g|_G}{\hat{d}(1, g)} \le \frac{K|g|_G}{|E_0| + |E_1| + |E_2|} \le \frac{K|g|_G}{|g|_G/2KD + |E_2|} \le 2K^2D \le 2K^2\dim(X_{\mathrm{NR}})$$

Consider the collection of subwords g_i of g provided by Lemma 5.46. Partition their indices:

$$J^+ := \{i \mid d_{Q_i}(1, g_i) \ge c|g_i|_G\} \qquad J^- := \{i \mid d_{Q_i}(1, g_i) < c|g_i|_G\}$$

In view of (25), we claim:

$$\frac{|g|_G}{4KD} \le \sum_{i\in J^+} d_{Q_i}(1, g_i) \tag{26}$$

If not, then by Lemma 5.46 (2):

$$\frac{|g|_G}{4KD} \le \sum_{i\in J^-} d_{Q_i}(1, g_i) < \sum_{i\in J^-} c|g_i|_G \le c(1 + \lambda_G/2)D|g|_G$$

This would give the contradiction:

$$3(\lambda_G + 2)KD^2 \le 3(\lambda_G + 2)K\dim(X_{\mathrm{NR}})^2 < 1/c < 2(\lambda_G + 2)KD^2$$

Next, suppose $|g_i|_G < R$ for every $i \in J^+$. Then, since $d_{Q_i}(1, g_i) \le d_1(1, g_i) \le K|g_i|_G < KR$, the bound (26) gives the estimate:

$$\frac{|g|_G}{4KD} \le \sum_{i\in J^+} d_{Q_i}(1, g_i) \le KR|J^+| \le KR|E_2|$$

This estimate combined with Proposition 5.47 (1) gives the contradiction:

$$4K^3R\dim(X_{\mathrm{NR}}) < r_n = \frac{|g|_G}{\hat{d}(1,g)} \leq \frac{4K^2DR|E_2|}{\hat{d}(1,g)} \leq \frac{4K^3DR|E_2|}{|E_0|+|E_1|+|E_2|} \leq 4K^3DR \leq 4K^3R\dim(X_{\mathrm{NR}})$$

Conclude that there exists some $i \in J^+$ such that $|g_i|_G \geq R$. By definition of J^+, we have $d_{Q_i}(1, g_i) \geq c|g_i|_G$. Since Q_i is a conjugate of P_i, and P_i is pre-wide, Q_i is also pre-wide. Thus, Proposition 5.45 says that G contains a nontrivial element normalizing some pre-wide parabolic subgroup P. Such an element fixes the cone vertex $\rho(P)$ of $\widehat{\Sigma}$ corresponding to the unique wide parabolic subcomplex whose stabilizer is $N_W(P)$, as described in Lemma 5.24, so this contradicts the assumption that every non-trivial element of G acts loxodromically on $\widehat{\Sigma}$. □

AI use

The authors conceived the arguments, developed the proofs, and wrote the entirety of this article. Over the course of the project several versions of ChatGPT5 were used as interactive sounding boards to explore possible proof strategies and test arguments.

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